

2027 Formula Bharat Rules Booklet

PART 1: INTRODUCTION

Formula Bharat is an exclusive student design competition which focuses on safety, good engineering practice, engineering design, education and creating opportunities for participating students.

Throughout this entire booklet, “Formula Bharat” shall be referred to as “FB”.

Formula Bharat is hosted by Curiosum Tech Private Limited.

PART 2: ADAPTATION TO FORMULA STUDENT RULES

The following rule book is an adaptation to the FS-Rules_2026_v1.1 rules booklet, published by Formula Student Germany. The purpose of the adaptation is to eliminate the requirement for a separate addendum for the Formula Bharat competition. Changes, inserts and modifications have been highlighted accordingly so that participating teams will be aware of those required for the Formula Bharat competition.

Teams who have participated at previous Formula Bharat competitions, are encouraged to thoroughly familiarize themselves with the new version as there have been several changes implemented by Formula Student Germany.

Any mention of a competition handbook may be referred to as the competition website.

Legend:

- Changes issued by FS2026 Rules (by FSG): **Text in blue**
- Updates by FB: **Text in green**
- DV specific additions: **Text in Purple**
- Eliminations: Text crossed out

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CHANGELOG

Rule	Ver	By	Change
A1.2.6	2027 1.1	FB	Addition - Driverless Cup (DC)
A1.2.7	2027 1.1	FB	Addition - Driverless Cup (DC)
A1.2.8	2027 1.1	FB	Addition - Driverless Cup (DC)
A1.2.9	2027 1.1	FB	Addition - Driverless Cup (DC)
A1.3	2027 1.1	FB	Addition - Driverless Cup (DC)
A2.3.2	2027 1.0	FB	Updated as per FB2027
A3.8.2	2027 1.1	FB	Addition - Driverless Cup (DC)
A4.3.1	2027 1.0	FB	Clarified that students from any academic discipline are eligible to participate
A4.9	2027 1.1	FB	Addition - Autonomous System Responsible (ASR)
A5.1.1	2027 1.1	FB	Addition - BPEFS, TVSD, ASRQ and ASF
A5.3.1	2027 1.1	FB	Addition - BPEFS, TVSD, ASRQ and ASF
A5.3.2	2027 1.1	FB	Addition - BPEFS, TVSD, ASRQ and ASF
A5.3.3	2027 1.0	FB	Clarified penalties for individual session
A5.6.2	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
A5.6.3	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
A5.6.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
A5.6.5	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
A6.5.3	2026 1.1	FSG	Clarified rules regarding hazardous work in the pit
A6.7.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
T1.2.1	2026 1.1	FSG	Clarified how to prove compliance to fire retardant standard
T1.4	2027 1.1	FB	Addition - Driving Mode Definitions
T2.3.2	2026 1.1	FSG	Clarified rule
T3.3.4	2026 1.1	FSG	Prohibited use of e.g.FDM in steering, brake and suspension system
T3.4.2	2026 1.1	FSG	Require that adhesive film, used to bond fiber layers and core, must be documented
T3.4.3	2026 1.1	FSG	Clarified definition of parallel fibers
T3.4.7	2026 1.1	FSG	Prohibited using unidirectional fibers in outer layer of primary structure
T3.5.3	2026 1.1	FSG	Added rig compliance, as used with in SES, to the rules
T3.5.4	2026 1.1	FSG	Clarified where derived material properties from single test panel may not be used
T3.11.5	2026 1.1	FSG	Added limitation for maximum (total) size of cut-out in area of front hoop bracing and front bulk head support
T3.14.3	2026 1.1	FSG	Added keep out zone between TS and rear impact structure
T3.15.3	2026 1.1	FSG	Clarified use of (steel) critical fasteners
T3.15.6	2026 1.1	FSG	Simplified requirement on bolts for primary structure attachments between panels/plates, no'smaller ,but more bolts may be used' anymore

Rule	Ver	By	Change
T3.16.7	2026 1.1	FSG	Clarified front bulk head measurement and condition for no testing of standard FSAE IAs
T3.16.7	2027 1.0	FB	Clarified 24Mpa requirement is for “standard honeycomb IA”
T3.18.2	2026 1.1	FSG	Clarified when copy of intended structure for IA test fixture is required, and test fixture (i.e. front bulkhead, front hoop bracing and front bulk head support representative structures) condition after test
T3.18.4	2026 1.1	FSG	Clarified that using standard IA may still require physical testing if non crushable objects are in front view of the AIP and are not designed to detach or move rearward of the AIP during a frontal impact
T3.19.2	2026 1.1	FSG	Require that sensors and aerodynamic devices are attached rearwards of the AIP
T4.12	2027 1.0	FB	Addition - Driver accommodation
T5.7.2	2026 1.1	FSG	Clarified head restraint size
T6.1.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
T6.1.10	2026 1.1	FSG	Included pedal face to the rule
T6.1.12	2026 1.1	FSG	Require positive stop for adjustable pedals
T7.2.7	2026 1.1	FSG	Relaxed temperature rating
T7.3.2	2026 1.1	FSG	Clarified rule
T8.2.1	2026 1.1	FSG	Changed height restriction
T9.1.6	2027 1.0	FB	Section Removed
T10.1.1	2026 1.1	FSG	Clarified rule
T10.1.2	2026 1.1	FSG	Require steel critical fasteners with marked strength classification Clarified that OEM parts must be unmodified
T10.2.3	2026 1.1	FSG	Clarified that OEM parts must be unmodified
T10.2.7	2026 1.1	FSG	Clarified positive locking exception for brake disc floaters
T11.1.4	2026 1.1	FSG	Clarified that OEM parts must be unmodified
T11.1.7	2027 1.1	FB	Addition
T11.5.2	2026 1.1	FSG	Changed in accordance with FSAE rules
T11.5.4	2026 1.1	FSG	Clarified rule
T11.6.4	2026 1.1	FSG	Clarified rule
T11.6.9	2027 1.0	FB	Removed and added in IN section with clarification
T11.9.1	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
T11.10	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
T11.10.3	2026 1.1	FSG	Clarified rule
T11.13.1	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
T12.2.1	2027 1.0	FB	Updated as per Indian Universities
T12.2.2	2026 1.1	FSG	Added additional requirements for location of university name
T13.3.2	2026 1.1	FSG	Clarified rule
T13.3.1	2026 1.1	FSG	Added new rule for driver communication equipment
T14	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)

Rule	Ver	By	Change
T15	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
CV1.2.2	2027 1.0	FB	Section Removed
CV1.2.3	2027 1.0	FB	Section Removed
CV1.2.4	2027 1.0	FB	Section Removed
CV1.3.3	2026 1.1	FSG	Clarified rule
EV1.2.2	2026 1.1	FSG	Specified high current path applies only to TS circuitry
EV2.2.2	2026 1.1	FSG	Restricted the maximum TS current
EV2.2.3	2026 1.1	FSG	Clarified that maximum current applies for recuperation
EV3.1.1	2026 1.1	FSG	Relaxed requirements for insulating TS enclosures
EV4.3.5	2027 1.0	FB	Clarified rule
EV4.6.5	2026 1.1	FSG	Specified type of allowed over current protection
EV4.9.5	2027 1.0	FB	Added requirement for discharge circuit
EV4.11.3	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
EV4.11.5	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
EV5.3.3	2026 1.1	FSG	Clarified rule
EV5.3.4	2026 1.1	FSG	Clarified rule
EV5.4.7	2026 1.1	FSG	Clarified rule
EV5.4.8	2026 1.1	FSG	Clarified that LED voltage indicators must always illuminate at full brightness
EV5.5.1	2026 1.1	FSG	Clarified attachment points of TSAC
EV5.5.2	2026 1.1	FSG	Clarified impact protection of TSAC
EV5.5.7	2026 1.1	FSG	Introduced specific requirements for TSAC structural parts, manufactured using additive processes
EV5.5.9	2026 1.1	FSG	Clarified attachment points of TSAC
EV5.5.11	2026 1.1	FSG	Limited number of cells mechanically attached in series if no stress analysis is provided to 6
EV5.5.15	2026 1.1	FSG	Clarified attachment points of TSAC
EV5.5.15	2027 1.0	FB	Minimum 4 attachment points required
EV5.8	2026 1.1	FSG	Updated requirements for cell temperature sensor placement
EV5.8.12	2026 1.1	FSG	Relaxed requirements regarding loss of a measurement connection
EV5.8.13	2026 1.1	FSG	Require that it is always possible to disconnect one of each measurements during technical inspection
EV7.1.6	2026 1.1	FSG	Clarified rule
EV7.1.9	2027 1.0	FB	Addition - related to handcart
EV7.1.10	2027 1.0	FB	Addition - TSMPs required for charger
EV7.1.11	2027 1.0	FB	Addition - requirement and location for indicators, switches, buttons, and connectors of the charger
EV7.2.2	2026 1.1	FSG	Require that open SDC during charging must prevent (re)-activation of TS
IN1.1.1	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
IN1.2.2	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
IN1.2.7	2026 1.1	FSG	Shortened rule

Rule	Ver	By	Change
IN1.4.1	2026 1.1	FSG	ASR only required for Autonomous System Inspection
IN1.5.1	2026 1.1	FSG	Replaced sensor adjustment by the concept of sensor repositioning
IN1.5.1	2026 1.1	FSG	Addition - allowed replacement of LV batteries
IN1.5.1	2026 1.1	FSG	Addition - Clarified how to handle modifications that are not listed
IN2.1.1	2026 1.1	FSG	Addition - related to fire extinguishers
IN4.1.2	2027 1.0	FB	Clarified requirement and test for BSPD current sensor
IN6	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
IN9.2.1	2026 1.1	FSG	Added that fans must be switched on during Rain Test
IN9.2.7	2026 1.1	FSG	Added that TS must stay on during full duration of Rain Test
IN11.1.4	2026 1.1	FSG	Addition - Procedure of the Brake Test
IN11.1.7	2026 1.1	FSG	Added maximum tire pressure for Brake Test
IN11.2	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
IN12.1.8	2026 1.1	FSG	Vehicle needs to be able to execute Inspection Mission during Post Inspection
IN12.1.9	2026 1.1	FSG	Vehicle needs to be able to reach R2D at least 15 min after any dynamic event
S1.1.1	2026 1.1	FSG	Clarified BPP objective
S1.1.2	2026 1.1	FSG	Restricted what can be considered as part of the business model
S1.1.3	2026 1.1	FSG	Teams must show how the product is derived from the current vehicle
S1.1.4	2026 1.1	FSG	Allowed teams to assign role to the judges
S1.4	2026 1.1	FSG	Addition - Executive & Financial Summary
S1.6.2	2026 1.1	FSG	Updated how time limit is handled
S1.8.1	2026 1.1	FSG	Updated BPP scoring categories wrt Executive & Financial Summary
S1.8.1	2027 1.2	FB	Scoring Corrected
S1.8.2	2026 1.1	FSG	Clarified penalty for not meeting BPP requirements
S2.3.3	2026 1.1	FSG	Clarified format of the change log
S2.4.5	2026 1.1	FSG	Clarified what is considered a fastener
S2.5.2	2026 1.1	FSG	Defined CCBOM system for 2027
S2.5.2	2027 1.0	FB	Corrected year to 2027
S2.5.8	2026 1.1	FSG	Clarified rule
S2.5.9	2026 1.1	FSG	Clarified how modified bought parts are treated wrt. carbon footprint
S3.2	2026 1.1	FSG	Replaced EDR and DSS with new TVSD
S3.3.7	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
S3.3.8	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
S3.6	2027 1.0	FB	Updated ED scoring
D1.1.10	2026 1.1	FSG	Added definition of turning diameter according to Oxford English Dictionary
D1.3.1	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D1.5.1	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D2.1.2	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D2.2.8	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)

Rule	Ver	By	Change
D2.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D2.5	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D2.6	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D2.7	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D4.2.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D4.2.5	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D5.2.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D6.1.3	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D6.2.3	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D6.2.4	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D8	2027 1.1	FB	Addition - Requirements for Driverless Cup (DC)
D9	2026 1.1	FSG	Dynamic events scoring Standardised scoring and moved to its own section Unified \scoring formula and adapted Tmax Changed [EFmax] definition and scoring formula
D9.1.1	2027 1.2	FB	Formula corrected

ABBREVIATIONS

AIP	Anti Intrusion Plate	IAD	Impact Attenuator Data
AIR	Accumulator Isolation Relay	IA	Impact Attenuator
AMS	Accumulator Management System	IMD	Insulation Monitoring Device
AMI	Autonomous Mission Indicator	IRP	Inspection Responsible Person
APPS	Accelerator Pedal Position Sensor	LCA	Life Cycle Assessment
AS	Autonomous System	LPI	Low Pressure Injection
ASB	Autonomous System Brake	LV	Low Voltage
ASES	Accumulator Structural Equivalency Spreadsheet	LVMS	Low Voltage Master Switch
ASF	Autonomous System Form	LVS	Low Voltage System
ASMS	Autonomous System Master Switch	OC	Off-course
ASR	Autonomous System Responsible	OEM	Original Equipment Manufacturer
ASRQ	ASR Qualification	PCB	Printed Circuit Board
ASSI	Autonomous System Status Indicator	PS	Photographic Submissions
BOM	Bill of Material	R2D	Ready-to-drive
BOTS	Brake Over-Travel Switch	RES	Remote Emergency System
BPEFS	Business Plan Executive & Financial Summary	RMS	Root Mean Square
BPP	Business Plan Presentation Event	SCS	System Critical Signal
BPPV	Business Plan Pitch Video	SDC	Shutdown Circuit
BSPD	Brake System Plausibility Device	SE3D	Structural Equivalency 3D Model
CCBOM	Costed Carbonized Bill of Material	SES	Structural Equivalency Spreadsheet
CGS	Compressed Gas System	SVCR	Second Year Vehicle Compliance Report
CO ₂ e	Carbon Dioxide Equivalents	TMI	Team Member Information
CRD	Cost Report Documents	TPS	Throttle Position Sensor
CV	Internal Combustion Engine Vehicle	TS	Tractive System
DC	Driverless Cup	TSAC	Tractive System Accumulator Container
DI	Direct Injection	TSAL	Tractive System Active Light
DNA	Did Not Attempt	TSMP	Tractive System Measuring point
DNF	Did Not Finish	TSMS	Tractive System Master Switch
DOO	Down or Out	TVSD	Technical Vehicle System Documentation
DQ	Disqualified	USS	Unsafe Stop
DV	Driverless Vehicle	VSV	Vehicle Status Video
EBS	Emergency Brake System		
ECU	Electronic Control Unit		
EI	Flexural Rigidity		
ESF	Electrical System Form		
ESO	Electrical System Officer		
ESOQ	Electrical System Officer Qualification		
ESS	Emergency Stop Switch		
ETC	Electronic Throttle Control		
EV	Electric Vehicle		
FDM	Fused Deposition Modeling		
GWP	Global Warming Potential		
HPI	High Pressure Injection		

HV High Voltage
HVD High Voltage Disconnect

A ADMINISTRATIVE REGULATIONS

A1 COMPETITION OVERVIEW

A1.1 Competition Objective

A1.1.1 The competition challenges teams of university students to conceive, design, fabricate, develop and compete with small, formula style, race cars.

A1.2 Competition Procedure

A1.2.1 The competition is split into the following categories:

- Internal Combustion Engine Vehicle (CV)
- Electric Vehicle (EV)

A1.2.2 All vehicles must meet the requirements defined in chapters T and either EV or CV, depending on their drivetrain type.

A1.2.3 The competition starts with a series of technical inspections described in chapter IN to check the vehicle for safety and compliance with the rules.

A1.2.4 The competition is divided into a series of static and dynamic events described in chapters S and D.

A1.2.5 Maximum points are awarded as described in Table 3.

A1.2.6 Teams registered for Formula Bharat 2027 under the Electric Vehicle (EV) Category can choose to take part in the Driverless Cup (DC).

A1.2.7 An Electric Vehicle (EV) Team that is not registered under the EV category for FB2027 can still choose to participate only in the Driverless Cup (DC).

A1.2.8 The maximum achievable points for the DC are listed in table 3.

A1.2.9 The team with the most overall points will win the competition for its class or the DC, respectively.

A1.3 Competition Information

A1.3.1 The competition specific rules and information are defined on the competition website.

A1.3.2 The official language of the competition is English.

A1.3.3 Every organizer of a competition based on this document bears the sole responsibility.

	CV & EV	DC
Static Events:		
Business Plan Presentation	75 Points	-
Cost & Manufacturing	100 Points	-
Engineering Design	150 Points	150 Points
Dynamic Events:		
Skid Pad	75 Points	-
Driverless (DV) Skidpad	-	75 Points
Acceleration	75 Points	-
Driverless (DV) Acceleration	-	75 Points
Autocross	100 Points	-
Driverless (DV) Autocross	-	100 Points
Endurance	325 Points	-
Efficiency	100 Points	-
Trackdrive	-	200 Points
Overall	1000 Points	600 Points

ScoreTable 3: Maximum points awarded

A2 VEHICLE ELIGIBILITY

A2.1 Student Competition

- A2.1.1 Vehicles entered into the competition must be conceived, designed and maintained by the student team members without direct involvement from external professional engineers, racers, machinists or related professionals.
- A2.1.2 The student team may use any information from professionals or from academics as long as the information is given as a discussion of alternatives with their pros and cons.
- A2.1.3 Professionals may not make design decisions or drawings.
- A2.1.4 Students should perform fabrication tasks wherever possible.

A2.2 First Year Vehicles

- A2.2.1 A vehicle may only be used for one year, counting from the first day onsite of its first competition.
- A2.2.2 To be classified as new, a vehicle must have a newly manufactured chassis.
- A2.2.3 If there is any question about whether or not the car is in fact a first year vehicle, it will be the sole responsibility of the team to produce such evidence as the organizers or judges may require.

A2.2.4 In the following case, the existing chassis may be used for one more consecutive year:

- Changing the powertrain from CV → EV

The team must inform the officials ahead of time if they intend to use this exception.

A2.3 Second Year Vehicles

A2.3.1 This rule is only applicable to teams participating in the Electric Category at Formula Bharat. The second year vehicle must be modified (if required) to comply with the rule book of that competition year.

A2.3.2 There will be no penalties applied for participating as a second year vehicle.

Note: For the **FB2027** competition, first year vehicles that participated at **FB2026** will be considered as a second year vehicle.

A2.3.3 Teams participating with a second year vehicle are required to submit a Second Year Vehicle Compliance Report (SVCR). The report must contain proof that the competition vehicle is not older than one year. Information presented must include details of the previous competitions attended by competition vehicle, details of the previous competition vehicle as in predecessor (if any), photographic details to prove significant changes in chassis as between competition vehicle and predecessor, any approved technical inspection documents etc. The competition vehicle will only be approved as a second year vehicle once the SVCR is approved.

A2.3.4 This rule does not apply to teams participating in the Combustion Category.

A3 RULES OF CONDUCT

A3.1 General Officials Authority

A3.1.1 The officials reserve the right to revise the schedule of the competition and/or interpret or modify the competition rules at any time and in any manner that is, in their sole judgment, required for safe and efficient operation.

A3.1.2 All team members are required to cooperate with, and follow all instructions from the officials.

A3.1.3 Official announcements shall be considered part of these rules.

A3.1.4 All guidelines and clarifications posted on the competition website for the current season including any competition handbook are considered part of these rules.

A3.1.5 Questions concerning the meaning or intent of the rules will be resolved by the officials.

A3.2 Official Instructions

A3.2.1 Failure of a team member to follow an instruction or command directed specifically to that team or team member will result in 25 penalty points being deducted from the team's overall score.

A3.3 Arguments with Officials

- A3.3.1 Argument with, or disobedience to, any official will result in the team being eliminated from the competition.

A3.4 Unsportsmanlike Conduct

- A3.4.1 In the event of unsportsmanlike conduct, 25 penalty points will be deducted from the team's overall score. A second violation will result in expulsion of the team from the competition.

A3.5 Violations of Intent

- A3.5.1 Violation of the intent of a rule will be considered a violation of the rule itself.
- A3.5.2 Any parts, devices or software fragments designed with the intent to violate a rule, will be considered as a violation.

A3.6 Questions about the Rules

- A3.6.1 Questions about the rules may be asked to the officials.
- A3.6.2 The frequently asked questions (FAQ) section on the competition website must be checked before submitting a question.
- A3.6.3 The officials will only answer questions that are not already answered in the rules or FAQs or that require new or novel interpretation.
- A3.6.4 Refer to the competition website for specific directions on how to submit a rules question.

A3.7 Protests

A3.7.1 Preliminary Review– Required

If a team has a question about scoring, judging, policies or any official action it must be brought to the officials' attention within the announced protest period for an informal preliminary review before a protest can be filed.

A3.7.2 Cause for Protest

A team may protest any rule interpretation, score or official action which they feel has caused some actual, non-trivial, harm to their team, or has had a substantive effect on their score.

A3.7.3 Protest Format and Forfeit

All protests must be filed in writing and presented to the officials by the team captain. In order to have a protest considered, a team must post a 25 point protest bond which will be forfeited if their protest is rejected.

- A3.7.4 The decision of the officials regarding any protest will be in a written form and is final.

A3.8 Penalties

- A3.8.1 Rule violations result in at least 20 penalty points, unless explicitly defined otherwise. The penalty points will be deducted from the team's overall score.
- A3.8.2 Overall penalties apply to CV and EV and **DC score**, except for penalties concerning events which are not part of the respective category class or cup.

A4 GENERAL REQUIREMENTS FOR TEAMS & PARTICIPANTS

A4.1 Teams per University

- A4.1.1 A university may register a CV team and an EV team.
- A4.1.2 For the purposes of registering and competing, a university's CV team and EV team are considered to be separate and independent entities.
- A4.1.3 Teams which are formed with members from two or more universities are treated as a single team. In such a case, the team must inform the organizers upon registration as to which university will be represented.
- A4.1.4 In the case where multiple colleges located in multiple cities or districts are grouped under a common University, either autonomous or through affiliation, the team must register under their College name.

A4.2 Team Members

- A4.2.1 A team member may only be part of one team, work on one vehicle and take part in static and dynamic events for only one team.
- A4.2.2 Each team must have one team member identified as the team captain. The team captain is the main contact person for the officials during the registration process and the competition. The team captain must have a 'current' student status at the attended College or University, at the time of the competition. It is the team captain's responsibility to communicate any competition updates to the team members.
- A4.2.3 Persons who have previously attended any official event as a judge are not allowed to participate as team members.
- A4.2.4 Each team is also required to identify two additional team members during the registration process who will serve as communication liaisons between the team and the competition, should the team captain not be available. These additional team members must have a 'current' student status at the attended College or University, at the time of the competition.
- A4.2.5 Any changes in the team's leadership roster can only be communicated to the competition by the team captain or by either of the two registered additional team members via the email addresses provided during registration.

A4.3 Student Status

- A4.3.1 Team members must be enrolled as degree seeking undergraduate or graduate students in any university. **Students from any academic discipline are eligible to participate.** Team

members who have graduated within the eleven month period prior to the competition remain eligible to participate.

A4.3.2 Students who have already graduated from a degree program and are seeking another degree of the same level or below are not allowed to participate.

A4.3.3 Students seeking a PhD degree/PhD students or equivalent are not allowed to participate.

A4.3.4 Team members must have Student Membership level status for the competition season, procured from Membership by Formula Bharat.

A4.4 Age

A4.4.1 Team members must be at least 16 years of age.

A4.5 Driver's License

A4.5.1 Team members who will drive a competition vehicle at any time during a competition must present a valid, government issued driver's license for passenger cars, containing a photograph.

A4.5.2 Designated drivers for the team are required to bring a hardcopy of both sides of their license to the competition site. Upon registration, they will be required to sign a bond and pay a fee that shall be collected by the venue. The amount for the fee shall be mentioned on the competition website closer to the event date.

A4.6 Insurance

A4.6.1 Each participant, including faculty advisors and volunteers, must provide proof of valid private liability & individual health insurance for the competition.

A4.7 Liability Waiver

A4.7.1 All onsite participants, including faculty advisors and volunteers must sign a liability waiver which can be found on the competition website.

A4.7.2 All onsite participants under 18 years of age must submit the Under 18 liability waiver prior to the competition. This shall be made available via the competition website.

A4.8 [EV ONLY] Electrical System Officer (ESO)

A4.8.1 Every participating team has to appoint one to four ESOs for the competition.

A4.8.2 The ESOs are responsible for all electrical work carried out on the vehicle during the competition.

A4.8.3 The ESOs are the only persons in the team who may declare the vehicle electrically safe, in order for work to be performed on any system of the vehicle by the team.

A4.8.4 The ESOs must be valid team members, which means that they must have student status (see A4.3).

A4.8.5 The ESOs must be contactable by phone at all times during the competition.

- A4.8.6 An ESO must accompany the vehicle whenever it is operated or moved around at the competition site.
- A4.8.7 If only one ESO is named by the team, this ESO may not be a driver.
- A4.8.8 An appropriate Electrical System Officer Qualification (ESOQ) requires the proof of a practical and theoretical training for working on the TS. Regarding content, the training should be on the basis of DGUV 209-093 stage 2E/3E (<https://fsg.one/4i5SP>) and held by an external expert. Teams receiving training from providers or programs other than those mentioned on the competition website must provide details of the training which include details of the program, topics covered within the training, format of training, training provider credentials etc. Additionally, the certificate issued must contain the level of training received (as in, 1E / 2E/ 3E).
- A4.8.9 Each team must have a minimum of one ESO trained in 3E. Only ESOs with 3E training are allowed to work on live high voltage systems.
- A4.8.10 It is recommended that all other team members receive an 1E training instruction. This can be provided by an external provider or from one of the team's certified ESO itself.

A4.9 Autonomous System Responsible (ASR)

- A4.9.1 To operate the AS, the participating team must appoint at least one ASR for the event. This person is responsible for all autonomous operations of the vehicle during the event which includes any work on the AS as well as racing and testing.
- A4.9.2 For vehicles with an electric drivetrain the ASR must fulfill A4.3 and therefore replaces the ESO. The team may register additional ESOs for Accumulator Inspection and work on TS only. The sum of registered ASRs and ESOs must not exceed four persons.
- A4.9.3 The ASR is the only person in the team who is allowed to declare the AS safe, so that the vehicle may be operated in manual or autonomous mode.
- A4.9.4 The ASR must be a valid team member, i.e. they must have student status, see A4.2.6.
- A4.9.5 The ASR must accompany the vehicle whenever it is operated or moved around at the event site.
- A4.9.3 If only one ASR is named by the team, this ASR must not be a driver.
- A4.9.3 The ASR must be properly qualified to handle the AS and to understand and deal with problems and failures. As ASR Qualification (ASRQ), a bachelor degree in computer science, electrical engineering, mechatronics, automation engineering, robotics or similar, i.e. comparable study content or progress, is a sufficient qualification. The qualification certificate needs to be an official university document and contain information on completed courses.

A4.10 Faculty Advisor

- A4.10.1 Each team is expected to have a Faculty Advisor appointed by the university (or college, where applicable). The Faculty Advisor is expected (however not mandatory) to accompany the team to the competition. In the case that the Faculty Advisor cannot attend, the Team

- Captain will be considered by competition officials to be the official university representative.
- A4.10.2 Faculty Advisors may advise their teams on general engineering and engineering project management theory.
- A4.10.3 Faculty Advisors may not design any part of the vehicle nor directly participate in the development of any documentation or presentation.
- A4.10.4 Additionally, Faculty Advisors may not fabricate nor assemble any components nor assist in the preparation, maintenance, testing or operation of the vehicle. In brief – Faculty Advisors may not design, build or repair any part of the car.
- A4.10.5 The appointed Faculty Advisor must be the participating institution's Faculty member and the individual should have completed their undergraduate studies 3 years prior to assuming such a post.

A5 DOCUMENTATION & DEADLINES

A5.1 Required Documents and Forms

- A5.1.1 The following documents and forms must be submitted by the action deadlines defined on the competition website:
- IAD, SES, SE3D, BPPV, [BPEFS](#), [TVSD](#), CRD (BOM, CEXP, CSUP), PS, VSV, TMI,
 - [EV ONLY] ASES, FMEA, ESOQ, SVCR, ESF-1, ESF-2
 - [DC ONLY] [ASRQ](#), [ASF](#)

A5.2 Submission

- A5.2.1 Uploaded documents may only be viewed by members of the submitting team, authorized judges and officials.
- A5.2.2 By submitting documents via the competition website, the team agrees that they may be reproduced and distributed by the officials, in both complete and edited versions, for educational purposes.
- A5.2.3 Documents and videos that are largely incomplete or not readable will be considered as not submitted.
- A5.2.4 Teams must submit documents in the prescribed method as mentioned on the competition website for both template and non-template driven documents.
- A5.2.5 Discovery of any forged documentation by the team, such as datasheets with incorrect values, submissions with incorrect data / images, etc. shall result in de-registration of the team from the competition.

A5.3 Penalties

- A5.3.1 The following table indicates penalties and consequences apply to **late or non-submissions**:

Document	Late Submission: Points penalty per 24h	Penalty deduction from?	Non-Submission: After 7 days
ESOQ [EV]	10	Overall	DR – IN & D
ARSQ [DC]	10	Overall	DR – IN & D
FMEA [EV]	10	Overall	DR – IN & D
SVCR [EV]	10	Overall	DR – IN & D
SES	10	Overall	DR – IN & D
ASES [EV]	10	Overall	DR – IN & D
SE3D	10	Overall	DR – IN & D
ESF-1 [EV]	10	Overall	DR – IN & D
ESF-2 [EV]	10	Overall	DR – IN & D
ASF [DC]	10	Overall	DR – IN & D
IAD	10	Overall	DR – IN & D
BPEFS	10	BPP	DR – BPP, 0 pts BPP
BPPV	10	BPP	DR – BPP, 0 pts BPP
TVSD	10	ED	DR – ED, 0 pts ED
PS	10	Overall	DR – IN & D
VSV	10	Overall	DR – IN & D
TMI	-	-	DR - Competition
CRD (BOM)	10	CR	DR – CR, 0 pts CR
CRD (CEXP)	10	CR	DR – CR, 0 pts CR
CRD (CSUP)	10	CR	DR – CR, 0 pts CR

Legend: DR – De-registered; IN – Technical Inspection events; D – Dynamic events; BPP – Business Plan Presentation event; ED – Engineering Design event; CR – Cost & Manufacturing event

Note 1: TMI – No submission of TMI by deadline will result in team members' inability to receive electronic certificates and inability to participate at the on-site event.

Note 2: No response to official's communication on delayed or non-submission of the documents shall result in de-registration of the team **from the competition**.

A5.3.2 Corrected files must be uploaded within 168 hours of receiving a reviewer notice. Certain documents may not require approval. The following table indicates penalties and consequences apply to delayed submission towards **correction requests**, post the grace period mentioned above:

Document	Late Correction Submission: Points penalty per 24h	Penalty deduction from?	Non-Submission: After 7 days
ESOQ [EV]	5	Overall	DR – IN & D
ARSQ [DC]	5	Overall	DR – IN & D
FMEA [EV]	5	Overall	DR – IN & D
SVCR [EV]	5	Overall	DR – IN & D
SES	5	Overall	DR – IN & D
ASES [EV]	5	Overall	DR – IN & D
SE3D	5	Overall	DR – IN & D
ESF-1 [EV]	5	Overall	DR – IN & D
ESF-2 [EV]	5	Overall	DR – IN & D
ASF [DC]	5	Overall	DR – IN & D

Document	Late Correction Submission: Points penalty per 24h	Penalty deduction from?	Non-Submission: After 7 days
IAD	5	Overall	DR – IN & D
BPEFS	5	BPP	DR – BPP, 0 pts BPP
BPPV	5	BPP	DR – BPP, 0 pts BPP
TVSD	5	ED	DR – ED, 0 pts ED
PS	5	Overall	Tier 2
VSV	5	Overall	Tier 2
TMI	-	-	-
CRD (BOM)	5	CR	DR – CR, 0 pts CR
CRD (CEXP)	5	CR	DR – CR, 0 pts CR
CRD (CSUP)	5	CR	DR – CR, 0 pts CR

Note 1: Simultaneous requests for different parts within one document or form will be penalized independently of each other.

Note 2: Unless the first revision is requested from the competition officials, uploading a revision before such request for a document would automatically make the latest submitted document the ‘first submission’, and any late submission penalties as per [A5.3.1](#) shall apply.

Note 3: If the review panel observes consistent neglect of certain changes by the team, a flat penalty of 20 points will be imposed for that specific document upon the initial instance. Subsequent occurrences will incur additional penalty points each time they are repeated.

Note 4: Comments made by the reviewer on the PS, or during the PS Team Calls must be responded to via the originally submitted PS only. Concerns raised must be addressed appropriately, before three weeks of the competition dates, for the teams Tier level to be assigned (Refer to [A5.10](#)).

Note 5: Corrections requested before the initial deadline are treated as if they were requested at the initial deadline.

A5.3.3 The following table indicates penalties and consequences apply to incorrect file naming and formatting. Please note tables in [A5.3.1](#) and [A5.3.2](#) for column of points’ deduction (i.e. Overall or Event-Specific with each respective document)

- Incorrect file type: 5 penalty points
- Incorrect file naming: 2 penalty points
- Incorrect template: Considered as non-submission
- Exceeding page / slide limit (this includes a non-required cover page as well): 2 penalty points
- Exceeding pitch or presentation time limit: 10 penalty points
- (For non-template files) Missing Team Number, Team Name, Institution Name, Event Name, Document Name in Header and Page number, Total Number of pages in Footer: 0-5 penalty points
- Missed Business Plan Presentation Session with less than 168 hour notice: 50 points
- Missed Cost and Manufacturing Session with less than 168 hour notice: 50 points
- Missed Engineering Design Session with less than 24 hour notice: 50 points

- Missed slot booking for virtual judging session (BPP Prelims / ED Prelims / Cost and Manufacturing Prelims: DR from particular event (Business Plan Presentation / Engineering Design / Cost and Manufacturing)

Note: While there is no penalty for file size, teams are requested to keep to the minimum possible size to ensure that judges and reviewers can easily download or review the file on any device and to also ensure that the team avoids uploading issues.

- A5.3.4 The competition website will state the last day for approval for certain documents. Any final document revisions must be submitted at least two weeks before the last day of approval. Teams which do not have the listed documents approved by the stated deadline on the competition website will be automatically de-registered from the specific event or from the competition, as listed in the table in [A5.3.1](#).

A5.4 De-registration

- A5.4.1 A team which is de-registered from the competition has a single chance to apply for a place on the waiting list. A team which is de-registered from specific events in the competition has a single chance to appeal their status.

- A5.4.2 To apply, the team must complete the following within 24 hours after the de-registration notification:

- Submit an informal application to the officials
- Correct the reason for de-registration (e.g. upload of a document)

- A5.4.3 If the application is positively confirmed by the officials, the team will be

- Placed at the end of the waiting list
- Receive 20 penalty points

- A5.4.4 Applications will be declined if the rework is inadequate or de-registration was caused by misbehavior of the team.

A5.5 Photographic Submissions (PS)

- A5.5.1 All teams must submit images showing the various views of the vehicle driving prior to the competition. The checklist of views required, including the method of submission shall be specified on the competition website.

- A5.5.2 Teams that attend other events with their vehicle, in proximity to the next competition, must ensure that the photos for submission are captured before attending those events to prevent the situation of missing the submission deadline.

- A5.5.3 Time stamped photos of chassis fabrication should be added in PS.

A5.6 Vehicle Status Video (VSV)

- A5.6.1 All teams must submit a video showing the vehicle driving prior to the competition. The video must be submitted before the deadline specified on the competition website.

- A5.6.2 [DC ONLY] The vehicle must instead drive in autonomous mode. No manual mode VSV is necessary.

A5.6.3 The video must show the following sequences in the described order:

- Standing still
- 360° video around the vehicle (close up, may overlap with the following two bullet points)
- [EV ONLY] Activate TS
- [CV ONLY] Start engine
- Straight driving, minimum distance is 30 m, minimum speed is 10 km/h for vehicles with driver
- Full stop
- 180° cornering
- Straight driving back to start point, minimum speed is 10 km/h for vehicles with driver
- [DC ONLY] *At the end of the VSV, the vehicle must be stopped by an emergency brake maneuver, see T15.*
- Standing still, [CV ONLY] engine off / [EV ONLY] TS deactivated

A5.6.4 The video must fulfill the following criteria:

- Continuous video from a third person view - no assembled sequences
- Vehicle must be clearly visible (light, video resolution, landscape mode, frames and frequency)
- Vehicle must run under its own power
- Driving must follow [A6.4.4](#)
- Vehicle must be presented in ready-to-race conditions incl. body work
- No crossing in front of the vehicle after [EV ONLY] TS activated / [CV ONLY] engine started
- [EV ONLY] Tractive System Active Light (TSAL), [DC ONLY] ASSI, and brake light, must be clearly visible in the video
- [EV ONLY] Ready-to-drive (R2D) sound must be audible in the video
- [DC ONLY] *In addition to the third person view, an onboard view and a visualization of the vehicle's environment perception and path planning must be shown in split screen. All parts must be time synchronized.*
- Must not exceed a length of 60 seconds.
- File format must be .pdf.

A5.6.5 *If a team fails only the [DC ONLY] part, it will only be de-registered from the DC.*

A5.6.6 The VSV must be uploaded on a video sharing platform of your choice (Youtube, Vimeo etc.). Please do not upload on Google Drive. The video link must be publicly accessible. If uploaded on Youtube, the privacy setting of the video must be changed to 'Unlisted' so that only those with the link can view the video and it will not be displayed publicly on your channel as well. The link to the VSV should be submitted within a document consisting of only 1 (one) A4 page. The document must only include the VSV link. No additional external links to be added.

A5.6.7 A team which uploads a VSV with a previous year vehicle will be de-registered from the competition.

A5.7 Team Member Information (TMI)

A5.7.1 All team members who plan on attending the competition on-site or who would like to receive participation certificates, must have Student Membership level for the competition season, procured via the Membership by Formula Bharat (MFB) platform.

- A5.7.2 All team members must provide details towards the Team Member Information (TMI) submission available on the competition website by the specified deadline. This information includes but is not limited to Full Name, Contact Email, Phone Number, Age, Year of Study, Medical Emergency contact information (which must be next of kin) etc.
- A5.7.3 The TMI submission criteria will be made available on the competition website approximately two months prior to the event.
- No new team member entries shall be allowed past the deadline or on-site.
- A5.7.4 The team captain is responsible for confirming participation of team members, [ON-SITE ONLY] for providing their insurance information and uploading insurance scans, under 18 liability waivers' scans (if any) and for providing driver license information.
- A5.7.5 Confirmed participants with incorrect or incomplete insurance information (including scanned copies) shall have their participation status revoked on-site at the competition.
- A5.7.6 There is no restriction on the number of team members who may participate on a team.

A5.8 Participation Certificates

- A5.8.1 Participants listed on the TMI with complete information and who have submitted all required documents are eligible:
- For a competition participation certificate, provided the team has not been de-registered from Inspection (IN) & Dynamics (D).
 - For a static event participation certificate, provided the team has participated in all static events and has been de-registered from Inspection (IN) & Dynamics (D).
- A5.8.2 Teams that withdraw at any point are not eligible for participation certificates.
- A5.8.3 The competition organizers shall not be distributing printed (hardcopy) versions of the certificates. The team may request a signed and stamped letter addressed to their institution to confirm the list of team members who attended and participated at the event.

A5.9 Teams Categorization

- A5.9.1 The competition organizers reserve the right to classify participating teams into Tier 1 and Tier 2 lines for technical inspection purposes, prior to the competition. This classification is based on the submission and quality of the Photographic Submissions (PS), Vehicle Status Video (VSV) and approval status of documents pertaining to IN and D events.
- A5.9.2 Notice for teams categorized as Tier 2 due to poor submissions shall be done by the date specified on the competition website.
- A5.9.3 Tier 1 teams will get prioritized at the Technical Inspection queue over Tier 2 teams. Comprehensive details on arrival time and registration shall be provided in the competition website prior to the competition dates.

A5.10 Negative Scoring

- A5.10.1 Should the team's total score or individual static score result in a negative score (as in below 0 points) due to late submission ([A5.3](#)) or delayed correction requests ([A5.4](#)) penalties, the specific result will be capped at 0 points.

A6 GENERAL RULES

A6.1 Arrival and Removal of Vehicle from the Site

- A6.1.1 Teams that do not have a 'DR - IN & D' status, must arrive with their vehicle to the competition site within the first two days of the competition during operational hours and must register for queuing at technical inspection. Teams that do not do so will be automatically de-registered from further participation in IN and D events.
- A6.1.2 [EV ONLY] Teams must present their accumulator for inspection no later than before the end of the second day of accumulator inspection, as within operational hours, as mentioned in the competition schedule. Teams that do not do so will be automatically de-registered from further participation in Accumulator Inspection (IN3), and subsequently in certain IN events and all D events.
- A6.1.3 Teams may remove their vehicle ([EV ONLY] vehicle without tractive system accumulator) from the competition site during the competition if no technical inspection stickers have been received. Once a team has received a competition technical inspection sticker, they are not allowed to leave the competition site. Doing so without appropriate approvals will result in removal of the sticker and disqualification from the competition.
- A6.1.4 Teams that remove their vehicle from the competition during the competition can only do so during the site's operational hours, as per the competition schedule, and will have to make their own provisions in moving the vehicle in and out of the competition site. Refer to the competition handbook for further details on restrictions on movement of vehicles off-site during the competition days.

A6.2 Forfeit for Non-Appearance

- A6.2.1 It is the responsibility of each team to be in the right place at the right time.
- A6.2.2 If a team is not present and ready to compete at the scheduled time, they forfeit an attempt at that event.

A6.3 Team and Driver Briefings

- A6.3.1 All team captains or designated team representatives must attend the team captains' briefing for that day.
- A6.3.2 Drivers who have missed attending the driver's briefing for the day will not be allowed to drive the car at any event on that particular day.

A6.4 Testing and Work Safety

- A6.4.1 Competition organizers are not responsible for the use of the vehicles outside of their competition.

- A6.4.2 The competition officials disassociate themselves from all activities of the teams besides their own competition and associated events.
- A6.4.3 All teams are advised to follow common practices and common sense when working on the vehicle and when operating the vehicle, before, during and after a competition.
- A6.4.4 The following listed requirements are considered the minimum for a testing/operating environment to qualify as safe. Following these guidelines does not guarantee safety under all circumstances:
- Driver wearing full protection gear according to T13.3
 - Working TSAL, IMD, AMS, ASSI, RES, EBS, APPS/brake pedal plausibility check, APPS, and ETC plausibility check if applicable
 - Rules compliant chassis and mounted impact attenuator
 - No other passenger cars, trucks etc. being parked or driven on the same premise at the same time, unless the areas are clearly separated
 - No running under low visibility conditions
 - No running at speeds above typical event speeds
 - No running in areas where crashing into obstacles at the height of the driver's head is possible, such that parts of the vehicle may pass below an obstacle, but the driver's head can be trapped between the obstacle and the main hoop for example.
- A6.4.5 Organizers reserve the right to disqualify a team registered for their competition in case of unsafe driving behavior, especially if the reputation of the competition, sponsors and other teams is compromised. Examples of what could be seen as "unsafe driving behaviour" are wheel-to-wheel racing or racing in unsuitable events, e.g. hill climb races or drag races, depending on the respective circumstances. This also includes any questionable material posted on social media by registered participating teams under the competition.
- A6.5 Onsite Working Safety**
- A6.5.1 Everyone in the dynamic area and everybody working on the vehicle must wear appropriate, closed-toed shoes.
- A6.5.2 When using metal cutting equipment, eye protection is required for the operator as well as any team member assisting. When operating loud tools, hearing protection is required.
- A6.5.3 Work that generates hazardous debris and dust, e.g. sanding or cutting of carbon fiber, must be performed in the designated power tools area. This excludes minor tasks, e.g. drilling a small hole, if safety precautions for the operator and nearby personnel are observed and all debris and dust are immediately cleaned up.
- A6.5.4 When jacking up the vehicle a safe and stable support device rated for the load must be used.
- A6.5.5 The use of motorcycles, quads, scooters, skateboards, rollerblades or similar mobility devices by team members and spectators in any part of the competition area is prohibited. Mobility devices used to assist with an individual's medical disability is accepted.
- A6.5.6 The use of self-propelled pit carts, tool boxes, tire carriers or similar motorized devices in any part of the competition site is prohibited.

A6.6 Prohibited Substances

- A6.6.1 Team members must not be under the influence of alcohol (0.0 ‰), cannabis, or any other recreational drug while on the competition site.
- A6.6.2 Violators will be disqualified immediately. Repeated offense results in the entire team being disqualified immediately.
- A6.6.3 Prescription medications and over-the-counter drugs used as directed for medical purposes are exempt from [A6.6.1](#). However, team members must not be impaired in a manner that could compromise safety.

A6.7 Vehicle Movement

- A6.7.1 A Practice Area for testing and tuning vehicles is available.
- A6.7.2 Vehicles must not move under their own power anywhere other than on the practice or competition tracks.
- A6.7.3 [EV ONLY] The detachable handle or key of the Tractive System Master Switch (TSMS) must be completely removed and kept by an ESO. The lockout/tagout function of the TSMS (see [EV6.2.2](#)) must be used.
- A6.7.4 [DC ONLY] For vehicles with an AS, the detachable handle or key of the ASMS must be completely removed and kept by an ASR. The lockout/tagout function of the ASMS, see [T14.5.7](#), must be used.
- A6.7.5 Vehicles must be pushed at a normal walking pace by means of a “pushbar” (see [T13.1](#)) and with a team member in the cockpit wearing the required driver equipment (helmet included) as defined in [T13.3](#) except for arm restraints and balaclava.
- A6.7.6 The team member in the cockpit must have full control of steering and braking.
- A6.7.7 When the pushbar is attached to the vehicle, the engine/tractive system must remain switched off.
- A6.7.8 Vehicles with wings are required to have two team members walking on either side of the front wing whenever the vehicle is being pushed.

A6.8 [CV ONLY] Engine Running

- A6.8.1 Running engines is not allowed in the pits.
- A6.8.2 Engine running is allowed in the engine test area and in the dynamic area, when the following conditions are met:
- The vehicle must have passed mechanical inspection.
 - The vehicle must be jacked up using a jack (see [T13.2](#)).
 - A driver, wearing required driver equipment (see [T13.3](#)), must be seated in the cockpit.
 - A fire extinguisher must be immediately available.
 - Driven wheels can only remain fitted to the vehicle if they will not rotate.
 - No one is allowed under the vehicle while the engine is running.
- A6.8.3 Cranking engines in pits is not allowed.

A6.9 Fueling and Oil

- A6.9.1 Fueling may only take place at the fuel station and must be conducted by officials only.
- A6.9.2 Open fuel containers are not permitted at the competition.
- A6.9.3 Waste oil must be taken to the fuel station for disposal.

A6.10 [EV ONLY] Working on the Vehicle

- A6.10.1 All activities require the TSAL to be green.
- A6.10.2 A prominent manual sign indicating the “TSAL green” state must be present whenever the LVS is switched off and the requirements for an only green TSAL according to [EV4.10](#) are met.
- A6.10.3 All activities on the vehicle, except when working on an active TS, require the High Voltage Disconnect (HVD) to be removed.

A6.11 [EV ONLY] Working on the Tractive System

- A6.11.1 Activities on the Tractive System (TS), except for the accumulator ([see A6.12](#)) must take place in the pit.
- A6.11.2 All activities require the attendance of the ESO.
- A6.11.3 For activities on the inactive TS, the following procedure must be carried out:
- The vehicle must be barred from anyone not involved in the work, by using barrier tape.
 - The Tractive System Master Switch (TSMS) must be switched off.
 - It must be assured that the TS cannot be restarted, by, at a minimum, using the lockout/tag out of the TSMS.
 - A check for zero-potential must be carried out.
 - A sign that declares the vehicle is electrically safe must be installed. The name of the ESO who is supervising the activities must be noted on the sign. This ESO is the only person who may remove the sign and the barrier.
- A6.11.4 In case of measurements on the active TS or an activation of the TS in the designated areas for purposes, the following steps must be followed:
- The vehicle must be barred from anyone not involved in the work, by using barrier tape.
 - The vehicle must be jacked up and the driven wheels removed.
 - One team member must be prepared to push a shutdown button at any time.
 - The TS must only be activated for as long as necessary.
 - Appropriate insulated tools and equipment must be used.
 - Safety glasses with side shields and compliant safety gloves must be worn by all participating team members when parts of the TS are exposed.
 - No other work on the vehicle is permitted when the TS is active.
- A6.11.5 If the TSAL is flashing red or malfunctioning, the TS is considered as active.

A6.11.6 There must be at least one team member present, who is not directly involved in the work, but who could assist in case of an incident.

A6.11.7 Turning ON the TS inside pits is restricted. Working on the vehicle with TS ON is only allowed in the designated areas specified by the officials.

A6.12 [EV ONLY] Working on Tractive System Accumulators

A6.12.1 Opening or working on accumulator containers is only allowed in the provided work places in the charging area ([see A6.13](#)).

A6.12.2 All activities require the attendance of an ESO.

A6.12.3 Whenever the accumulator containers are opened, the cell segments must be separated with the maintenance plugs ([see EV5.4.4](#)).

A6.12.4 Appropriate insulated tools and equipment must be used.

A6.12.5 Safety glasses with side shields and compliant safety gloves must be worn by all participating team members.

A6.12.6 There must be at least one team member present, who is not directly involved in the work conducted on the accumulator, but who could assist in case of an incident.

A6.12.7 Moving accumulator cells and/or accumulator segment(s) around at the competition site is only permitted if they are inside a completely closed accumulator container.

A6.12.8 All parts and modules of the TSAC that are not currently being worked on must be covered at least according to IPxxB while working on the accumulator container.

A6.12.9 Additional safety measures may be included in the respective competition handbook or on the competition website.

A6.12.10 The TSAC and any spare cells have to be stored in a designated area specified by competition officials.

A6.12.11 The TSAC must be stored in designated areas specified by competition officials after the competition's daily operational hours, even after a vehicle passes technical inspections. The TSAC shall not be stored in the team's pit.

A6.13 [EV ONLY] Charging

A6.13.1 There will be a separated charging area on the competition site. Charging TS accumulators is only allowed inside this area.

A6.13.2 Accumulators must be removed from the vehicle and placed on the accumulator container hand cart ([see EV8.1](#)) for charging.

A6.13.3 No grinding, drilling, etc. is allowed in the charging area.

A6.13.4 At least one team member who has knowledge of the charging process must stay with the accumulator(s) during charging.

A6.14 Trash Cleanup

A6.14.1 Cleanup of trash and debris is the responsibility of the teams.

- The team's work area should be kept uncluttered.
- At the end of the day, each team must clean all debris from their area and help with maintaining a clean paddock.

A6.14.2 Teams must remove all of their material and trash when leaving the site at the end of the competition.

A6.14.3 Teams that abandon furniture, or that leave a paddock that requires special cleaning, will be billed for removal and/or cleanup costs.

T GENERAL TECHNICAL REQUIREMENTS

T1 DEFINITIONS

T1.1 Chassis Definitions

- T1.1.1 Chassis – The fabricated structural assembly that supports all functional vehicle systems. This assembly may be a single welded structure, multiple welded structures or a combination of composite and welded structures.
- T1.1.2 Bodywork – the outermost surface of the chassis or any fairing parts and covers.
- T1.1.3 Cockpit – The volume which accommodates the driver which is defined by the top of the vehicle, the floor closeout, the inner side of the bodywork, the front bulkhead and the firewall.
- T1.1.4 Chassis member – A minimum representative single piece of uncut, continuous tubing or equivalent structure.
- T1.1.5 Front bulkhead – A planar structure that defines the forward plane of the chassis and provides protection for the driver's feet (in front view, together with the AIP, covers the driver's feet).
- T1.1.6 Front bulkhead support (FBHS) – A structure that defines the side of the chassis from front bulkhead back to the top of the upper side impact structure and the front hoop.
- T1.1.7 Front hoop – A roll bar located above the driver's legs, in proximity to the steering wheel.
- T1.1.8 Impact Attenuator (IA) – A deformable, energy absorbing device located forward of the front bulkhead.
- T1.1.9 Main hoop – A roll bar located alongside or just behind the driver's torso.
- T1.1.10 Monocoque – A chassis made of composite material.
- T1.1.11 Node-to-node triangulation – An arrangement of chassis members projected onto a plane, where a co-planar load applied in any direction, at any node, results in only tensile or compressive forces in the chassis members as shown in figure 1 (below).

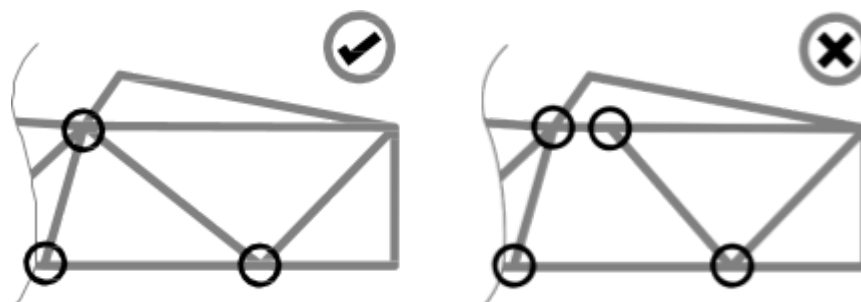


Figure 1: Node-to-node triangulation of chassis members (left correct and right incorrect).

- T1.1.12 Primary structure – The primary structure is comprised of the following components:
- Main hoop
 - Front hoop
 - Roll hoop braces and supports
 - Impact structures as mentioned under T3.14
 - Anti Intrusion Plate
 - Front bulkhead
 - Front bulkhead support system
 - All chassis members, guides and supports that transfer load from the driver's restraint system into the above mentioned components of the primary structure
- T1.1.13 Roll hoops – Both the front hoop and the main hoop are classified as “roll hoops”
- T1.1.14 Roll hoop bracing – The structure from a roll hoop to the roll hoop bracing support.
- T1.1.15 Roll hoop bracing supports – The structure from the lower end of the roll hoop bracing back to the roll hoop(s).
- T1.1.16 Rollover protection envelope – the union of
- the primary structure and any additional structures fixed to the primary structure which meet the minimum specification defined in T3.2 or equivalent
 - the plane from the top of the main hoop to the top of the front hoop
 - the plane from the top of the main hoop to the rearmost structure according to T3.2 or equivalent
- Example depicted in figure 2.
- T1.1.17 Side impact structure – The area of the side of the chassis between the front hoop and the main hoop and from the chassis floor to the height as required in T3.14 above the lowest inside chassis point between front hoop and main hoop.
- T1.1.18 Surface envelope – the surface envelope is the surface of the union of the rollover protection envelope, see T1.1.16, and the volume defined by
- the top of the main hoop and the outside edges of the four tires
 - a plane from the top of the main hoop to the top of the front bulkhead
 - a plane from the top of the main hoop to the the rearmost structure according to T3.2 or equivalent
 - a projection in side view of the lower boundaries of the rollover protection envelope

Example depicted in figure 3.

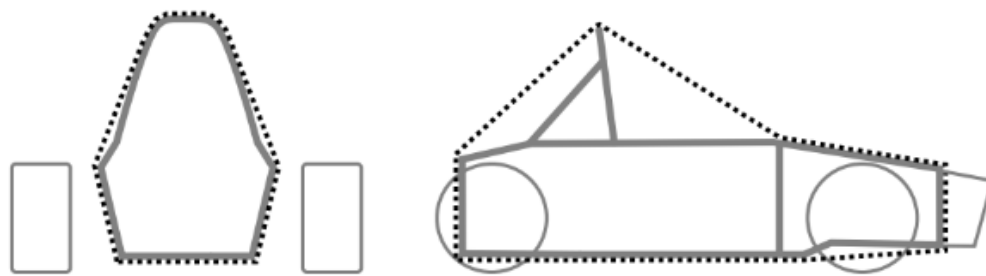


Figure 2: Rollover protection envelope T1.1.16, thick lines are primary structure members according to T3.2

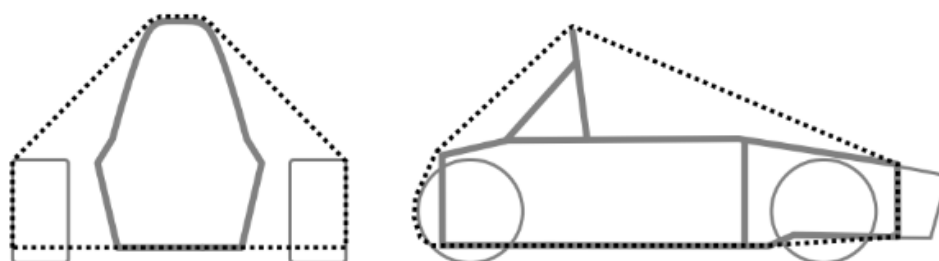


Figure 3: Surface Envelope T1.1.18

T1.2 Material Definitions

T1.2.1 Fire Retardant – A material meeting one of the following standards:

- UL94 V-0 for the minimum used material thickness
- FAR 25.853(a)(1)(i)
- For foams only: UL94 HF-1 and UL94 VTM-0

Equivalent standards are only accepted, if the team shows equivalence and this is approved by the officials before the event. [Compliance of alternative materials to above standards can be shown through testing with appropriate documentation in the SES/ASES and/or at Technical Inspection.](#)

T1.2.2 Coolant - A substance used for heat transfer by convection.

T1.3 Electrical Definitions

T1.3.1 Direct Connection – Two devices or circuits are directly connected if the connection is not routed through any common PCB and does not include any devices or functionality other than overcurrent protection or connectors.

T1.4 Driving Mode Definitions

T1.4.1 Manual Mode– a vehicle is in manual mode when driven by a human driver. In this case the ASMS must be off (AS deactivated).

T1.4.2 Autonomous Mode– a vehicle is in autonomous mode when the AS is activated. When a vehicle is in autonomous mode, there must be no person inside the vehicle.

T2 GENERAL DESIGN REQUIREMENTS

T2.1 Vehicle Configuration

- T2.1.1 The vehicle must be designed and fabricated in accordance with good engineering practices.
- T2.1.2 The vehicle must be open-wheeled, single seat and open cockpit (a formula style body) with four wheels that are not in a straight line.
- T2.1.3 Open wheel vehicles must satisfy the following (see also figure 4):
- The wheel/tire assembly must be unobstructed when viewed from the side.
 - No part of the vehicle may enter a keep-out-zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm behind the outer diameter of the front and rear tires in the side view of the vehicle, with steering straight ahead. This keep-out zone extends laterally from the outside plane of the wheel/tire to the inboard plane of the wheel/tire assembly.

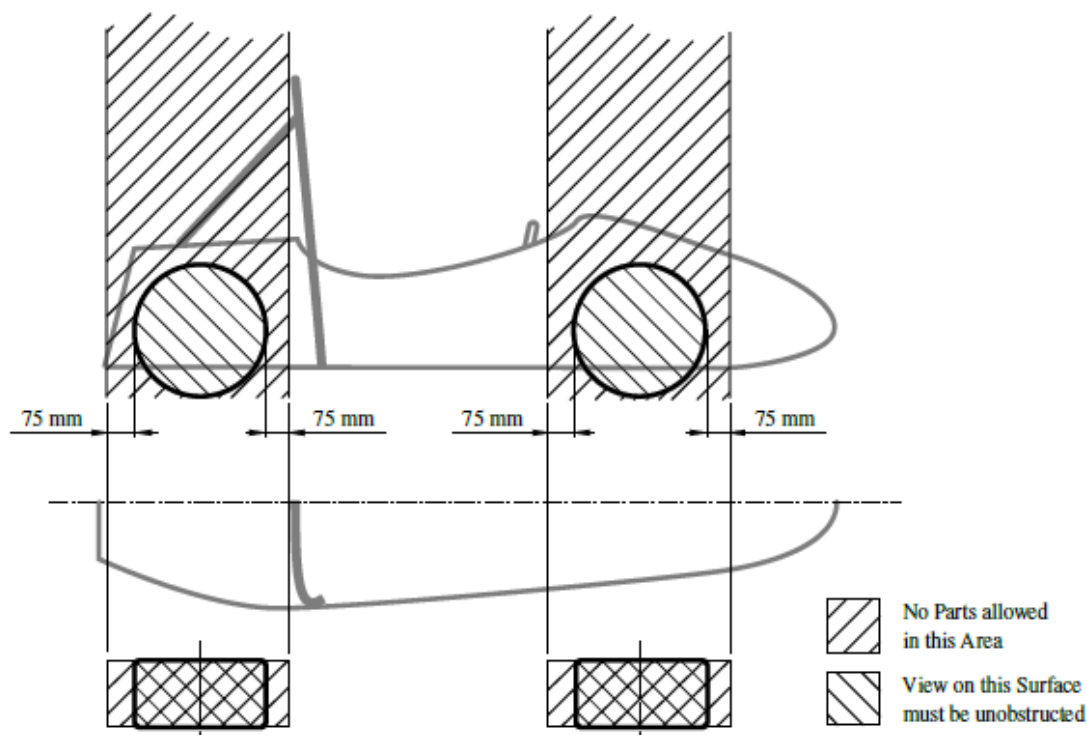


Figure 4: Keep-out-zones for the definition of an open-wheeled vehicle.

T2.2 Ground Clearance

- T2.2.1 The minimum static ground clearance of any portion of the vehicle, other than the tires, including a driver, must be 30 mm. If an active suspension system is installed, the static ground clearance is measured in the lowest adjustable position.
- T2.2.2 Sliding skirts or other aerodynamic devices that by design, fabrication or as a consequence of moving, contact the track surface are prohibited.

T2.3 Bodywork

- T2.3.1 There must be no openings through the bodywork into the driver compartment other than that required for the cockpit opening. Minimal openings around the front suspension and steering system components are allowed.
- T2.3.2 In any side [section](#) view in front of the cockpit opening and outside the area defined in [T8.2](#) all parts of the bodywork must have no external concave radii of curvatures. Any gaps between bodywork and other parts must be reduced to a minimum.
- T2.3.3 Enclosed chassis structures and structures between the chassis and the ground must have two venting holes of at least 25 mm diameter in the lowest part of the structure to prevent accumulation of liquids. Additional holes are required when multiple local lowest parts exist in the structure.
- T2.3.4 The bodywork in front of the front wheels with the tangent of surface above 45° relative to the forward direction must have a radius of at least 38 mm along the top, sides and bottom of all affected edges.

T2.4 Suspension

- T2.4.1 The vehicle must be equipped with fully operational front and rear suspension systems including shock absorbers and a usable wheel travel of at least 50 mm and a minimum jounce of 25 mm with driver seated.
- T2.4.2 All suspension mounting points must be visible at technical inspection, either by direct view or by removing any covers.

T2.5 Wheels

- T2.5.1 Any wheel mounting system that uses a single retaining nut must incorporate a device to prevent loosening of the nut and the wheel. A second nut (“jam nut”) does not meet these requirements.
- T2.5.2 Wheel lug bolts and studs must be made of steel or titanium. The team must be able to show good engineering practice and provide adequate strength by calculations. Wheel lug bolts, drive pegs and studs must not be hollow.
- T2.5.3 Aluminum wheel nuts may be used, but they must be hard anodized and in pristine condition.
- T2.5.4 The radial clearance between any non-rotating part and the inside of the rim must be at least 5 mm in static condition at any steering angle and any ride height.

T2.6 Tires

- T2.6.1 Vehicles must have two types of tires as follows:
- Dry tires - The tires on the vehicle when it is presented for technical inspection are defined as its “dry tires”.
 - Wet tires - Wet tires may be any size or type of treaded or grooved tire provided:

- The tread pattern or grooves were molded in by the tire manufacturer or were cut by the tire manufacturer or their appointed agent. Any grooves that have been cut must have documentary proof that it was done in accordance with these rules.
- There is a minimum tread depth of 2.4 mm.

T2.6.2 Tires on the same axle must have the same manufacturer, size and compound.

T2.6.3 Tire warmers are not allowed.

T2.6.4 Special agents that increase traction may not be added to the tires or track surface.

T2.6.5 A team can choose to use a single set of tires which are designed for “dry” and “wet” conditions. If so, the tires need to be treaded or grooved compliant to T2.6.1 in order to be deemed as “fit” to run under “wet” conditions. Tires in violation of T2.6.1 or with no treads/grooves are not considered “wet” tires, and hence a vehicle with these tires cannot race if the track is deemed “wet”. However, these tires can be used as dry tires.

T2.6.6 Tires that are specifically constructed by manufacturers to be run in wet conditions can be used only under "Damp" or "Wet" operating conditions (D3.2.1) of the track and cannot be used as dry tires.

T2.7 Steering

T2.7.1 Steering systems using cables or belts for actuation are prohibited.

T2.7.2 The steering wheel must directly mechanically actuate the front wheels.

T2.7.3 The steering system must have positive steering stops that prevent the steering linkages from locking up. The stops must be placed on the rack and must prevent the tires and rims from contacting any other parts. Steering actuation must be possible during standstill.

T2.7.4 Allowable steering system free play is limited to a total of 7° measured at the steering wheel.

T2.7.5 The steering wheel must be attached to the column with a quick disconnect. The driver must be able to operate the quick disconnect while in the normal driving position with gloves on.

T2.7.6 The steering wheel must be no more than 250 mm rearward of the front hoop. This distance is measured horizontally, on the vehicle centerline, from the rear surface of the front hoop to the forward most surface of the steering wheel with the steering in any position.

T2.7.7 The steering wheel must have a continuous perimeter that is near circular or near oval. The outer perimeter profile may have some straight sections, but no concave sections.

T2.7.8 In any angular position, the top of the steering wheel must be no higher than the top-most surface of the front hoop.

T2.7.9 The steering rack must be mechanically attached to the primary structure and all stationary parts must be within the rollover protection envelope, see T1.1.16.

T2.7.10 Joints between all components attaching the steering wheel to the steering rack must be mechanical and visible at technical inspection. Bonded joints are allowed in accordance with T3.2.8.

- T2.7.11 Rear wheel steering, which can be electrically actuated, is permitted if mechanical stops limit the range of angular movement of the rear wheels to a maximum of 6°. This must be demonstrated with a driver in the vehicle and the team must provide the equipment for the steering angle range to be verified at technical inspection.

T2.8 Wheelbase, Track and Rollover Stability

- T2.8.1 The vehicle must have a wheelbase of at least 1525 mm.
- T2.8.2 The smaller track of the vehicle (front or rear) must be no less than 75 % of the larger track.
- T2.8.3 The track and center of gravity of the vehicle must combine to provide adequate rollover stability. This is tested by the tilt test IN6.

T3 GENERAL CHASSIS DESIGN

T3.1 General Requirements

- T3.1.1 Among other requirements, the vehicle's structure must include:

- Two roll hoops that are braced
- A front bulkhead with support system and IA
- Side impact structures

T3.2 Minimum Material Requirements

- T3.2.1 Table 4 shows the minimum requirements for the members of the primary structure if made from steel tubing.

Item or application	Minimum wall thickness	Minimum cross sectional area	Minimum area moment of inertia
Main and front hoops, shoulder harness mounting bar	2.0 mm	173 mm ²	11 320 mm ⁴
Impact structures, front bulkhead, roll hoop bracing, driver's restraint harness attachment (except as noted above)	1.2 mm	119 mm ²	8509 mm ⁴
Front bulkhead support, main hoop bracing supports	1.2 mm	91 mm ²	6695 mm ⁴

Table 4: Minimum Material Requirements

- T3.2.2 Steel tubing has to be made from unalloyed carbon steel with a maximum content of 0.3% carbon, 1.7% manganese and 0.6% of any other element. All other steel grades are considered alternative materials and require additional testing and documentation (see T3.3).

T3.2.3 Except for inspection holes, any holes drilled in any part which is a member of the primary structure must be considered in the SES.

T3.2.4 The steel properties used for the calculations in the SES must be the following, or what is written in the material property documentation, whichever is lower:

Non-welded strength for continuous material calculations:

- Young's Modulus (E) = 200 GPa
- Yield Strength (S_y) = 305 MPa
- Ultimate Strength (S_u) = 365 MPa

Welded strength for discontinuous material such as joint calculations:

- Yield Strength (S_y) = 180 MPa
- Ultimate Strength (S_u) = 300 MPa

T3.2.5 Any tubing with a wall thickness less than 1.2 mm or a minimum area moment of inertia less than 6695 mm⁴ is considered non-structural and will be ignored when assessing compliance to any rule regarding the vehicle structure.

T3.2.6 If a member of the primary structure (except for the roll hoops) is a bent tube or made from multiple tubes an additional tube must support it. This support tube must:

- Have its attachment point at the position along the bend tube where it deviates farthest from a straight line connecting both ends.
- Be of the same dimension as the supported tube(s).
- Terminate at a node of the chassis.
- Be angled no more than 30° from the plane of the supported tube(s).

T3.2.7 Any welded seams shape must not be mechanically altered in any way.

T3.2.8 Where bonded joints are applicable and governed by the rules, a 50% reduction shall be applied to all nominal and tested adhesive values. Bonded joints in the primary structure must be documented in the SES.

T3.3 Alternative Materials

T3.3.1 Alternative materials may be used for all parts of the primary structure and the tractive system accumulator container with the following exceptions:

- The main hoop and the main hoop bracing must be steel
- The front hoop must be metal
- Any welded structures of the primary structure must be steel
- However, the front hoop may be an aluminum welded structure

T3.3.2 If any other materials than steel tubing are used in the primary structure or the tractive system accumulator container, physical testing is required to show equivalency to the minimum material properties for steel in T3.2.

T3.3.3 If alloyed steel as defined by T3.2.2 is used, the team has to include tests and documentation in the SES to show structural equivalency. This may include, but is not limited to:

- Receipts and data sheets of the used tubing materials
- Documentation about welding processes and filler materials
- Documentation about heat treatments
- Tests showing adequate strength and elongation at break in the welded condition.

T3.3.4 Parts of the steering, brake and suspension system must not be manufactured using additive processes that produce anisotropic strength properties (e.g. consumer-grade FDM). Additively manufactured parts are permitted only if proven to have equivalent strength in all load directions.

T3.4 Composite Structures

T3.4.1 If composite structures are used in the primary structure and/or the TSAC:

- The Flexural Rigidity (EI) of that structure must be calculated with the tools and formulas in the SES and/or Accumulator Structural Equivalency Spreadsheet (ASES).
- The EI must be calculated as the EI of a flat panel about its neutral axis.
- The structure used for flat panel calculations must have the same composition as the structure used in the primary structure or the TSAC.
- The actual geometry and curvature of the panel may only be considered if the flat panel EI equivalency exceeds 60 % of the minimum requirements (T3.2)
- Actual geometry calculations are not permitted for the side impact structure, front bulkhead and TSAC.

T3.4.2 If composite materials are used in the primary structure or the TSAC, the SES must include:

- Material type(s)
- Cloth weights
- Resin type
- Fiber orientation
- Number of layers
- Core material
- Lay-up technique
- Adhesive film type(s)
- 3-point-bend test and shear test data (the AIP is exempt from these tests, but must comply with T3.16.4)

T3.4.3 For any laminate in the primary structure and/or the TSAC, the maximum weight content of parallel fibers, relative to the weight of all fibers in the laminate, is 50%. All fibers whose orientation fall within any 20° window (+/-10°) are considered as parallel

T3.4.4 If an asymmetrical lay-up is used in the primary structure, the thinner skin must have a thickness of at least 40 % of the thicker skin or 1 mm whichever is lower.

T3.4.5 Wherever backing plates are required, they must be fully supported by the structure they are attached to.

T3.4.6 Backing plates must have a continuous perimeter that is near circular or near oval. The outer perimeter profile may have some straight sections, but no concave sections. Backing plates must not have any cut-outs within their outside perimeter except for the holes for bolts.

T3.4.7 Unidirectional fibers are not allowed in the outermost structural layers of a laminate used in the primary structure.

T3.5 Laminate Testing

T3.5.1 If composite materials are used for any part of the primary structure or the tractive system accumulator container the team must:

- Build a representative test panel which must measure exactly 275 mm × 500 mm that has the same design, laminate and fabrication method as used for the respective part of the primary structure represented as a flat panel. The sides of the test panel must not be laminated (core material must be visible).
- Perform a 3-point bending test on this panel

The data from these tests and pictures of the test samples and test setup must be included in the SES. In the pictures, the following must be identifiable: distance between the two supports, dimensions of the load applicator and test sample marking as per T3.5.4. The test results must be used to derive strength and stiffness properties used in the SES formula for all laminate panels.

Representative test panels for parts of the TSAC may use smaller dimensions, provided that the panel core thickness is 5 mm or smaller. This representative test panel must then measure 150 mm×275 mm. In this case, the distance between the two test panel supports must be at least 200 mm and the load applicator must have a radius of at least 5 mm. T3.5.5 and T3.5.6 do not apply.

T3.5.2 If a panel represents side impact structure it must be proven that it has at least the same properties as two steel tubes meeting the requirements for side impact structure tubes for buckling modulus, yield strength and absorbed energy.

T3.5.3 A baseline 3-point bending test, performed with steel tube(s), must demonstrate a minimum rig compliance of 85% and be documented in the SES.

T3.5.4 Composite structures with different core thicknesses but otherwise identical construction may use material properties derived from a single test panel, except for the front bulkhead and side impact structure. The panel with the thicker core must be tested and the structure using derived material properties must not use a core thickness of less than 66% of the tested panel.

T3.5.5 When a laminate is not quasi-isotropic, i.e. has equal strength and stiffness in the 0°, 90° and ±45° direction, the results from the 3 point bending test will be assigned to the 0° lay-up direction. To show equivalence in the SES, the 0° lay-up direction must be used and oriented accordingly in the chassis.

T3.5.6 The test samples must be presented at technical inspection. All samples must be marked with the following non-removable (e.g.: permanent marker or engraving, but no sticker) information: laminated structure acronym and date of testing.

T3.5.7 The distance between the two test panel supports must be at least 400 mm.

T3.5.8 The load applicator used to test any panel or tube must be metallic and have a radius of 50 mm.

T3.5.9 The load applicator must overhang the test piece to prevent edge loading.

T3.5.10 There must be no material between the load applicator and the test piece.

- T3.5.11 Perimeter shear tests must be completed which measure the force required to push or pull a 25 mm diameter flat punch through a flat laminate sample. The sample must be at least 100 mm × 100 mm. Core and skin thicknesses must be identical to those used in the actual chassis structure and be manufactured using the same materials and processes. If an asymmetrical lay-up is used, the thinner skin must face the punch.
- T3.5.12 The test fixture must support the entire sample, except for a 32 mm hole aligned co-axially with the punch. The sample must not be clamped to the fixture.

T3.6 Structural Documentation

- T3.6.1 All teams must submit a Structural Equivalency Spreadsheet (SES), Structural Equivalency 3D Model (SE3D) and [EV ONLY] ASES.
- T3.6.2 The SES and [EV ONLY] ASES spreadsheet form can be downloaded from the competition website.
- T3.6.3 The SE3D must contain a three dimensional CAD model of the chassis including all members of the primary structure and their mechanical attachment details.
- [MONOCOQUE ONLY] The SE3D must include the inner, outer skin and core thicknesses.
 - [MONOCOQUE ONLY] If the actual EI of a structure is used in the SES, this must be highlighted and included as a measurable instance.
 - [EV ONLY] The TSACs and their attachment must be included.
 - [CV ONLY] The air intake system, the fuel tank and filler neck must be included.
- T3.6.4 Vehicles must be fabricated in accordance with the materials and processes described in the SES.
- T3.6.5 Teams must bring a copy of the approved SES to technical inspection.

T3.7 Roll Hoops

- T3.7.1 Both roll hoops must be securely integrated to the primary structure using node-to-node triangulation or equivalent joining methods.
- T3.7.2 The minimum radius of any bend, measured at the tube centerline, must be at least three times the tube outside diameter. Bends must be smooth and continuous with no evidence of crimping or wall failure. The minimum area moment of inertia ([see T3.2](#)) must be maintained in all areas, including the bends of the manufactured tubes.
- T3.7.3 In a plane perpendicular to the longitudinal axis of the vehicle and through the lower endpoints of the roll hoop, no part of the primary structure may lie below 30 mm of the endpoints of the roll hoop.
- T3.7.4 Roll hoops attached to a composite primary structure must be mechanically attached at the top and bottom of both sides of the structure and at intermediate locations if needed to show equivalency. The lower roll hoop tubing attachment points must be within 50 mm of the endpoints of the roll hoop.
- T3.7.5 Mounting plates welded to the roll hoops must be at least 2 mm thick steel or 3 mm thick aluminum, dependent on the roll hoop material.

T3.7.6 Both roll hoops must have one 4.5 mm hole in a non-critical straight location and its surface at this point must be unobstructed for at least 180°.

T3.8 Main Hoop

T3.8.1 The main hoop must be constructed of a single piece of uncut, continuous, closed section steel tubing.

T3.8.2 In the side view the portion of the main hoop which is above its upper attachment point to the side impact structure must be inclined less than 10° from vertical.

T3.8.3 In the side view any bends in the main hoop above its upper attachment point to the primary structure must be braced to a node of the main hoop bracing support structure with tubing meeting the requirements of main hoop bracing.

T3.8.4 In the side view any portion lower than the upper attachment point to the side impact structure must be inclined either forward or not more than 10° rearward.

T3.9 Front Hoop

T3.9.1 The front hoop must be constructed of a continuous and closed section.

T3.9.2 If the front hoop is made from more than one piece it must be supported by node-to-node triangulation or an equivalent construction.

T3.9.3 In the side view, no part of the front hoop can be inclined more than 20° from vertical.

T3.9.4 If the front hoop is a welded construction made from multiple aluminum profiles, the equivalent yield strength must be considered in the as-welded condition unless the team demonstrates and shows proof that it has been properly solution heat treated and artificially aged. The team must supply sufficient documentation proving the appropriate heat treatment process was performed.

T3.9.5 The front hoop requires six attachment points, two on each side connecting to the front bulkhead support structures and two connecting to the front hoop bracing, and must therefore show equivalency to 180 kN, as follows from [T3.15.1](#) and [T3.11.4](#).

T3.9.6 Fully laminating the front hoop to the monocoque is acceptable. Fully laminating means that the hoop has to be encapsulated with laminate around its whole circumference, see figure 5. Equivalence to [T3.7.4](#) must be shown in the SES. The laminate encapsulating the front hoop must overlap by at least 25 mm on each side. It must have the same lay-up as the laminate that it is connecting to.

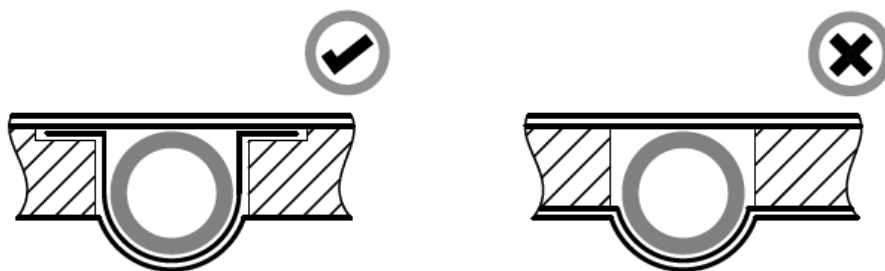


Figure 5: Front hoop laminating requirements

T3.10 Main Hoop Bracing

- T3.10.1 The main hoop must be supported to the front or the rear by bracing tubes on each side of the main hoop.
- T3.10.2 In the side view the main hoop and the main hoop braces must not lie on the same side of a vertical line coincident with the top of the main hoop.
- T3.10.3 The main hoop braces must be attached to the main hoop no lower than 160 mm below the top-most surface of the main hoop. The included angle formed by the main hoop and the main hoop braces must be at least 30°.
- T3.10.4 The main hoop braces must be straight.
- T3.10.5 The lower ends of the main hoop braces must be supported back to the upper attachment point of the main hoop to the side impact structure and to the lower attachment point of the main hoop to the side impact structure by a node-to-node triangulated structure or equivalent composite structure.
- T3.10.6 If any item which extends outside of the primary structure is attached to the main hoop braces, additional bracing is required to prevent bending loads in a rollover situation.

T3.11 Front Hoop Bracing

- T3.11.1 The front hoop bracing attaches on each side of the front hoop as well as the structure forward of the driver's feet. A minimum of two tubes without any bends must be straight on a line in the side view of the frame and must have a minimum distance of 100 mm between each other at the front hoop.
- T3.11.2 The front hoop bracing structure must be attached no lower than 50 mm below the top-most surface of the front hoop, see figure 6.
- T3.11.3 If the front hoop is inclined more than 10° to the rear, additional braces extending rearwards are required.
- T3.11.4 Composite front hoop bracing structures and their attachments cannot be counted towards the front bulkhead support structures and vice-versa for the structural equivalency documentation.
- T3.11.5 **Openings or reductions in effective panel height within a composite front hoop bracing or a composite front bulkhead support must not exceed a total area of 625 cm².**

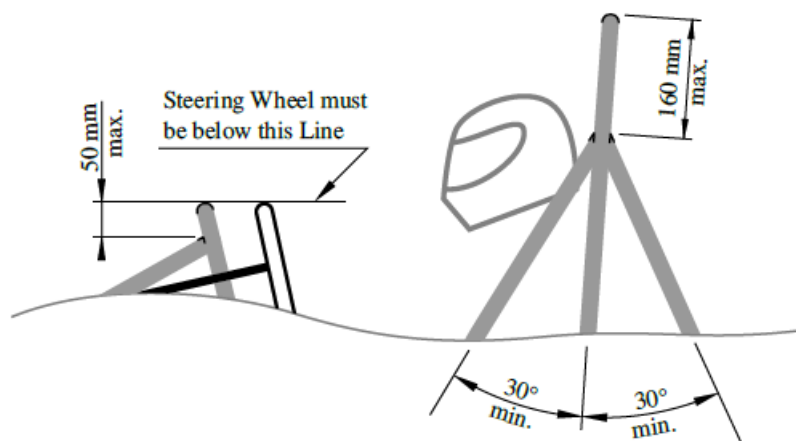


Figure 6: Front hoop bracing, main hoop bracing and steering wheel requirements

T3.12 Front Bulkhead

- T3.12.1 Any alternative material used for the front bulkhead must have a perimeter shear strength equivalent to a 1.5 mm thick steel plate.
- T3.12.2 If the front bulkhead is part of a composite structure and is modeled as an "L" shape, the EI of the front bulkhead about the vertical and lateral axes must be equivalent to a steel tube meeting the requirements for the front bulkhead. The length of the section perpendicular to the bulkhead may be a maximum of 25 mm measured from the rearmost face of the bulkhead.

T3.13 Front Bulkhead Support

- T3.13.1 The front bulkhead must be supported back to the front hoop by a minimum of three tubes on each side; an upper member, a lower member and diagonal bracing to provide triangulation.
- The upper support member must be attached to the front bulkhead a maximum of 50 mm below the top-most surface of the front bulkhead, and attached to the front hoop a maximum of 50 mm below the upper side impact member. If the attachment point of the upper member is greater than 100 mm above the upper side impact member, node-to-node triangulated bracing is required to transfer load to the main hoop.
 - The lower support member must be attached to the base of the front bulkhead and the base of the front hoop.
 - The diagonal bracing must triangulate the upper and lower support members' node-to-node.
- T3.13.2 If the front bulkhead support is part of a composite structure, it must have equivalent EI to the sum of the EI of the six baseline steel tubes that it replaces and it must not be counted towards the composite front hoop bracing structures for the structural equivalency documentation (i.e. T3.11.4).
- T3.13.3 The EI of the vertical side of the front bulkhead support structure must be equivalent to at least the EI of one baseline steel tube that it replaces.

T3.13.4 The perimeter shear strength of the monocoque laminate in the front bulkhead support structure must be at least 4 kN.

T3.14 Impact Structures

T3.14.1 The side impact structure must consist of at least three steel tubes ([see T3.2](#)) on each side of the cockpit (figure 7).

- The upper member must connect the main hoop and the front hoop.
- The upper member must be at a height between 240 mm and 320 mm above the lowest inside chassis point between the front and main hoop.
- The lower member must connect the bottom of the main hoop and the bottom of the front hoop.
- The diagonal member must triangulate the upper and lower member between the roll hoops node-to-node.

T3.14.2 Other impact structures, see [CV1.3.2](#), [EV4.4.2](#) and [EV5.5.2](#), must be fully triangulated structures and consist of at least three steel tubes, see [T3.2](#), on each side and rearward of the component that needs to be protected.

- If the component projects outwards from the roll hoops to the side, the front of the component must also be protected.
- The upper member must not be higher than 320 mm above the lowest inside chassis point between the front and main hoop.

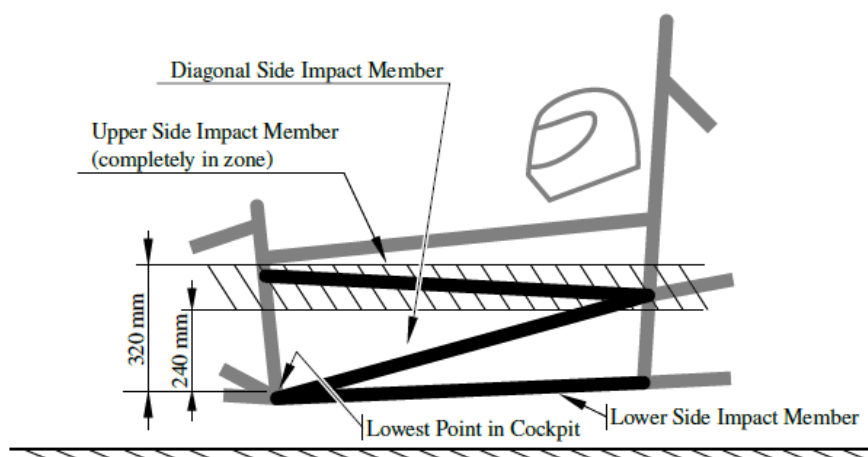


Figure 7: Side impact structure

T3.14.3 No part of the TSAC, motor(s) and their mountings may be within 25mm of the most rear impact structure of the vehicle. The TSAC must not be attached directly to the rear impact structure; however non-structural components such as cooling ducts are permitted.

T3.14.4 If the side impact structure is part of a composite structure, the following is required:

- The region of the structure up to a height of 320 mm above the lowest inside chassis point between the front and main hoop must have an EI equal to the three baseline steel tubes that it replaces. An example for side impact structure is given in figure 8.
- The vertical impact structure must have an EI equivalent to two baseline steel tubes and half the horizontal floor must have an EI equivalent to one baseline steel tube.

- The vertical impact structure must have an absorbed energy equivalent to two baseline steel tubes, exceeding 65 J.
- The perimeter shear strength must be at least 7.5 kN.

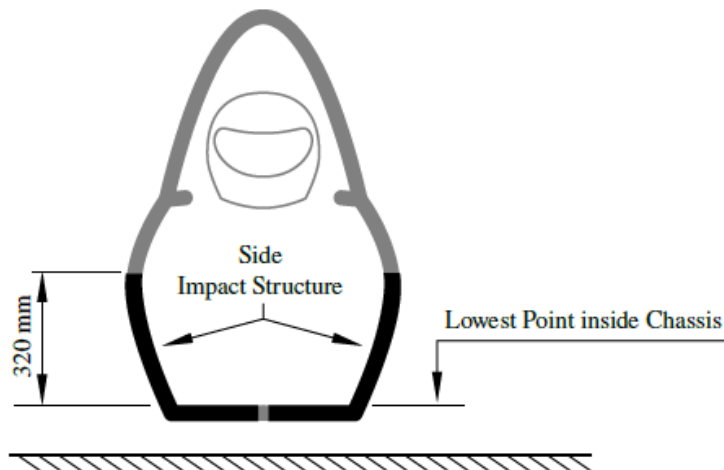


Figure 8: Side impact structure monocoque

T3.15 Bolted Primary Structure Attachments

- T3.15.1 If two parts of the primary structure are bolted together, each attachment point between the two parts must be able to carry a load of 30 kN in any direction.
- T3.15.2 Data obtained from the laminate perimeter shear strength test must be used to prove that adequate shear area is provided.
- T3.15.3 Each attachment point requires a minimum of two 8mm metric grade 8.8 [critical fasteners](#), see [T10.1](#), and steel backing plates with a minimum thickness of 2mm.
- T3.15.4 For the attachment of front hoop bracing, main hoop bracing and main hoop bracing support to the primary structure the use of one 10 mm metric grade 8.8 bolt is sufficient, if the bolt is on the centerline of the tube, see figure 9.

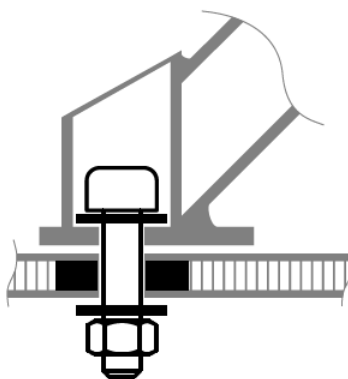


Figure 9: Bolted roll hoop bracing support

- T3.15.5 When using bolted joints within the primary structure, no crushing of the laminate core material is permitted.
- T3.15.6 For the AIP to front bulkhead attachment, and if two panels or plates of the primary structure are bolted together, for each 200mm of reference perimeter a minimum of one 8mm metric grade 8.8 bolt(s) must be used, rounded up to the next integer. The bolts must be evenly distributed over the circumference using good engineering practices. The reference perimeter is the outside perimeter of the attached part at the connection. The bolts are considered critical fasteners, must comply with T10 and require steel backing plates with a minimum thickness of 2mm.
- T3.15.7 For each bolt and its corresponding insert of T3.15.6 an exception in favor of blind inserts may be given, if physical testing is documented in the SES and/or ASES. Each bolt and its corresponding insert must withstand a force of 15 kN in both the axial pull-out and the weakest radial direction.
- T3.16 Impact Attenuator (IA)**
- T3.16.1 Each vehicle must be equipped with an IA assembly consisting of an IA and AIP.
- T3.16.2 The IA must be:
- Installed forward of the front bulkhead.
 - At least 100 mm high and 200 mm wide for a minimum distance of 200 mm forward of the front bulkhead.
 - No portion of the required 100 X 200 X 200 mm³ volume of the IA can be positioned more than 350 mm above the ground
 - Not able to penetrate the front bulkhead in the event of an impact.
 - Attached securely and directly to the AIP.
 - Not part of the non-structural bodywork.
 - Designed with a closed front section.
 - Cannot be wider or higher than the AIP.
- T3.16.3 The baseline design for the AIP is a 1.5 mm solid steel or 4.0 mm solid aluminum plate.
- If the IA and AIP (IA assembly) are bolted to the front bulkhead, it must be the same size as the outside dimensions of the front bulkhead.
 - If it is welded to the front bulkhead, it must extend at least to the centerline of the front bulkhead tubing in all directions.
 - The AIP must not extend past the outside edges of the front bulkhead.
- T3.16.4 Alternative AIP designs are permissible if equivalency to T3.16.3 is proven by physical testing as in T3.18.2.
- T3.16.5 The IA may be attached to the AIP by a minimum of four 8 mm metric grade 8.8 bolts that are considered critical fasteners and must comply with T10. Attachment(s) using adhesive must be able to carry a load of 60 kN in any direction.
- T3.16.6 The attachment of the IA assembly must be designed to provide an adequate load path for transverse and vertical loads in the event of off-center and off-axis impacts. Segmented foam attenuators must have the segments bonded together to prevent sliding or parallelogramming.

T3.16.7 A team may use one of the “standard” FSAE IAs without physical testing, provided that all of the following is fulfilled:

- A1.5 mm solid steel AIP is used, that is welded along its full perimeter to a steel bulkhead or a 4mm solid aluminium AIP is used, that is bolted to any bulkhead with a minimum of eight 8mm metric grade 8.8 bolts.
- adhesive used to mount the “**standard honeycomb IA**” to the AIP must have a shear strength of at least 24 MPa.
- T3.19.4 is fulfilled regarding aerodynamic device and sensor positions relative to the AIP.
- If the front bulkhead has an overall width greater than 400mm and/or an overall height greater than 350mm, a diagonal or X-bracing that is a front bulkhead support tube or an approved equivalent per T3.2, or equivalent for monocoque bulkheads, must be included.
- If the “standard” honeycomb IA is used, the IA must be of pre-crushed type.

T3.16.8 If the standard IA is used, but does not comply with the requirements of T3.16.7, physical testing must be carried out to prove that the AIP does not permanently deflect more than 25 mm.

T3.17 Impact Attenuator Data Requirement

T3.17.1 All teams must submit an IA data report using the Impact Attenuator Data (IAD) template provided at the competition website.

T3.18 Impact Attenuator Test Requirements

T3.18.1 The IA assembly, when mounted on the front of a vehicle with a total mass of 300 kg and impacting a solid, non-yielding impact barrier with a velocity of impact of 7 m/s, must meet the following requirements:

- Decelerate the vehicle at a rate not exceeding 20 g average and 40 g peak.
- The energy absorbed in this event must meet or exceed 7350 J.
- Teams using the standard IA are not required to submit test data with their IAD report, but all other requirements must be included.

Equivalent (higher) test velocities are only allowed to accommodate for a lower total testing mass, as long as the energy absorbed is 7350 J or more. If these requirements cannot be met, a team must use the standard IA.

T3.18.2 During the IA test:

- The IA must be attached to the AIP using the intended vehicle attachment method.
- The IA assembly must be attached to a test fixture that has geometry representative of the intended primary structure and equal or higher stiffness and strength. When no baseline design for the AIP is used, see T3.16.3, the test fixture must be a copy of the intended primary structure (i.e. materials, lay-up, joining methods).
- There must be at least 50 mm clearance rearwards of the AIP to the test fixture.
- No part of the AIP may permanently deflect more than 25 mm beyond the position of the AIP before the test.
- The representative test fixture of the intended primary structure may not be structurally compromised (i.e. no cracks) after the test.

- T3.18.3 Teams using IAs (typically structural noses) directly attached to the front bulkhead, which shortcut the load path through the bulk of the AIP, must conduct an additional test. This test must prove that the AIP can withstand a load of 120 kN (300 kg multiplied by 40 g), where the load applicator matches the minimum IA dimensions.
- T3.18.4 Vehicles with aerodynamic devices and/or sensors in front of the front bulkhead must not exceed the peak deceleration of T3.18.1 for the combination of their IA assembly and the non-crushable object(s). One of the following three methods may be used to prove the design does not exceed 120 kN:
- Physical testing of the IA assembly including any non-crushable object(s) or **geometrically** representative dummies thereof in front of the AIP. When non-crushable object(s) are located in front view of the AIP, physical testing is required.
 - Combining the peak force from physical testing of the IA assembly with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.
 - Combining the “standard” IA peak load of 95 kN with the failure load for the mounting of the non-crushable object(s), calculated from fastener shear and/or link buckling.
- T3.18.5 Dynamic testing (sled, pendulum, drop tower, etc.) of the IA may only be conducted at a dedicated test facility. This facility may be part of the university, but must be supervised by professional staff. Teams are not allowed to design their own dynamic test apparatus.
- T3.18.6 When using acceleration data from the dynamic test, the average deceleration must be calculated based on the raw unfiltered data. If peaks above the 40 g limit are present in the data, a 100 Hz, 3rd order, low pass Butterworth (–3 dB at 100 Hz) filter may be applied.
- T3.18.7 Conducting an IA test by using a manned student vehicle prototype to drive (powered or unpowered) into any structure, will result in de-registration of the team from the competition.

T3.19 Non-Crushable Objects

- T3.19.1 All non-crushable objects (e.g. pedals, master cylinders, hydraulic reservoirs) must be rearward of the rearmost plane of the front bulkhead and at least 25 mm behind the AIP at any time, except for environment perception sensors, aerodynamic devices and their mountings.
- T3.19.2 **Sensor and aerodynamic devices may extend forward of the AIP, but their chassis attachment point must be located rearward of the AIP**

T4 COCKPIT

T4.1 Cockpit Opening

- T4.1.1 The size of the cockpit opening needs to be sufficient for the template shown on the left in figure 10 to pass vertically from the opening below the top of the side impact structure when held horizontally. The template may be moved fore and aft.
- T4.1.2 If the side impact structure is not made of tubes, the template must pass until it is 320 mm above the lowest inside chassis point between the front and main hoop.

- T4.1.3 The steering wheel, seat and all padding may be removed for the template to fit. Any other parts may only be removed if they are integrated with the steering wheel.

T4.2 Cockpit Internal Cross Section

- T4.2.1 The cockpit must provide a free internal cross section sufficient for the template shown on the right in figure 10 to pass from the cockpit opening to a point 100 mm rearwards of the face of the rearmost pedal in an inoperative position. The template may be moved up and down. Adjustable pedals must be in their most forward position.

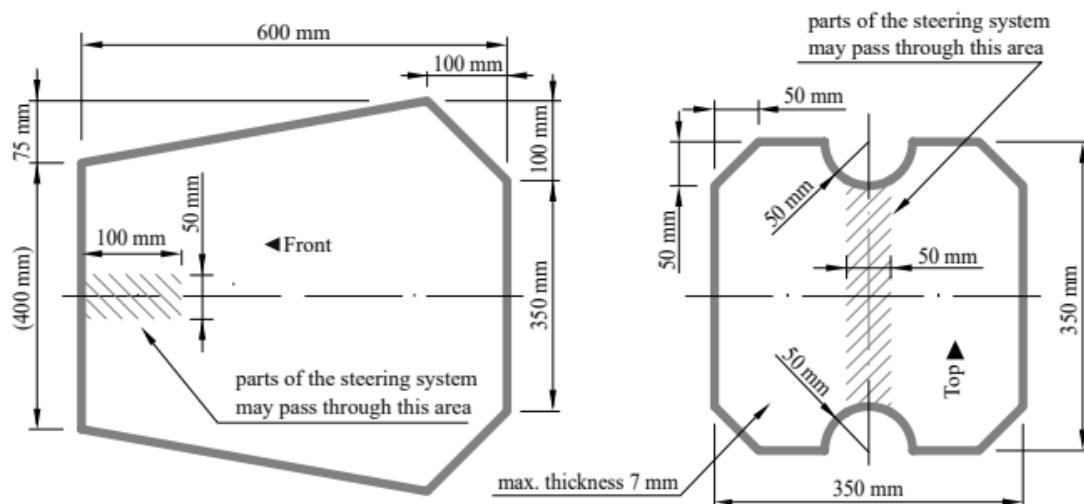


Figure 10: Cockpit opening template (left) and cockpit internal cross section template (right)

- T4.2.2 The steering wheel and any padding that can be removed without the use of tools while the driver is seated may be removed for the template to fit.
- T4.2.3 The driver's feet and legs must be completely contained within the primary structure when the driver is seated normally and the driver's feet are touching the pedals. In the side and front views, any part of the driver's feet or legs must not extend above or outside of this structure.

T4.3 Percy (95th percentile male)

- T4.3.1 When seated normally and restrained by the driver's restraint system, the helmet of a 95th percentile male and all of the team's drivers must, see figure 11:
- Be a minimum of 50 mm away from the straight line drawn from the top of the main hoop to the top of the front hoop.
 - Be a minimum of 50 mm away from the straight line drawn from the top of the main hoop to the lower end of the main hoop bracing if the bracing extends rearwards.
 - Be no further rearwards than the rear surface of the main hoop if the main hoop bracing extends forwards.
- T4.3.2 The 95th percentile male is represented by a two dimensional figure consisting of two circles of 200 mm diameter (one representing the hips and buttocks and one representing the shoulder region) and one circle of 300 mm (representing the head with helmet).

T4.3.3 The two 200 mm circles are connected by a straight line measuring 490 mm. The 300 mm circle is connected by a straight line measuring 280 mm with the upper 200 mm circle.

T4.3.4 The figure has to be positioned in the vehicle as follows, see figure 12:

- The seat adjusted to the rearmost position
- The pedals adjusted to the front-most position
- The bottom 200 mm circle is placed on the seat bottom. The distance between the center of the circle and the rearmost actuation face of the pedals must be minimum 915 mm.
- The middle circle positioned on the seat back
- The upper 300 mm circle is positioned 25 mm away from the head restraint.

T4.4 Side Tubes

T4.4.1 If there is any chassis member alongside the driver at the height of the neck of any of the drivers in the team, a metal tube or piece of sheet metal must be attached to the chassis to prevent the driver's shoulders from passing under that chassis member.

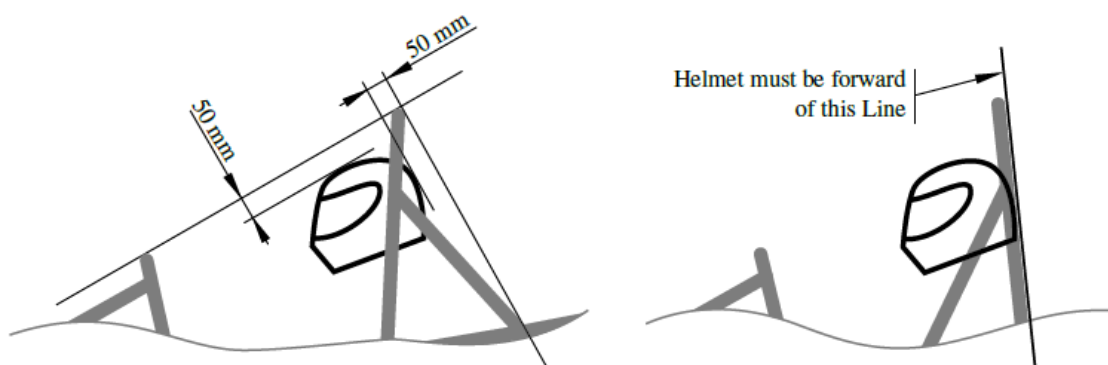


Figure 11: Minimum helmet clearance

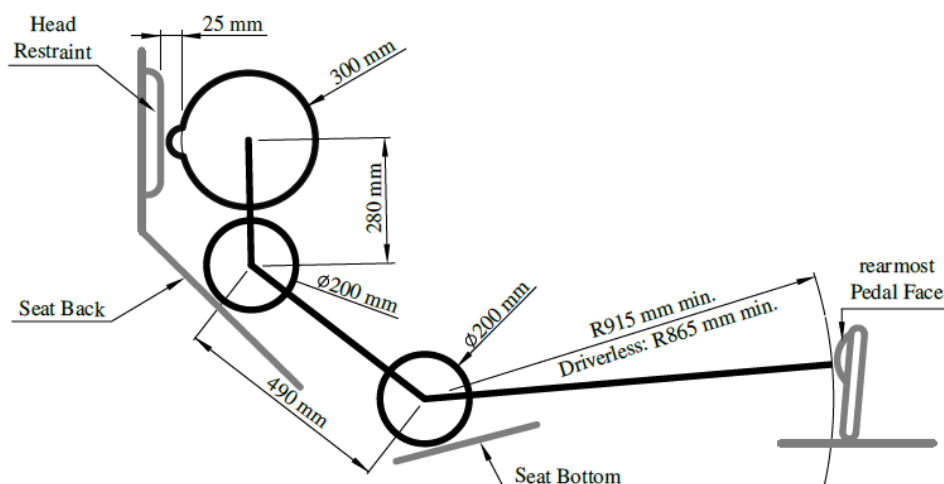


Figure 12: Percy placement

T4.5 Driver's Harness Attachment

- T4.5.1 Any harness attachment to a monocoque must be using one 10 mm metric grade 8.8 bolt or two 8 mm metric grade 8.8 bolts (or bolts of an equivalent norm) and steel backing plates with a minimum thickness of 2 mm.
- T4.5.2 Any harness that is fastened to the primary structure using brackets must use two 8 mm metric grade 8.8 or stronger fasteners.
- T4.5.3 It must be proven that the attachments for shoulder and lap belts can support a load of 13 kN and the attachment points of the anti-submarine belts can support a load of 6.5 kN.
- T4.5.4 If the lap belts and anti-submarine belts are attached less than 100 mm apart, these must support a total load of 19.5 kN.
- T4.5.5 If the belts are attached to a laminated structure or the mounting brackets and tabs are not made from steel at least 1.6 mm thick, physical testing is required. The following requirements must be met:
- Load is applied to a test sample representing the tubular or laminated structure and must use the same brackets and tabs
 - Edges of the test fixture supporting the sample must be a minimum of 125 mm from the load application point.
 - The width of the shoulder harness test sample must not be any wider than the shoulder harness panel height used to show equivalency for the shoulder harness mounting bar.
 - Designs with attachments near a free edge may not support the free edge during the test.
 - Harness loads must be tested with the worst case for the range of the angles specified for the driver's harness.

T4.6 Driver's Seat

- T4.6.1 The lowest point of the driver's seat must in the side view not extend below the upper face of the lowest side impact structure member or have a longitudinal tube (or tubes) that meet the material requirements for the side impact structure (T3.2), passing underneath the lowest point of the seat.
- T4.6.2 Adequate heat insulation must be provided to ensure that the driver is not able to contact any parts of the vehicle with a surface temperature above 60°C. The insulation may be external to the cockpit or incorporated with the driver's seat or firewall. The design must address all three types of heat transfer with the following minimum requirements between the heat source and the part that the driver could contact:
- (a) Conduction insulation by:
 - i. No direct contact, or
 - ii. A heat resistant, conduction insulation material with a minimum thickness of 8 mm.
 - (b) Convection insulation by a minimum air gap of 25 mm.
 - (c) Radiation insulation by:
 - i. A solid metal heat shield with a minimum thickness of 0.4 mm or
 - ii. Reflective foil or tape when combined with T4.6.2.a.ii.

T4.7 Floor Closeout

- T4.7.1 All vehicles must have a floor closeout made of one or more panels, which separate the driver from the ground.
- T4.7.2 The closeout must extend from the front bulkhead to the firewall.
- T4.7.3 The panels must be made of a solid, non-brittle material.
- T4.7.4 If multiple panels are used, gaps between panels may not exceed 3 mm.

T4.8 Firewall

- T4.8.1 A firewall must separate the driver compartment from all components of the fuel supply system, hydraulic fluid (except brake system and dampers), flammable liquids, the low voltage battery and any TS component (see EV1.1.1).
- T4.8.2 The firewall must cover any straight line between the parts mentioned in T4.8.1 and any part of the tallest driver below a plane 100 mm above the bottom of the helmet.
- T4.8.3 The firewall must be a non-permeable surface made from a rigid, fire resistant material (see T1.2.1) which must be rigidly mounted to the vehicle's structure.
- T4.8.4 Any firewall must seal completely against the passage of fluids, especially at the sides and the floor of the cockpit.
- T4.8.5 Pass-throughs for wiring, cables, etc. are allowed if grommets, cable glands or connectors are used to seal the pass-through.
- T4.8.6 Multiple panels may be used to form the firewall but must overlap at least 5 mm and be sealed at the joints. Any sealing material must not be vital to the structural integrity of the firewall.
- T4.8.7 [EV ONLY] The TS firewall between driver and TS components must be composed of two layers:
 - One solid layer, facing the TS side, must be made of aluminum with a thickness of at least 0.5 mm. This part of the TS firewall must be grounded according to EV3.1.
 - The second layer, facing the driver, must be made of an electrically insulating and fire retardant material (see T1.2.1). The second layer must not be made of CFRP.
 - The thickness of the second layer must be sufficient to prevent penetrating this layer with a 4 mm wide screwdriver and 250 N of force.

A sample of the tractive system firewall must be presented at technical inspection.
- T4.8.8 [EV ONLY] Conductive parts (except for the chassis) may not protrude through the TS firewall or must be properly insulated on the driver's side. The driver must not be able to touch uninsulated firewall mounting points while operating the vehicle.
- T4.8.9 [EV ONLY] TS parts outside of the envelope (see EV4.4.3) do not need a firewall.

T4.9 Accessibility of Controls

- T4.9.1 All vehicle controls must be operated from inside the cockpit without any part of the driver, e.g. hands, arms or elbows, being outside the vertical planes tangent to the outermost surface of the side impact structure.

T4.10 Driver Visibility

- T4.10.1 The driver must have adequate visibility to the front and sides of the vehicle. Seated in a normal driving position, the driver must have a minimum field of vision of 100° to either side. The required visibility may be obtained by the driver turning their head and/or the use of mirrors.
- T4.10.2 If mirrors are required to meet [T4.10.1](#), they must remain in place and be adjusted to enable the required visibility throughout all dynamic events.

T4.11 Driver Egress

- T4.11.1 All drivers must be able to exit to the side of the vehicle in less than 5 s with the driver in the fully seated position, hands in the driving position on the connected steering wheel (in all possible steering positions) and wearing the required driver equipment as in [T13.3](#). The egress time will stop when the driver has both feet on the ground.

T4.12 Driver Accommodation

- T4.11.1 The vehicle must be able to accommodate drivers of sizes ranging from 5th percentile female up to 95th percentile male.
- Accommodation includes driver position, driver controls, and driver equipment.

T5 DRIVER RESTRAINT SYSTEM**T5.1 Definitions**

- T5.1.1 6-point system – Consists of a two-piece lap belt (minimum width 50 mm), two shoulder straps (minimum width 75 mm) and two leg or anti-submarine straps (minimum width 50 mm).
- T5.1.2 7-point system – Same as the 6-point system except it has three anti-submarine straps.
- T5.1.3 Upright driving position – Position with a seat back angled at 30° or less from the vertical as measured along the line joining the two 200 mm circles of the 95th percentile male template as defined in [T4.3](#) and positioned per [T4.3.4](#).
- T5.1.4 Reclined driving position – Position with a seat back angled at more than 30° from the vertical as measured along the line joining the two 200 mm circles of the 95th percentile male template as defined in [T4.3](#) and positioned per [T4.3.4](#).

T5.2 Belts - General

- T5.2.1 All drivers must use a 6-point or 7-point restraint harness meeting the following specifications:

- All driver restraint systems must meet SFI Specification 16.1, SFI Specification 16.5, SFI Specification 16.6, FIA specification 8853/2016.
- The belts must bear the appropriate dated labels.
- The material of all straps must be in perfect condition.
- There must be a single metal-to-metal latch type quick release for all straps.
- All lap belts must incorporate a tilt lock adjuster (“quick adjuster”). A tilt lock adjuster in each portion of the lap belt is highly recommended. Lap belts with “pull-up” adjusters are recommended over “pull-down” adjusters.
- Vehicles with a “reclined driving position” must have either anti-submarine belts with tilt lock adjusters (“quick adjusters”) or have two sets of anti-submarine belts installed.
- The shoulder harness must be the “over-the-shoulder type”. Only separate shoulder straps are permitted (i.e. “Y”-type shoulder straps are not allowed). The “H”-type configuration is allowed.
- The shoulder harness straps must be threaded through the three bar adjusters in accordance with the manufacturer’s instructions.
- When a HANS device is used by the driver, FIA certified 50 mm wide shoulder harnesses are allowed.

T5.2.2 SFI spec harnesses must be replaced following December 31st of the 2nd year after the date of manufacture as indicated by the label. FIA spec harnesses must be replaced following December 31st of the year marked on the label.

Note: Teams competing at Formula Bharat may extend the use of the harness until January 31st of the 3rd year after the date of manufacture as indicated by the label.

T5.2.3 The restraint system must be worn tightly at all times.

T5.3 Belt, Strap and Harness Installation - General

T5.3.1 The lap belt, shoulder harness and anti-submarine strap(s) must be securely mounted to the primary structure. This structure and any guide or support for the belts must meet the minimum requirements of T3.2.

T5.3.2 Harnesses, belts and straps must not pass through a firewall, i.e. all harness attachment points must be on the driver’s side of any firewall.

T5.3.3 The attachment of the driver’s restraint system to a monocoque structure requires an approved SES per T3.6. The lap belts and anti-submarine belts must not be routed over the sides of the seat. Where the belts or harness pass through a hole in the seat, the seat must be rolled or grommited to prevent chafing of the belts.

T5.4 Lap Belt Mounting

T5.4.1 The lap belt must pass around the pelvic area below the anterior superior iliac spines (the hip bones).

T5.4.2 The lap belts should come through the seat at the bottom of the sides of the seat to maximize the wrap of the pelvic surface and continue in a straight line to the anchorage point.

T5.4.3 In the side view, the lap belt must be capable of pivoting freely by using either a shouldered bolt or an eye bolt attachment.

- T5.4.4 With an “upright driving position”, in the side view the lap belt must be at an angle of between 45° and 65° to the horizontal.
- T5.4.5 With a “reclined driving position”, in the side view the lap belt must be between an angle of 60° and 80° to the horizontal.
- T5.4.6 In “upright driving position” the centerline of the lap belt at the seat bottom should be between 0 mm to 76 mm forward of the seat back to seat bottom junction as in figure 13 (below). In “reclined driving position” it may also be rearward of the junction point.

T5.5 Shoulder Harness

- T5.5.1 The shoulder harness must be mounted behind the driver to a structure that meets the requirements of the primary structure. However, it cannot be mounted to the main hoop bracing or attendant structure without additional bracing to prevent loads being transferred into the main hoop bracing.
- T5.5.2 If the harness is mounted to a tube that is not straight, the joints between this tube and the structure to which it is mounted must be reinforced in the side view by triangulation tubes to prevent torsional rotation of the harness mounting tube. Supporting calculations are required. Analysis method: Use 7 kN load per attachment and the range of angles in T5.5.5, calculate that the bent shoulder harness bar triangulation stresses are less than as welded yield strength T3.2.4 for combined bending and shear and does not fail in column buckling. If the team chooses not to perform the strength analysis T3.2.6 will apply.
- T5.5.3 The strength of any shoulder harness bar and bracing tubes must be proven in the relevant tab of the team’s SES submission.
- T5.5.4 The shoulder harness mounting points must be between 180 mm and 230 mm apart, measured center to center.
- T5.5.5 From the driver’s shoulders rearwards to the mounting point or structural guide, the shoulder harness must be between 10° above the horizontal and 20° below the horizontal as in figure 13.

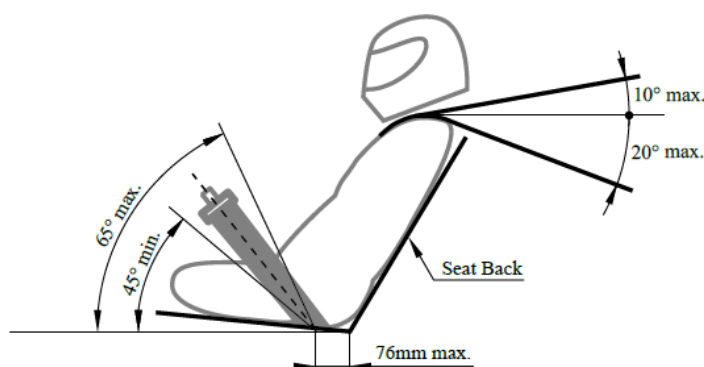


Figure 13: Lap belt and shoulder harness mounting

T5.6 Anti-Submarine Belt Mounting

T5.6.1 The anti-submarine belts of a 6 point harness should be mounted in one of the following setups:

- With the belts going vertically down from the groin, or angled up to 20° rearwards. The anchorage points should be approximately 100 mm apart.
- With the anchorage points on the primary structure at or near the lap belt anchorages, the driver sitting on the anti-submarine belts and the belts coming up around the groin to the release buckle.

T5.7 Head Restraint

T5.7.1 A head restraint must be provided on the vehicle to limit the rearward motion of the driver's head.

T5.7.2 The head restraint must:

- Be vertical or near vertical in the side view.
- Be padded with an energy absorbing material with a minimum thickness of 40 mm that meets either the SFI 45.2 standard, or is listed in the FIA technical list no 17 as a type B material for single seater cars.
- [Include a rectangular area of at least a minimum width and height of 150mm.](#)
- Be located so that for each driver:
 - The restraint is no more than 25 mm away from the back of the driver's helmet, with the driver in their normal driving position.
 - The contact point of the back of the driver's helmet on the head restraint is no less than 50 mm from any edge of the head restraint.

T5.7.3 The head restraint, its attachment and its mounting must withstand a force of 890 N applied in the rearward direction at any point on its surface.

T5.8 Roll Bar Padding

T5.8.1 Any portion of the roll bar, roll bar bracing or chassis which might be contacted by the driver's helmet must be covered with a minimum thickness of 12 mm of padding which meets SFI spec 45.1 or FIA 8857-2001.

T5.9 Driver's Leg Protection

T5.9.1 All moving suspension and steering components and other sharp edges inside the cockpit between the front hoop and a vertical plane 100 mm rearward of the pedals, must be shielded with solid material.

T5.9.2 Covers over suspension and steering components must be removable to allow inspection of the mounting points.

T6 BRAKE SYSTEM

T6.1 Brake System - General

- T6.1.1 The vehicle must be equipped with a hydraulic brake system that acts on all four wheels and is operated by a single control.
- T6.1.2 The brake system must have two independent hydraulic circuits such that in the case of a leak or failure at any point in the system, effective braking power is maintained on at least two wheels. Each hydraulic circuit must have its own fluid reserve, either by the use of separate reservoirs or by the use of a dammed reservoir.
- T6.1.3 The brake system must be sealed to prevent leakage.
- T6.1.4 While parts of the brake system are disconnected, e.g. by a quick-release decoupling
- [EV ONLY] the vehicle must not be able to enter R2D mode, see [EV4.11.7](#).
 - the ASB must not pass its initial checkup sequence, see T15.3.1, in autonomous mode.
- T6.1.5 A single brake acting on a limited-slip differential is acceptable.
- T6.1.6 “Brake-by-wire” systems are prohibited.
- T6.1.7 Unarmored plastic brake lines are prohibited.
- T6.1.8 The brake system must be protected from failure of the drivetrain (see [T7.3.2](#)) from touching any movable part and from minor collisions.
- T6.1.9 In the side view any portion of the brake system that is mounted on the sprung part of the vehicle must not project below the lower surface of the chassis.
- T6.1.10 Any part of the brake system must be within the surface envelope, see [T1.1.18](#).
- T6.1.11 The brake pedal, [including the load transferring pedal face](#) and its mounting must be designed to withstand a force of 2 kN without any failure of the brake system or pedal box. This may be tested by pressing the pedal with the maximum force that can be exerted by any official when seated normally.
- T6.1.12 The brake pedal, including pedal face, must be fabricated from steel or aluminum or machined from steel, aluminum or titanium.
- T6.1.13 [Adjustable pedals must have a positive stop to prevent loosening from the chassis.](#)
- T6.1.13 [EV ONLY] The first 90% of the brake pedal travel may be used to regenerate brake energy without actuating the hydraulic brake system. The remaining brake pedal travel must directly actuate the hydraulic brake system, but brake energy regeneration may remain active.

T6.2 Brake Over-Travel Switch (BOTS)

- T6.2.1 A brake pedal over-travel switch must be installed on the vehicle as part of the shutdown circuit, as in [EV6](#) or [CV4.1](#). This switch must be installed so that in the event of a failure in at least one of the brake circuits the brake pedal over travel will result in the shutdown

circuit being opened. This must function for all possible brake pedal and brake balance settings without damaging any part of the vehicle.

T6.2.2 Repeated actuation of the switch must not close the shutdown circuit, and it must be designed so that the driver cannot reset it.

T6.2.3 The BOTS must be a push-pull, push-rotate or flip type mechanical switch. It may consist of a series connection of switches.

T6.3 Brake Light

T6.3.1 The vehicle must be equipped with one brake light that is illuminated if and only if

- the hydraulic brake system is actuated
- [EV ONLY] or the electric brake system is actuated (see [EV2.2.3](#))

T6.3.2 The brake light must

- be a red light.
- be clearly visible from the rear.
- meet requirements according to T11.10.

T6.3.3 In the side view, the brake light must be oriented vertical or near vertical and mounted between the wheel centerline and driver's shoulder level. Viewed from the back it should be positioned approximately at the vehicle's centerline.

T7 POWERTRAIN

T7.1 Transmission and Drive

T7.1.1 Any transmission and drivetrain may be used.

T7.1.2 Movement of the vehicle without a person in the vehicle and with the master switch(es) in the off position must be possible.

T7.2 Coolant Fluid and System Sealing

T7.2.1 [CV ONLY] Water-cooled engines must only use plain water.

T7.2.2 [EV ONLY] TS components may only use plain water, air or oil as the coolant.

T7.2.3 Cooling systems using plain water (except outboard wheel motors and their cooling hoses) must have a heat resistant (Permanently rated for at least 100 °C), rigid and rigidly mounted cover which covers any part of the tallest driver below a plane 100 mm above the bottom of the helmet.

T7.2.4 Any cooling or lubrication system must be sealed to prevent leakage.

T7.2.5 Separate catch cans must be employed to retain fluids other than plain water from any vents for the cooling system or combustion engine lubrication system. Each catch-can must have a minimum volume of 10% of the fluid being contained or 900 ml whichever is greater.

- T7.2.6 Any vent for other systems containing liquid lubricant or coolant must have a catch-can with a minimum volume of 10% of the fluid being contained or 100 ml, whichever is greater.
- T7.2.7 All parts of the engine cooling, motor(s) cooling and lubrication system, including their mountings, must be rated for at least 100°C or the temperatures the respective fluid may reach, whichever is higher.
- T7.2.8 Catch cans must be rigidly mounted to the chassis and located rearwards of the firewall below the driver's shoulder level.
- T7.2.9 Any catch can must vent through a hose with a minimum internal diameter of 3 mm down to the bottom level of the chassis and must exit outside the bodywork.

T7.3 Drive Train Shields and Guards

- T7.3.1 The lowest point of any lubrication system can only be lower than the line between the lowest point of the main hoop and the lowest chassis member behind the lubrication system if it is protected from hitting the ground by a structure mounted directly to the chassis.
- T7.3.2 Exposed rotating final drivetrain parts, such as gears, clutches, chains and belts must be fitted with scatter shields. Scatter shields and their mountings must:
- Be constructed of non-perforated 2 mm steel or 3 mm aluminum alloy 6061-T6.
 - Be attached with 6 mm metric grade 8.8 or stronger and must comply with T10.1.
- T7.3.3 Scatter shields for high-speed rotating final drivetrain parts (such as electric motors, clutches, sprockets, gears etc.) that have an OEM casing that do not comply with T7.3.2 may be used, if material is added to achieve the minimum required thickness.
- T7.3.4 [EV ONLY] When an electrical motor casing is rotating around the stator or is perforated, a scatter shield must be included around the motor. This scatter shield must be at least 1 mm thick and made from aluminum alloy 6061-T6 or steel.
- T7.3.5 Scatter shields for chains and belts must be centered on the centerline of the chain or belt and remain aligned with the chain or belt under all conditions. The minimum width of the scatter shield should be at least three times the width of the chain or belt. The minimum material requirements are:
- For metallic chains and belts: 2 mm steel.
 - For non-metallic chains and belts: 3 mm aluminum alloy 6061-T6.
- T7.3.6 All fasteners attaching scatter shields and guards must be 6 mm metric grade 8.8 or stronger and must comply with T10.1.
- T7.3.7 Finger guards are required to cover any drivetrain parts that spin while the vehicle is stationary with the engine running. Finger guards may be made of lighter material, sufficient to resist finger forces. Mesh or perforated material may be used but must prevent the passage of a 12 mm diameter object through the guard.

T8 AERODYNAMIC DEVICES

T8.1 Definition Aerodynamic Device

T8.1.1 A specifically designed structure mounted on the vehicle to guide the airflow around the vehicle, increasing the downforce acting on the vehicle and/or lowering its drag. The mounting of this structure is not regarded as an aerodynamic device, unless it is intentionally designed to be one.

T8.2 Restrictions for Aerodynamic Devices

T8.2.1 Height restrictions:

- All aerodynamic devices forward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position, must be lower than 500 mm from the ground.
- All aerodynamic devices in front of the front axle and extending further outboard than the most inboard point of the front tire/wheel must be lower than 250 mm from the ground.
- All aerodynamic devices rearward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position must be lower than 1.1 m from the ground.

T8.2.2 Width restrictions:

- All aerodynamic devices lower than 500 mm from the ground and further rearward than the front axle, must not be wider than a vertical plane touching the most outboard point of the front and rear wheel/tire.
- All aerodynamic devices higher than 500 mm from the ground, must not extend outboard of the most inboard point of the rear wheel/tire.

T8.2.3 Length restrictions:

- All aerodynamic devices must not extend further rearward than 250 mm from the rearmost part of the rear tires.
- All aerodynamic devices must not extend further forward than 700 mm from the fronts of the front tires.

T8.2.4 All restrictions must be fulfilled with the wheels pointing straight and with any suspension setup with or without a driver seated in the vehicle.

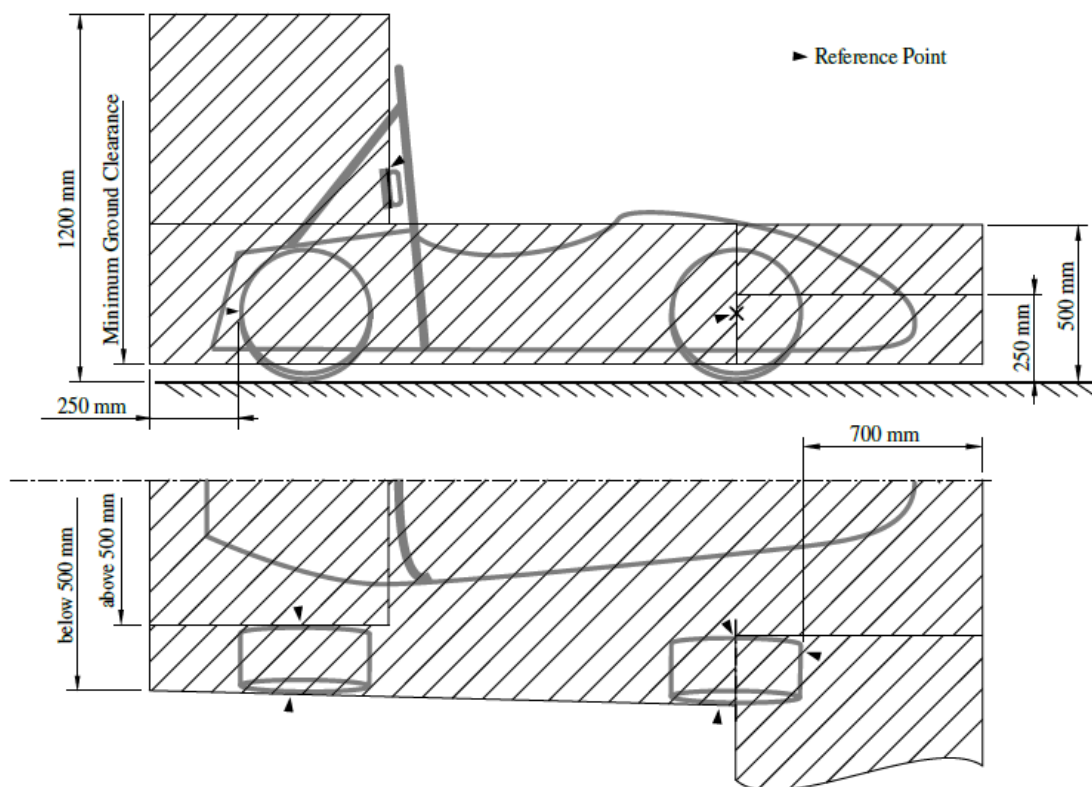


Figure 14: Maximum dimensions and positioning of aerodynamic devices. The positioning space is further restricted ([see T2.1](#)).

T8.3 Minimum Edge Radii of Bodywork and Aerodynamic Devices

T8.3.1 For all edges that could come into contact with any standing pedestrian without reaching to the vehicle, the minimum radius of the bodywork and aerodynamic devices is 3 mm for all forward facing edges and 1 mm for all other edges.

T8.4 Aerodynamic Devices Stability and Strength

T8.4.1 Any aerodynamic device must be able to withstand a force of 200 N distributed over a minimum surface of 225 cm² and not deflect more than 10 mm in the load carrying direction.

T8.4.2 Any aerodynamic device must be able to withstand a force of 50 N applied in any direction at any point and not deflect more than 25 mm.

T9 COMPRESSED GAS SYSTEMS AND HIGH PRESSURE HYDRAULICS

T9.1 Compressed Gas Cylinders and Lines

T9.1.1 Compressed Gas System (CGS) – any system that uses a compressed gas as an actuating medium, except for gas springs.

- T9.1.2 The working gas of any CGS must be non-flammable.
- T9.1.3 The pressure inside any CGS must not exceed 10 bar. Gas cylinders/tanks may exceed the 10 bar limit, if a pressure regulator which limits the output pressure to a maximum of 10 bar is mounted directly onto them.
- T9.1.4 All parts of any CGS must be designed for the maximum possible operating pressure.
- T9.1.5 Gas cylinders/tanks and their pressure regulators must be of proprietary manufacture, certified and labeled as such.
- T9.1.6 All parts of any CGS and their mountings must be located within the rollover protection envelope, see T1.1.16 and must be protected from collision or damage.
- T9.1.7 Gas cylinders/tanks must be mounted according to T9.3.1 and their axis must not point at the driver.
- T9.1.8 Gas cylinders/tanks and their pressure regulators must be shielded from the driver. The shields must be steel or aluminium with a minimum thickness of 1 mm.
- T9.1.11 Gas cylinders/tanks must be insulated from any heat sources.

T9.2 High Pressure Hydraulic Pumps and Lines

- T9.2.1 The driver and anyone standing outside the vehicle must be shielded from any hydraulic pumps and lines with line pressures of 2100 kPa or higher. The shields must be steel or aluminum with a minimum thickness of 1 mm. Brake lines are not considered as high pressure hydraulic lines.

T9.3 Critical Component Mounting

- T9.3.1 Components, see T9.1.8 and EV5.5.8, themselves and their mountings must be able to withstand the following accelerations:
- 40 g in the longitudinal direction (forward/aft)
 - 40 g in the lateral direction (left/right)
 - 20 g in the vertical direction (up/down)

T10 FASTENERS

T10.1 Critical Fasteners

- T10.1.1 Critical fasteners are defined as bolts, nuts, and other fasteners utilized in the primary structure, the (autonomous) steering, (autonomous) braking, driver's harness, suspension systems and those specifically designated as critical fasteners in the respective rule.
- T10.1.2 All threaded critical fasteners must be at least 4 mm metric grade 8.8 ([unmodified](#) OEM parts 3 mm metric grade 8.8), equivalent size or of that specified in the referencing rule, whichever is larger. [The strength classification of critical fasteners must be identifiable.](#)

- T10.1.3 All threaded critical fasteners must be of the type hexagon bolts (ISO 4017, ISO 4014 or an equivalent standard) or socket head cap screws (ISO 4762, DIN 7984, ISO 7379 or an equivalent standard.) including their fine-pitch thread versions.
- T10.1.4 Any bolted joint in the primary structure and mounting of the TSAC to the chassis using either tabs or brackets, must have an edge distance ratio “e/D” of 1.5 or greater. “D” equals the hole diameter and “e” equals the distance from the hole centerline to the nearest free edge of the tab or bracket. Any tabs attaching suspension members to the primary structure are not required to meet this rule.
- T10.1.5 For steering and suspension systems, alternative fasteners are allowed if equivalency to T10.1.2 and T10.1.3 can be shown.

T10.2 Securing Fasteners

- T10.2.1 All critical fasteners must be secured from unintentional loosening by the use of positive locking mechanisms.
- T10.2.2 The following methods are accepted as positive locking mechanisms:
- Correctly installed safety wiring.
 - Cotter pins.
 - Nylon lock nuts (ISO 7040, ISO 10512, EN 1663 or an equivalent standard) for locations where no temperature rating above 80 °C is required.
 - Prevailing torque lock nuts (DIN 980, ISO 7042 or equivalent, and jet nuts or K-nuts).
 - Locking plates.
 - Tab washers.
- Any locking mechanism based on pre-tensioning or an adhesive is not considered a positive locking mechanism.
- T10.2.3 Snap or retaining rings according to DIN 471, DIN 472, or equivalent standard are allowed in **unmodified** OEM applications or for securing bearings or springs or brake disc floaters given that they do not bear any loads under normal driving conditions. The groove must be in pristine condition and manufactured according to the standard of the snap or retaining ring.
- T10.2.4 A minimum of two full threads must project from any lock nut.
- T10.2.5 All spherical rod ends and spherical bearings on the steering or suspension must be in double shear or captured by having a screw/bolt head or washer with an outer diameter that is larger than the spherical bearing housing inner diameter.
- T10.2.6 Adjustable tie-rod ends must be constrained with a jam nut to prevent loosening.
- T10.2.7 **Other positive locking mechanisms than specified can only be accepted for brake disc floaters, given they are unmodified OEM applications and installed according to the manufacturer’s specifications.**

T10.3 Self-made Fasteners

- T10.3.1 Teams that use self-made fasteners will need to present documentation with appropriate reasoning, calculations and testing during the mechanical inspection. A technical inspector

may request for additional documentation should the material presented be deemed questionable.

T10.3.2 Queries over self-made fasteners can be made directly to the Rules Committee prior to the competition.

T11 ELECTRICAL COMPONENTS

T11.1 Low Voltage System (LVS)

T11.1.1 The Low Voltage System (LVS) is defined as

- [CV ONLY] all electrical circuits of the vehicle.
- [EV ONLY] every electrical part that is not part of the TS (see EV1.1.1)

T11.1.2 The maximum permitted voltage that may occur between any two electrical connections in the LVS is 60 VDC or 50 VAC RMS.

T11.1.3 All LVS parts must be adequately insulated.

T11.1.4 [CV ONLY] The following systems are excluded from the LVS voltage limit (see T11.1.2):

- High voltage systems for ignition
- High voltage systems for injectors
- Voltages internal to **unmodified** OEM charging systems designed for <60 VDC output.

T11.1.5 [EV ONLY] The LVS must not use orange wiring or conduit.

T11.1.6 [EV ONLY] The LVS must be grounded to the chassis.

T11.1.7 Batteries to operate a real-time clock as well as battery-operated driver communication and onboard cameras are not part of the LVS system

T11.2 Master Switches

T11.2.1 Master switches (see T11.3 and EV6.2) must be a mechanical switch of the rotary type, with a red, removable handle. The handle must have a width of at least 50 mm and must only be removable in an electrically open position. The handle must have the car number mentioned on it. They must be direct acting, i.e. they must not act through a relay or logic.

T11.2.2 Master switches must be located on the right side of the vehicle, in proximity to the main hoop, at the 95th percentile male driver's shoulder height, as defined in T4.3, and be easily actuated from outside the vehicle.

The center of any master switch must not be mounted lower than the vertical distance of the template's (see T4.3) middle circle center to the ground surface multiplied by 0.8.

T11.2.3 The "ON" position of the switch must be in the horizontal position and must be marked accordingly. The "OFF" position of the master switch must also be clearly marked.

T11.2.4 Master switches must be rigidly mounted to the vehicle and must not be removed during maintenance.

T11.2.5 [EV ONLY] Master switches must be mounted next to each other.

T11.3 Low Voltage Master Switch (LVMS)

T11.3.1 An LVMS according to T11.2 must completely disable

- [EV ONLY] power to the LVS
- [CV ONLY] power from the Low Voltage (LV) battery and the alternator to the LVS

T11.3.2 The LVMS must be mounted in the middle of a completely red circular area of ≥ 50 mm diameter placed on a high contrast background.

T11.3.3 The LVMS must be marked with “LV” and a symbol showing a red spark in a white edged blue triangle.

T11.4 Shutdown Buttons

T11.4.1 A system of three shutdown buttons must be installed on the vehicle.

T11.4.2 Each shutdown button must be a push-pull or push-rotate mechanical emergency switch where pushing the button opens the shutdown circuit (see EV6.1 and CV4.1).

T11.4.3 One button must be located on each side of the vehicle behind the driver’s compartment at approximately the level of the driver’s head. The minimum allowed diameter of the shutdown buttons on both sides of the vehicle is 40 mm. The buttons must be easily reachable from outside the vehicle.

T11.4.4 One shutdown button serves as a cockpit-mounted shutdown button and must

- have a minimum diameter of 24 mm
- be located in easy reach of a belted-in driver
- be alongside of the steering wheel and unobstructed by the steering wheel or any other part of the vehicle

T11.4.5 The international electrical symbol consisting of a red spark on a white-edged blue triangle must be affixed in close proximity to each shutdown button.

T11.4.6 Shutdown buttons must be rigidly mounted to the vehicle and must not be removed during maintenance.

T11.4.7 Shutdown buttons must be red.

T11.5 Inertia Switch

T11.5.1 An inertia switch must be part of the shutdown circuit (see EV6.1 and CV4.1) such that an impact will result in the shutdown circuit being opened. The inertia switch must latch until manually reset.

T11.5.2 The device must trigger due to a longitudinal impact load which decelerates the vehicle at between $\leq 8g$ and $\leq 11g$ depending on the duration of the deceleration.

T11.5.3 The device must not include any semiconductor components.

T11.5.4 The device must be rigidly attached to the vehicle and installed according to manufacturer specification to the vehicle. [It must be removable to test functionality.](#)

T11.5.5 Inertia switch should be directly acting. External relay circuits are prohibited.

T11.6 Brake System Plausibility Device (BSPD)

T11.6.1 A standalone non-programmable circuit, the BSPD, must open the shutdown circuit (see [EV6.1](#) and [CV4.1](#)) when hard braking occurs, whilst

- [EV ONLY] ≥ 5 kW power is delivered to the motors.
- [CV ONLY] the throttle position is more than 25 % over idle position.

The shutdown circuit must remain open until power cycling the LVMS or the BSPD may reset itself if the opening condition is no longer present for more than 10 s.

T11.6.2 The action of opening the shutdown circuit must occur if the implausibility is persistent for more than 500 ms.

T11.6.3 The BSPD must be directly supplied ([see T1.3.1](#)) from the LVMS ([see T11.3](#)).

T11.6.4 Standalone is defined as there is no additional functionality implemented on all required PCBs. The interfaces must be reduced to the minimum necessary signals, i.e. power supply, required sensors and the SDC. Supply and sensor signals must not be routed through any other devices before entering the BSPD. [If other systems use the sensors in parallel, the team must show at Technical Inspection they do not interfere with the BSPD.](#)

T11.6.5 To detect hard braking, a brake system pressure sensor must be used. The threshold must be chosen such that there are no locked wheels and the brake pressure is ≤ 30 bar.

T11.6.6 [EV ONLY] To measure power delivery, a DC circuit current sensor only must be used. The threshold must be chosen to an equivalent of ≤ 5 kW for maximum TS voltage.

T11.6.7 It must be possible to separately disconnect each sensor signal wire for technical inspection.

T11.6.8 All necessary signals are System Critical Signal (SCS) ([see T11.9](#)).

~~T11.6.9~~ [Removed](#)

T11.6.10 [EV ONLY] The BSPD including all required sensors must not be installed inside the accumulator container.

T11.7 LV Batteries

T11.7.1 LV batteries are all batteries connected to the LVS.

T11.7.2 LV batteries must be securely attached to the chassis and located within the rollover protection envelope ([see T1.1.16](#)).

T11.7.3 Any wet-cell battery located in the driver compartment must be enclosed in a non-conductive, water proof (according to IPX7 or higher, IEC 60529) and acid resistant container.

T11.7.4 LV batteries must have a rigid and sturdy casing. For EV ONLY:

- The enclosure must be fire retardant (see [T1.2.1](#)).
- Must comply with [EV4.5.1](#).
- If the enclosure is in the TS loadpath; additionally, the enclosure and its mounting must comply with [EV5.5.8](#).

T11.7.5 Completely closed LV battery cases must have an overpressure relief. Venting gases must be separated from the driver by a firewall.

T11.7.6 Ungrounded terminals must be insulated.

T11.7.7 LV batteries must be protected from short circuits, not more than 100 mm from ungrounded terminals.

T11.7.8 Battery packs based on lithium chemistry other than lithium iron phosphate (LiFePO_4):

- Must have a fire retardant casing (see [T1.2.1](#)).
- Must include overcurrent protection that trips at or below the maximum specified discharge current of the cells.
- Must include over temperature protection of at least 30% of the cells, meeting [EV5.8.3](#), that trips when any cell leaves the allowed temperature range according to the manufacturer's datasheet, but not more than 60°C, for more than 1 s and disconnects the battery.
- Must include voltage protection of all cells that trips when any cell leaves the allowed voltage range according to the manufacturer's datasheet for more than 500 ms and disconnects the battery.
- It must be possible to display all cell voltages and measured temperatures, e.g. by connecting a laptop.
- Must meet [EV5.8.10](#).
- Signals needed to fulfill these requirements are SCS (see [T11.9](#)).

T11.8 Accelerator Pedal Position Sensor (APPS)

T11.8.1 T11.8 only apply for electric vehicles (see chapter EV), or internal combustion vehicles using Electronic Throttle Control (ETC) (see [CV1.6](#)).

T11.8.2 The APPS must be actuated by a foot pedal.

T11.8.3 Pedal travel is defined as the percentage of travel from fully released position to a fully applied position where 0% is fully released and 100% is fully applied.

T11.8.4 The foot pedal must return to the 0% position when not actuated. The foot pedal must have a positive stop preventing the mounted sensors from being damaged or overstressed. Two springs must be used to return the foot pedal to the 0% position and each spring must work when the other is disconnected. Springs in the APPS are not accepted as return springs.

T11.8.5 At least two separate sensors must be used as APPSs. The sensors may share the housing.

T11.8.6 If analog sensors are used, they must have different, non-intersecting transfer functions. A short circuit between the signal lines must always result in an implausibility according to [T11.8.9](#).

T11.8.7 The APPS signals are SCSs (see [T11.9](#)).

- T11.8.8 If an implausibility occurs between the values of the APPSs and persists for more than 100 ms
- [EV ONLY] The power to the motor(s) must be immediately shut down completely. It is not necessary to completely deactivate the tractive system, the motor controller(s) shutting down the power to the motor(s) is sufficient.
 - [CV ONLY] The power to the electronic throttle must be immediately shut down.
- T11.8.9 Implausibility is defined as a deviation of more than ten percentage points pedal travel between any of the used APPSs or any failure according to T11.9.
- T11.8.10 If three sensors are used, then in the case of an APPS implausibility, any two sensors that are plausible may be used to define the torque target and the 3rd APPS may be ignored.
- T11.8.11 It must be possible to separately disconnect each APPS signal wire to check all functionalities.
- T11.8.12 A fully released accelerator pedal must result in:
- [EV ONLY] A wheel torque of ≤ 0 N m
 - [CV ONLY] An idle position or lower throttle set-point. This may only be exceeded during a gearshift for a maximum of 500ms.

T11.9 System Critical Signals (SCSs)

- T11.9.1 SCS are defined as all electrical signals which
- Influence actions on the shutdown circuit (see CV4.1 and EV6.1).
 - Influence the wheel torque.
 - [EV ONLY] Influence indicators according to EV5.8.9, EV4.10 or EV6.3.7.
 - Influence indicators according to T14.10.4.
- T11.9.2 Any of the following SCS single failures must result in a safe state of all connected systems:
- (a) Failures of signals transmitted by cable:
 - Open circuit
 - Short circuit to ground
 - (b) Failures of analog sensor signals transmitted by cable:
 - Short circuit to supply voltage
 - (c) Failures of sensor signals used in programmable devices:
 - Implausibility due to out of range signals, e.g. mechanically impossible angle of an angle sensor.
 - (d) Failures of digitally transmitted signals by cable or wireless:
 - Data corruption (e.g. checked by a checksum)
 - Loss and delay of messages (e.g. checked by transmission time outs)

Signals might be a member of multiple signal classes, e.g. analog signals transmitted by cable might be a member of T11.9.2.a, T11.9.2.b and T11.9.2.c.

- If a signal failure is correctable, e.g. due to redundancy or worst case values, the safe state must be entered as soon as an additional non correctable failure occurs.
- T11.9.3** If a signal failure is correctable, e.g. due to redundancy or worst case values, the safe state must be entered as soon as an additional non correctable failure occurs.
- T11.9.4** The maximum allowed delay of messages according to T11.9.2.d must be chosen depending on the impact of delayed messages to the connected system, but must not exceed 500ms.
- T11.9.5** Safe state is defined depending on the signals as follows:
- signals only influencing indicators – Indicating a failure of its own function or of the connected system
 - low voltage battery signals – At least one pole is electrically disconnected from the rest of the vehicle
 - [EV ONLY] For all others signals – opened shutdown circuit and opened AIRs
 - [CV ONLY] For all others signals – opened shutdown circuit and stopped engine
- T11.9.6** Indicators according to T11.9.1 with safe state “illuminated” (e.g. absence of failures is not actively indicated) must be illuminated for 1s to 3s for visible check after power cycling the LVMS.
- T11.10 System status light**
- T11.10.1** Any system status light(s), see T6.3 and T14.9, must meet the following requirements:
- Black background.
 - Rectangular, triangular or near round shape.
 - Minimum illuminated surface of 15cm² with even luminous intensity.
 - Clearly visible in very bright sunlight.
 - If LED lights are used without a diffuser, they must not be more than 20mm apart.
 - If a single line of LED lights is used, the minimum length is 150mm.
- T11.11 Sensors, Cameras & Electrical Components Mounting**
- T11.11.1** All sensors, cameras and components must be securely mounted. For all mounts, T8.3.1 applies.
- T11.11.2** Sensors, cameras and components may not come into contact with the driver’s helmet under any circumstances.
- T11.11.3** All sensors, cameras and components (including housing, mounting and covers) must be positioned within the surface envelope, see T1.1.18, or within the box defined in T8.2.
- T11.11.4** Passive antennas which do not incorporate active electronic parts, e.g. amplifiers with the longest side <100 mm may additionally to T11.10.3 protrude from the surface envelope see T1.1.17, but not their mounting.
- T11.12 Fans and Turbines**
- T11.12.1** The maximum combined total rated power of any active devices designed to move air is 500 W, this includes cooling fans but does not apply to CV1.8.

T11.13 Legal & Work Safety

- T11.13.1 All sensors must fulfill the local legislative specifications (i.e. eye-protection classification for laser sensors, power limitation for radar sensors, etc.) in the country of competition.

T12 VEHICLE IDENTIFICATION

T12.1 Vehicle Number

- T12.1.1 Each vehicle will be assigned a number at the time of its entry into a competition. The number will have an alphabet located at the prefix: C for CV and E for EV.
- T12.1.2 Vehicle numbers must appear on the vehicle at the front and both sides as follows:
- Height: At least 150 mm high;
 - Font: Roman Sans-Serif characters. Italic, outline, serif, or shadow numbers are prohibited.
 - Stroke width and spacing between numbers: At least 20 mm.
 - Color: Either white numbers on a black background or black numbers on a white background.
 - Background shape: The number background must be one of the following: round, oval, square or rectangular. There must be at least 25 mm between the edge of the numbers and the edge of the background.
 - Clear: The numbers must not be obscured by parts of the vehicle.
 - Alignment: In front view the number at the front and in side view the numbers on the sides must be horizontally aligned.
 - Entire vehicle number, along with the prefix must appear on the vehicle.

T12.2 University Name

- T12.2.1 The university name (or college name, where applicable per [A4.1.4](#)) must be written fully. Only the following abbreviations in the university name are accepted if the city name is written fully:
- University → Uni
 - Indian Institute of Technology → IIT
 - National Institute of Technology → NIT
 - If the university officially uses an abbreviation in their proper name, this abbreviation is accepted.
- T12.2.2 The university name must be displayed and written in Roman Sans-Serif characters that are at least 50mm tall, in front of the mainhoop on both sides of the vehicle. The top of each character must not be more than 50mm below the top of the bodywork.
- T12.2.3 The characters must be clearly visible at a distance and placed on a high contrast background.

T13 VEHICLE AND DRIVER EQUIPMENT

T13.1 Pushbar

- T13.1.1 Each team must have a removable device (called the pushbar) that attaches to the rear of the vehicle and allows two people to push and pull the vehicle while standing erect behind the vehicle.
- T13.1.2 The pushbar must have a red color.
- T13.1.3 The university name (or college name, where applicable per [A4.1.4](#)) must be written on the push bar. The characters must be clearly visible and placed on a high contrast background.
- T13.1.4 The pushbar must be capable of slowing and stopping the forward motion of the vehicle and pulling it rearwards.
- T13.1.5 An approved fire extinguisher ([see T13.4.1](#)) must be mounted to the pushbar such that it is quickly accessible.
- T13.1.6 [EV ONLY] Two pairs of high-voltage insulating gloves and a multimeter with two 4 mm banana plug test leads rated for 600 V CAT III or better must be attached to the pushbar. The HV gloves must be protected by a case or similar means from mechanical damage, humidity and sunlight. It must be possible to open the case without using tools.

T13.2 Jacks

- T13.2.1 Each team must have one or two removable devices (jacks) that hold the vehicle, so that all driven wheels are at least 100 mm off the ground.
- T13.2.2 Lifting the vehicle and positioning of the device(s) must be done by the team in a safe way.
- T13.2.3 In the lifted position:
- The vehicle must be adequately supported and stable
 - It must be safe for a driver to enter and exit the vehicle, without additional devices.
 - The device(s) must not extend out of the vehicle's projected surface area.
- T13.2.4 On both sides of the vehicle, the devices pickup points must be indicated by orange triangles.
- T13.2.5 The university name (or college name, where applicable per [A4.1.4](#)) must be written on the jack. The characters must be clearly visible and placed on a high contrast background.

T13.3 Driver Equipment

- T13.3.1 The equipment specified below must be worn by the driver anytime while in the cockpit with the engine running or with the tractive system active for electric vehicles and anytime between starting a dynamic event and either finishing or abandoning a dynamic event. Removal of any driver equipment during the event will result in disqualification.
- T13.3.2 A well-fitting, [unmodified](#), closed face helmet that meets one of the following certifications and is labeled as such:

- Snell SA2020, EA2016, SA2025 or newer
- SFI 31.1/2015, 31.1/2020 or newer
- FIA 8860-2010, FIA 8860-2018, FIA 8859-2015, 8859-2024 or newer

Open faced helmets and off-road helmets (helmets without integrated eye shields) are not approved. Additionally, all helmets must have a double D-ring chin strap fastening system from the manufacturer. Helmets that have been modified or customized to add D-rings are not permitted. All helmets to be used in the competition must be presented during technical inspection where approved helmets will be stickered.

T13.3.3 If Frontal Head Restraint systems (FHR, HANS) are used, they must be certified to one of the following standards and be labeled as such:

- FIA 8858-2010
- FIA 8860-2004
- SFI 38.1

T13.3.4 A balaclava which covers the driver's head, hair and neck, made from acceptable fire resistant material as defined in T13.3.13, or a full helmet skirt of acceptable fire resistant material.

T13.3.5 A fire resistant one piece suit, made from a minimum of two layers that covers the body from the neck down to the ankles and the wrists. The suit must be certified to one of the following standards and be labeled as such:

- SFI 3-2A/5 (or higher)
- SFI 3.4/5 (or higher)
- FIA Standard 8856-2000
- FIA Standard 8856-2018

T13.3.6 Fire resistant underwear (long pants and long sleeve t-shirt). This fire resistant underwear must be made from acceptable fire resistant material as listed in T13.3.13 and must cover the driver's body completely from neck down to ankles and wrists.

T13.3.7 Fire resistant socks made from acceptable fire resistant material as defined in T13.3.13, that cover the bare skin between the driver's suit and the boots or shoes.

T13.3.8 Fire resistant shoes made from acceptable fire resistant material as defined in T13.3.13. The shoes must be certified to the standard and labeled as such:

- SFI Spec 3.3
- FIA Standard 8856-2000
- FIA Standard 8856-2018

T13.3.9 Fire resistant gloves made from acceptable fire resistant material as defined in T13.3.13.

Gloves of all leather construction or fire resistant gloves constructed using leather palms with no insulating fire resisting material underneath are not acceptable.

T13.3.10 Fire resistant clothing must not be older than 10 years, recognisable since no FIA hologram label present.

- T13.3.11 Arm restraints are required and must be worn such that the driver can release them and exit the vehicle unassisted regardless of the vehicle's position. Arm restraints must be commercially manufactured according to SFI Standard 3.3 or equivalent.
- T13.3.12 All driver equipment covered in T13.3 must be in good condition. Specifically, it must not have any tears, rips, open seams, areas of significant wear or abrasion or stains which might compromise fire resistant performance. Teams must be able to show that the clothing still meets the requirements of T13.3 if driver's clothing is embroidered. This requires, but is not limited to, that only the outer shell is perforated. The officials reserve the right to impound all non-approved driver equipment until the end of the competition.
- T13.3.13 Driver communication equipment connected to the vehicle must automatically disconnect during driver egress. The connector must be placed behind the drivers shoulder and the cable length reduced to the minimum.
- T13.3.14 For the purpose of this section some, but not all, of the approved fire resistant materials are: Nomex, Aramid, Belcotex and Indura.
- T13.3.15 T-shirts, socks or other undergarments made from nylon or any other synthetic material which will melt when exposed to high heat are prohibited.

T13.4 Fire Extinguishers

- T13.4.1 Each team must have at least two dry chemical/dry powder fire extinguishers with a minimum firefighting agent capacity of 0.9 kg.
- T13.4.2 The following are the minimum accepted ratings:
- USA, Canada and Brazil: 10BC or 1A 10BC
 - Europe: 34B or 5A 34B
 - Australia: 20BE or 1A 10BE
 - India: Dry chemical or dry powder ABC fire extinguishers are acceptable.
- Extinguishers of larger capacity (higher numerical ratings) are acceptable.
- T13.4.3 Aqueous Film Forming Foam (AFFF) fire extinguishers are prohibited. Halon extinguishers and systems are prohibited.
- T13.4.4 All extinguishers must be equipped with a manufacturer installed pressure/charge gauge.
- T13.4.5 Except for the initial inspection, one extinguisher must readily be available in the team's paddock area, and the second must accompany the vehicle wherever the vehicle is moved. Both extinguishers must be presented with the vehicle at technical inspection.
- T13.4.6 Hand held fire extinguishers are not permitted to be mounted on or in the vehicle.

T13.5 Camera Mounts

- T13.5.1 The mounts for video/photographic cameras must be of a safe and secure design:
- All camera installations must be approved at technical inspection.
 - Helmet mounted cameras are prohibited.

- The body of any camera or recording unit must be secured at a minimum of two points on different sides of the camera body. If a tether is used to restrain the camera, the tether length must be limited so that the camera cannot contact the driver.

T14 AUTONOMOUS SYSTEM (AS)

T14.1 Definitions

T14.1.1 Each vehicle must implement a full AS according to T14, to run in autonomous mode.

T14.1.2 Section Removed

T14.2 Data Logger

T14.2.1 The officials will provide a standardized data logger that must be installed during the event. Further specifications for the data logger and required hardware and software interfaces will be **shared on a separate document that will be published on the competition website.**

T14.2.2 The team needs to provide two sets of signals to the data logger:

- Basic set of signals as defined **on a separate document that will be published on the competition website.**
- Vehicle-individual set of signals that is monitored by the ASB to ensure redundancy and fault detection

T14.2.3 Section Removed

T14.3 Remote Emergency System (RES)

T14.3.1 Every vehicle must be equipped with a standard RES specified in **a separate document that will be published on the competition website.** The system consists of two parts, the remote control and the vehicle module.

T14.3.2 The RES must be purchased by the team.

T14.3.3 The RES has two functions:

- When the remote emergency stop button is pressed, it must open the SDC defined in T14.4.
- When the “Go” button is pressed, the preselected autonomous mission is started.

T14.3.4 The RES vehicle module must be directly, see T1.3.1, integrated in the vehicle’s SDC with one of its relays hard-wired in series to the shutdown buttons.

T14.3.5 The RES relay, which is integrated into the SDC, must be bypassed by a normally closed relay, when driving manually. The relay must:

- Be directly supplied, see T1.3.1, by the ASMS, see T14.5
- Have a safety certified forcibly guided or a mirrored normally open contact which is directly connected, see T1.3.1, in series to the ASMS.

T14.3.6 The antenna of the RES must be mounted unobstructed and without interfering parts in proximity, e.g. other antennas, etc.

T14.4 Shutdown Circuit

T14.4.1 The SDC may only be closed by the AS, if the following conditions are fulfilled:

- Manual Driving: Manual Mission is selected, the AS has checked that ASB is deactivated, i.e. no autonomous brake actuation possible.
- Autonomous Driving: Autonomous Mission is selected, ASMS is switched on and sufficient brake pressure is built up, i.e. brakes are closed.

T14.5 Autonomous System Master Switch (ASMS)

T14.5.1 Each vehicle must be equipped with an ASMS according to [T11.2](#).

T14.5.2 The ASMS must be mounted in the middle of a completely blue circular area of ≥ 50 mm diameter placed on a high contrast background.

T14.5.3 The ASMS must be marked with “AS”.

T14.5.4 The power supply of all steering and brake actuators must be switched by the LVMS and the ASMS. Other than stated in [T11.2.1](#), a relay may be used as part of the ASMS.

T14.5.5 When the ASMS is in “Off” position, the following must be fulfilled:

- No steering, braking and propulsion actuation can be performed by request of the autonomous system.
- The sensors and the processing units can stay operational.
- The vehicle must be able to be pushed as specified in [A6.7](#).
- It must be possible to operate the vehicle manually as a normal CV or EV.

T14.5.6 While the ASMS is switched to the “On” position

- no person may be (seated) inside the vehicle.
- no other manual steps other than activating the TS may be performed at the vehicle.

T14.5.7 The ASMS must be fitted with a “lockout/tagout” capability to prevent accidental activation of the AS. The ASR must ensure that the ASMS is locked in the off position whenever the vehicle is outside the dynamic area or driven in manual mode.

T14.6 Steering Actuation

T14.6.1 Steering system actuation, i.e. movement, must only happen if the vehicle is in R2D mode or during “AS Emergency”, see [T14.8.3](#).

T14.6.2 Manual steering must be possible without manual release steps, e.g. operating manual valves and/or (dis-)connecting mechanical elements, while the ASMS is switched “Off”.

T14.7 Actuator Decoupling and Sensor Repositioning

- T14.7.1 Actuators may be disconnected if the decoupling mechanism cannot block manual operation in any position.
- T14.7.2 Sensors according to [T11.11](#) may be repositioned to multiple fixed positions if full rules compliance is ensured in all fixed positions.
- T14.7.3 The mechanisms used for actuator decoupling, see [T14.7.1](#), or sensor repositioning, see [T14.7.2](#), must ensure that
- no parts except covers are removed from the vehicle.
 - covers are only removed from the vehicle temporarily during the process.
 - all parts remain in physical contact to the vehicle without using a tether line or similar, i.e. fasteners may be loosened but must not be fully undone.
 - they are securely locked in all (fixed) positions.

T14.8 Autonomous System Status Definitions

- T14.8.1 The EBS is considered to be “activated”, if the power supply path defined in [T15.2.2](#) is cut after passing the initial checkup sequence, see [T15.3.1](#). Brakes may only be released by operating the deactivation point(s) defined in [T15.1.7](#).
- T14.8.2 The EBS is no longer considered to be activated, if all brakes have been released using the deactivation point(s) defined in [T15.1.7](#).
- T14.8.3 The status of the AS must be determined according to the flowchart in Figure 15.

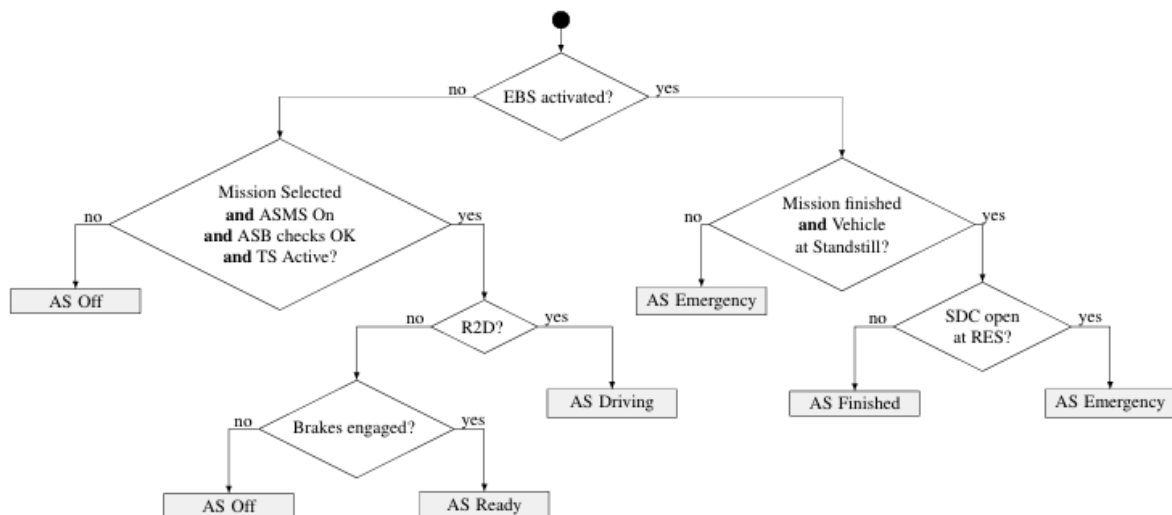


Figure 15: AS Status Flowchart

T14.8.4 In autonomous mode, R2D mode may only be entered by the “Go” signal from the RES, after the system has remained in “AS Ready” for at least 5s.

T14.8.5 In autonomous mode, the vehicle must not start moving, until the system has remained in “AS Driving” for at least 3s.

T14.9 Autonomous System Status Indicators (ASSIs)

T14.9.1 The vehicle must include three ASSIs that must indicate the status of the AS, as defined in T14.8, correlating to illumination as shown:

AS Off	AS Ready	AS Driving	AS Emergency	AS Finished
off	Yellow continuous	Yellow flashing	Blue flashing	Blue continuous

During “AS Driving” and “AS Emergency” the ASSIs must be flashing continuously with a frequency between 2Hz and 5Hz and a duty cycle of 50%. The ASSIs must not perform any other functions.

T14.9.2 One ASSI must be located on each side of the vehicle behind the driver’s compartment, in a region 160mm below the top of the main hoop and 600mm above the ground. The third ASSI must be located at the rear of the vehicle, on the vehicle centerline, in a region 160mm below the top of the main hoop and 100mm above the brake light.

T14.9.3 The entire surface of at least one ASSI must be visible from any angle of the vehicle from a point 1.60m vertically from ground level, within 3m horizontal radius from the top of the main hoop.

T14.9.4 Each ASSI must meet the requirements according to [T11.10](#).

T14.9.5 The status “AS Emergency” has to be indicated by an intermittent sound with the following parameters:

- on-/off-frequency: 1Hz to 5Hz
- duty cycle 50%
- sound level between 80dBA and 90dBA, fast weighting in a radius of 2m around the vehicle.
- duration between 8s and 10s after entering “AS Emergency”

T14.10 Autonomous Missions

T14.10.1 The AS must at least implement the following missions:

- [DC ONLY] Acceleration
- [DC ONLY] Skidpad
- [DC ONLY] Autocross
- [DC ONLY] Trackdrive
- EBS Test
- Inspection
- Manual Driving

- T14.10.2 The Inspection Mission is defined by slowly spinning the drivetrain and actuating the steering system with a sine wave while the vehicle is jacked up and all wheels are removed. After 25s to 30s the AS must transition to “AS Finished”.
- T14.10.3 It must be possible to select any mission without the use of an external device.
- T14.10.4 The selected mission must be indicated by the Autonomous Mission Indicator (AMI).
- T14.10.5 The AMI must be easily readable by any person standing next to the ASMS and can either be part of the dashboard or located next to the ASMS.

T14.11 Autonomous System Form (ASF)

- T14.11.1 Before the event, all teams must submit a clearly structured documentation of their entire AS, including ASB, called ASF. Template for the same will be shared on the competition website.

T15 AUTONOMOUS SYSTEM BRAKE (ASB)

T15.1 Technical Requirements

- T15.1.1 To run in autonomous mode, the vehicle must be equipped with an ASB that features an EBS, see [T15.2](#), as part of it.
- T15.1.2 All parts of the ASB and their mountings must be located within the rollover protection envelope, see [T1.1.16](#), with the following exceptions:
- brake system according to [T6](#)
 - deactivation point(s) according to [T15.1.8](#)
- T15.1.3 The TS is not considered to be a brake system.
- T15.1.4 Manual braking must always be possible. In case of manual and autonomous braking simultaneously, always the highest of both pressures must be applied to the brakes.
- T15.1.5 Master brake cylinders must not be connected in series.
- T15.1.6 The ASB may be part of the hydraulic brake system. For all components inside the vehicles brake circuit [T6](#) is applied. On all remaining pneumatic and hydraulic components [T9](#) applies.
- T15.1.7 The ASB must be designed so that it can be easily deactivated by two deactivation points.
- T15.1.8 Any deactivation point of the ASB must:
- work without the aid of electrical power
 - be in proximity to each other
 - be mounted within the surface envelope, see [T1.1.18](#)
 - either be mounted in proximity to the ASMS or on the top side of the vehicle between front bulkhead and front hoop close to the vehicles center line
 - be protected against unintended actuation, e.g. by being hit by a cone, while driving

- be operable by maximum two simple push/pull and/or turning actions, the order and direction of these actions must be shown next to the deactivation points.
- be marked with “Brake release”
- have a red handle

T15.2 Emergency Brake System (EBS)

T15.2.1 The EBS must only use passive systems with mechanical energy storage. Electrical power loss at EBS must lead to a direct emergency brake maneuver with the performance specified in T15.4.

T15.2.2 The EBS must be supplied by

- LVMS
- ASMS
- the normally open contact of the relay according to T14.3.5
- a relay which is supplied by the SDC, i.e. parallel to the AIRs, but must not be delayed.

T15.3 Functional Safety

T15.3.1 An initial check has to be performed to ensure that the ASB is able to

- operate as specified in T15.3.2 and T15.3.4
- build up brake pressure as expected

T15.3.2 After the initial check the ASB and its SCS must be continuously monitored for failures.

T15.3.3 The functionality required to achieve T15.3.1 and T15.3.2 must be implemented on a single PCB. Sensor signals may be routed and preprocessed through additional PCBs.

T15.3.4 The vehicle must automatically transition to the safe state, if:

- the functionality according to T15.2.1 cannot be ensured.
- an (additional) single point of failure would lead to total loss of brake capability.

T15.3.5 The safe state is the vehicle at a standstill, brakes engaged to prevent the vehicle from rolling, and an open SDC.

T15.3.6 To get to the safe state, the vehicle must perform an autonomous brake maneuver described in section T15.4 and IN11.2.

T15.4 Emergency Brake System Performance

T15.4.1 The system reaction time, i.e. the time between opening of the SDC and the start of the deceleration, must not exceed 200ms.

T15.4.2 The average deceleration must be greater than 10m/s² under dry track conditions.

T15.4.3 In case of a single failure the ASB should be designed to achieve at least half of the performance specified in T15.4.2.

T15.4.4 Whilst decelerating, the vehicle must remain in a stable driving condition.

CV INTERNAL COMBUSTION ENGINE VEHICLES

CV1 INTERNAL COMBUSTION ENGINE POWERTRAINS

CV1.1 Engine Limitation

CV1.1.1 The engine(s) used to power the vehicle must be a piston engine(s) using a four-stroke primary heat cycle with a displacement not exceeding 710 cm³ per cycle. Hybrid powertrains, such as those using electric motors running off stored energy, are prohibited.

CV1.2 Starter

CV1.2.1 Each vehicle must be equipped with an on-board starter, which must be used to start the vehicle.

CV1.3 Air Intake System

CV1.3.1 All parts of the engine air and fuel control systems (including the throttle and the complete air intake system, including the air filter and any air boxes) must lie within the surface envelope, see [T1.1.18](#).

CV1.3.2 Any portion of the air intake system that is less than 350 mm above the ground must be protected from impacts, see [T3.14](#). Impact protection must follow [T3.15](#) when having bolted attachments.

CV1.3.3 The intake manifold must be securely connected to the engine block or cylinder head with brackets and mechanical fasteners that are considered critical fasteners and must comply with T10. This connection must ensure that the intake manifold including the injectors is safely attached. Rubber bushings or hoses are not considered as securely attached.

CV1.3.4 Intake systems with significant mass or cantilever from the cylinder head must be supported to prevent stress to the intake system resulting from engine movement and chassis torsion.

CV1.4 Throttle

CV1.4.1 The vehicle must be equipped with a throttle body. The throttle body may be of any size or design.

CV1.4.2 The throttle must be actuated mechanically by a foot pedal, i.e. via a cable or a rod system (see [CV1.5](#)), or by an ETC system (see [CV1.6](#)).

CV1.4.3 Throttle position is defined as the percentage of travel from fully closed to fully open where 0% is fully closed and 100% is fully open. The idle position is the average position of the throttle body while the engine is idling.

CV1.4.4 The throttle system mechanism must be protected from debris ingress to prevent jamming.

CV1.5 Mechanical Throttle Actuation

- CV1.5.1 CV1.5 only applies if no ETC system is used.
- CV1.5.2 The throttle actuation system must use at least two return springs located at the throttle body, so that the failure of any one of the two springs will not prevent the throttle returning to the idle position.
- CV1.5.3 Each return spring must be capable of returning the throttle to the idle position with the other disconnected.
- CV1.5.4 Springs in the Throttle Position Sensor (TPS) are not acceptable as return springs.
- CV1.5.5 Throttle cables must be located at least 50 mm from any exhaust system component and out of the exhaust stream.
- CV1.5.6 Throttle cables or rods must have smooth operation and must not have the possibility of binding or sticking. They must be protected from being bent or kinked by the driver's foot during operation or when entering the vehicle.
- CV1.5.7 A positive pedal stop must be incorporated on the accelerator pedal to prevent over-stressing the throttle cable or actuation system.

CV1.6 Electronic Throttle Control (ETC)

- CV1.6.1 CV1.6 only applies if ETC is used.
- CV1.6.2 The team must be able to demonstrate the functionality of all safety features and error detections of the ETC system at technical inspection, see IN.
- CV1.6.3 The ETC system must be equipped with at least the following sensors:
- Accelerator Pedal Position Sensor (APPSs) as defined in T11.8
 - Two Throttle Position Sensor (TPSs) to measure the throttle position.
- CV1.6.4 All ETC signals are System Critical Signals (SCSs) (see T11.9).
- CV1.6.5 When power is removed, the electronic throttle must immediately close at least to idle position $\pm 5\%$. An interval of one second is allowed for the throttle to close to idle, failure to achieve this within the required interval must result in immediate disabling of power to ignition, fuel injectors and fuel pump. This action must remain active until the TPS signals indicate the throttle has returned to idle position $\pm 5\%$ for at least one second.
- CV1.6.6 If plausibility does not occur between the values of at least two TPSs and this persists for more than 100 ms, the power to the electronic throttle must be immediately shut down. Plausibility is defined as a deviation of less than ten percentage points between the sensor values as defined in CV1.4.3 and no detected failures as defined in T11.9.
- CV1.6.7 The electronic throttle must use at least two sources of energy capable of returning the throttle to the closed position. One of the sources may be the device that normally actuates the throttle, e.g. a DC motor, but the other device(s) must be a return spring that can return the throttle to the idle position in the event of a loss of actuator power.
- CV1.6.8 Springs in the TPSs are not acceptable as return springs.

- CV1.6.9 The power to the electronic throttle must be immediately shut down, as defined in [CV1.6.6](#), if the throttle position differs by more than 10 % from the expected target TPS position for more than 500 ms.

CV1.7 Intake System Restrictor

- CV1.7.1 In order to limit the power capability from the engine, a single circular restrictor must be placed in the intake system and all engine airflow must pass through the restrictor. The only allowed sequences of components are the following:
- a. For naturally aspirated engines, the sequence must be: throttle body, restrictor, and engine. (see figure 15)
 - b. For turbocharged or supercharged engines, the sequence must be: restrictor, compressor, throttle body, engine. (see figure 16)
- CV1.7.2 The maximum restrictor diameters which must be respected at all times during the competition are:
- a. Gasoline fueled vehicles - 20mm
- CV1.7.3 The restrictor must be located to facilitate measurement during the inspection process.
- CV1.7.4 The circular restricting cross section may not be movable or flexible in any way, e.g. the restrictor must not be part of the movable portion of a barrel throttle body.

CV1.8 Turbochargers and Superchargers

- CV1.8.1 The intake air may be cooled with an intercooler. Only ambient air may be used to remove heat from the intercooler system. Air-to-air and water-to air intercoolers are permitted. The coolant of a water-to-air intercooler system must be plain water without any additives.
- CV1.8.2 If pop-off valves, recirculation valves, or heat exchangers (intercoolers) are used, they may only be positioned in the intake system as shown in figure 16.
- CV1.8.3 Plenums anywhere upstream of the throttle body are prohibited. A “plenum” is any tank or volume that is a significant enlargement of the normal intake runner system.
- CV1.8.4 The maximum allowable internal diameter of the intake runner system between the restrictor and throttle body is 60 mm diameter, or the equivalent area of 2827 mm² if non-circular.

CV1.9 Crankcase / Engine Lubrication Venting

- CV1.9.1 Any crankcase or engine lubrication vent lines routed to the intake system must be connected upstream of the intake system restrictor.
- CV1.9.2 Crankcase breathers that pass through the oil catch tank(s) to exhaust systems, or vacuum devices that connect directly to the exhaust system, are prohibited.

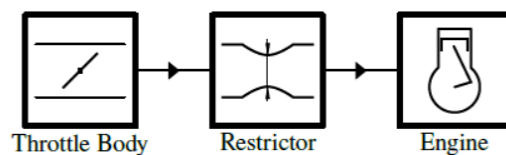


Figure 15: Intake configuration for naturally aspirated engines.

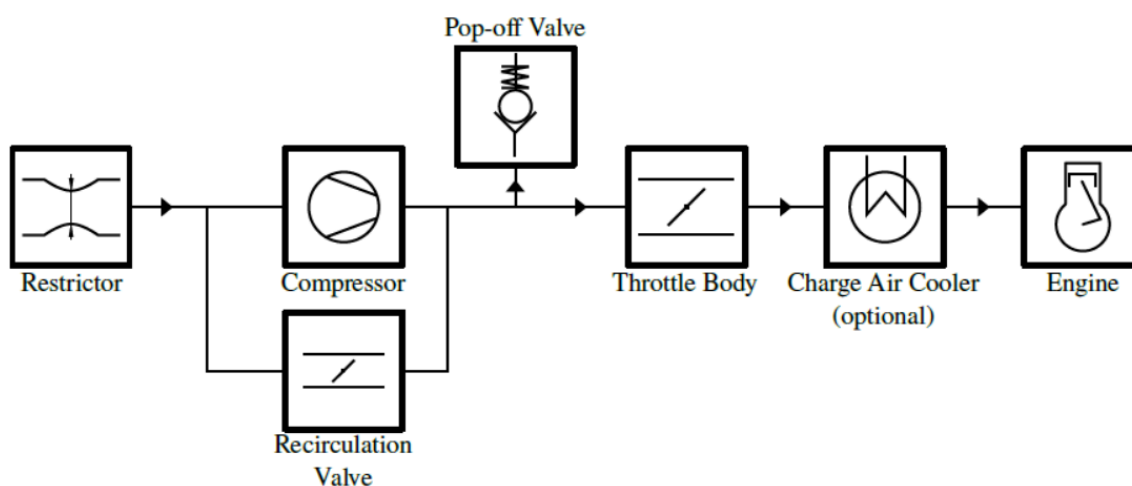


Figure 16: Intake configuration for turbocharged or supercharged engines.

CV2 FUEL AND FUEL SYSTEM

CV2.1 Fuel

- CV2.1.1 The available fuel types will be unleaded gasoline 91 Octane.
- CV2.1.2 The vehicles must be operated with the fuels provided at the competition.
- CV2.1.3 No agents other than fuel and air may be induced into the combustion chamber.
- CV2.1.4 The temperature of fuel introduced into the fuel system may not be changed with the intent to improve calculated efficiency.

CV2.2 Fuel System Location Requirements

- CV2.2.1 The fuel tank must be located within the rollover protection envelope (see [T1.1.16](#)), except the fuel filler neck if it is 350 mm above the ground.
- CV2.2.2 All parts of the fuel storage and supply system must lie within the surface envelope (see [T1.1.18](#))
- CV2.2.3 All parts of the fuel storage and supply system must be adequately protected against any heat sources and located at least 50 mm from any exhaust system component.

CV2.2.4 All parts of the fuel system which can come in contact with the fuel must be rated for permanent contact with fuel.

CV2.3 Fuel Tanks

CV2.3.1 The fuel tank is defined as the part of the fuel containment device that is in contact with the fuel. It may be made of a rigid material or a flexible material.

CV2.3.2 The fuel tank must be securely attached to the vehicle structure with mountings that allow some flexibility such that chassis flex cannot unintentionally load the fuel tank.

CV2.3.3 The fuel tank must not touch any part of the vehicle other than its mounting and parts of the fuel system at any time.

CV2.3.4 Any fuel tank that is made from a flexible material, for example a bladder fuel cell or a bag tank, must be enclosed within a rigid fuel tank container which is securely attached to the vehicle structure. Fuel tank containers (containing a bladder fuel cell or bag tank) may be load carrying.

CV2.3.5 The fuel system must have a provision for emptying the fuel tank. A drain hole or drain valve is required on fuel tanks such that the fuel can directly be drained to the ground or a container or pan kept below the vehicle. The drain hole needs to be easily accessible and should not require disassembly of the tank. If there is a diffuser or floor that runs between the fuel tank and the ground, a hose can be connected to the drain hole such that the drained fuel can be routed around the diffuser/floor and eventually to the ground level.

CV2.3.6 The fuel tank, by design, must not have a variable capacity.

CV2.4 Fuel Lines for Low Pressure Systems

CV2.4.1 Fuel lines between fuel tank and fuel rail and return lines must:

- have reinforced rubber fuel lines with an abrasion protection with a fuel hose clamp which has a full 360° wrap, a nut and bolt system for tightening and rolled edges to prevent the clamp cutting into the hose, or
- have metal braided hoses with crimped-on or reusable, threaded fittings.
- be rated for temperatures of at least 120°C.

CV2.4.2 For reinforced rubber fuel lines with hose & clamp style connections beaded or barbed hose fittings must be used.

CV2.4.3 Fuel lines must be securely attached to the vehicle and/or engine.

CV2.4.4 All fuel lines must be shielded from possible rotating equipment failure or collision damage.

CV 2.4.5 All fuel lines must be fitted in such a way that any leakage cannot result in the accumulation of fuel in the cockpit.

CV2.5 Fuel Injection System Requirements

Low Pressure Injection (LPI) fuel systems are those functioning at a pressure below 10 bar and High Pressure Injection (HPI) fuel systems are those functioning at a T10 bar pressure

or above. Direct Injection (DI) fuel systems are those where the injection occurs directly into the combustion chamber.

CV2.5.1 The following requirements apply to LPI fuel systems:

- The fuel lines must comply with CV2.4.
- The fuel rail must be securely attached to the engine cylinder block, cylinder head, or intake manifold with mechanical fasteners. The threaded fasteners used to secure the fuel rail are considered critical fasteners and must comply with T10.
- The use of fuel rails made from plastic, carbon fiber or rapid prototyping flammable materials is prohibited. However, the use of unmodified Original Equipment Manufacturer (OEM) Fuel Rails manufactured from these materials is acceptable.

CV2.5.2 The following requirements apply to HPI and DI fuel systems:

- All high pressure fuel lines must be stainless steel rigid line or Aeroquip FC807 smooth bore PTFE hose with stainless steel reinforcement and visible Nomex tracer yarn. Use of elastomeric seals is prohibited. Lines must be rigidly connected every 100 mm by mechanical fasteners to structural engine components.
- The fuel rail must be securely attached to the engine cylinder head with mechanical fasteners. The fastening method must be sufficient to hold the fuel rail in place with the maximum regulated pressure acting on the injector internals and neglecting any assistance from in-cylinder pressure acting on the injector tip. The threaded fasteners used to secure the fuel rail are considered critical fasteners and must comply with T10.
- The fuel pump must be rigidly mounted to structural engine components.
- A fuel pressure regulator must be fitted between the high and low pressure sides of the fuel system in parallel with the DI boost pump. The external regulator must be used even if the DI boost pump comes equipped with an internal regulator.
- Prior to the tilt test specified in IN6, engines fitted with mechanically actuated fuel pumps must be run to fill and pressure the system downstream of the high pressure pump.

CV2.6 Fuel Tank Filler Neck and Sight Tube

CV2.6.1 The fuel tank must have a filler neck which:

- has at least an inner diameter of 35 mm at any point between the fuel tank and the top of the fuel filler cap.
- is accompanied by a clear fuel resistant sight tube above the top of the fuel tank with a length of at least 125 mm vertical height for reading the fuel level, see figure 17.
- is made of material that is permanently rated for temperatures of at least 120 °C.

CV2.6.2 A clear filler neck tube may be used as a sight tube.

CV2.6.3 Above the lowest point of the sight tube, the filler neck must not be angled more than 30° from the vertical.

CV2.6.4 A permanent, non-moveable, clear and easily visible fuel level line must be located between 12 mm and 25 mm below the top of the visible portion of the sight tube. This line will be used as the fill line for the tilt test (IN6.1), and before and after the endurance test to measure the amount of fuel used during the endurance event.

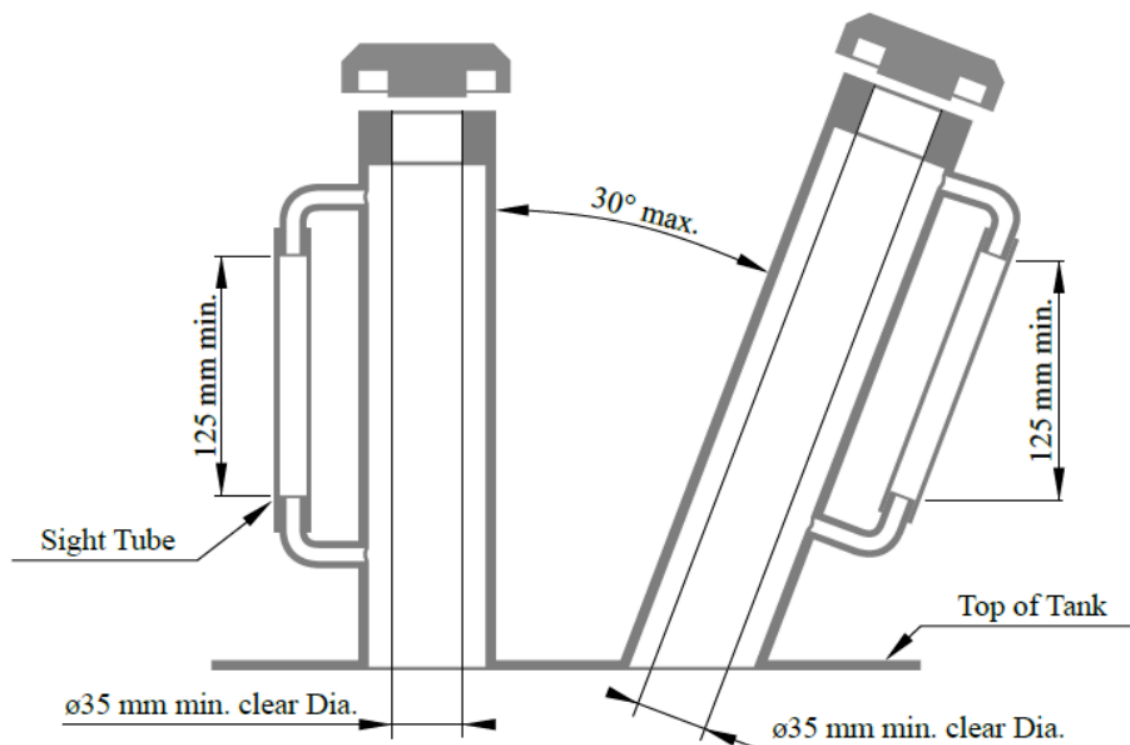


Figure 17: Minimum requirements fuel tank filler neck.

CV2.6.5 The filler neck opening must be directly accessible without removing any parts of the vehicle except for the fuel filler cap.

CV2.6.6 The filler neck must have a fuel filler cap that can withstand severe vibrations or high pressures such as could occur during a vehicle rollover event.

CV2.7 Tank Filling Requirement

CV2.7.1 The fuel tank must be capable of being filled to capacity without manipulating the tank or the vehicle in any manner. The fuel system must be designed in a way that during refueling of the vehicle on a level surface, the formation of air cavities or other effects that cause the fuel level observed at the sight tube to drop after movement or operation of the vehicle (other than due to consumption) is prevented.

CV2.7.2 The fuel system must be designed such that the spillage during refueling cannot contact the driver position, exhaust system, hot engine parts, or the ignition system.

CV2.8 Venting Systems

CV2.8.1 The fuel tank venting systems must be designed such that fuel cannot spill during hard cornering or acceleration.

CV2.8.2 All fuel vent lines must be equipped with a check valve to prevent fuel leakage when the tank is inverted. All fuel vent lines must exit outside the bodywork.

- CV2.8.3 Only check-valves for fuel vent application are permissible. The use of inline, one-way, or non-return one-way check-valves is strictly prohibited. The check-valve must facilitate the unrestricted entry and exit of air when in an upright position, and while in an inverted position, it must block the flow of fluids.

CV3 EXHAUST SYSTEM AND NOISE CONTROL

CV3.1 Exhaust System General

- CV3.1.1 The exhaust outlet must be routed to the side or rear of the vehicle and so that the driver is not subjected to fumes at any speed considering the draft of the vehicle.
- CV3.1.2 The exhaust outlet(s) must not extend more than 450 mm behind the centerline of the rear axle and shall be no more than 600 mm above the ground.
- CV3.1.3 Any exhaust components (headers, mufflers, etc.) that protrude from the side of the chassis in front of the rear axle must be shielded to prevent contact by persons approaching the vehicle or a driver exiting the vehicle. The temperature of the outer surface must not be harmful to a person touching it.
- CV3.1.4 The application of fibrous/absorbent material, e.g. “headerwrap”, to the outside of an exhaust manifold or exhaust system is prohibited.

CV3.2 Maximum Sound Level

- CV3.2.1 The maximum sound level test speed for a given engine will be the engine speed that corresponds to an average piston speed of 15.25 m/s. The calculated speed will be rounded to the nearest 500 rpm. The maximum permitted sound level up to this calculated speed is 110 dB(C), fast weighting.
- CV3.2.2 The idle test speed for a given engine will be up to the team and determined by their calibrated idle speed. If the idle speed varies then the vehicle will be tested across the range of idle speeds determined by the team. At idle the maximum permitted sound level is 103 dB(C), fast weighting.

CV4 SHUTDOWN SYSTEM

CV4.1 Shutdown Circuit

- CV4.1.1 The shutdown circuit directly controls all electrical power to the ignition, fuel injectors and all fuel pumps. It must act through a minimum of two mechanical relays. One relay for the fuel pump and at least one relay for injection and ignition.

An explanatory schematic of the required shutdown circuit, is shown in figure 18.

- CV4.1.2 The shutdown circuit is defined as a series connection of at least the LVMS ([see T11.3](#)), the BSPD ([see T11.6](#)), three shutdown buttons ([see T11.4](#)), the BOTS ([see T6.2](#)) and the inertia switch ([see T11.5](#)).

CV4.1.3 All circuits that are part of the shutdown circuit must be designed in a way that in the de-energized/disconnected state they open the shutdown circuit.

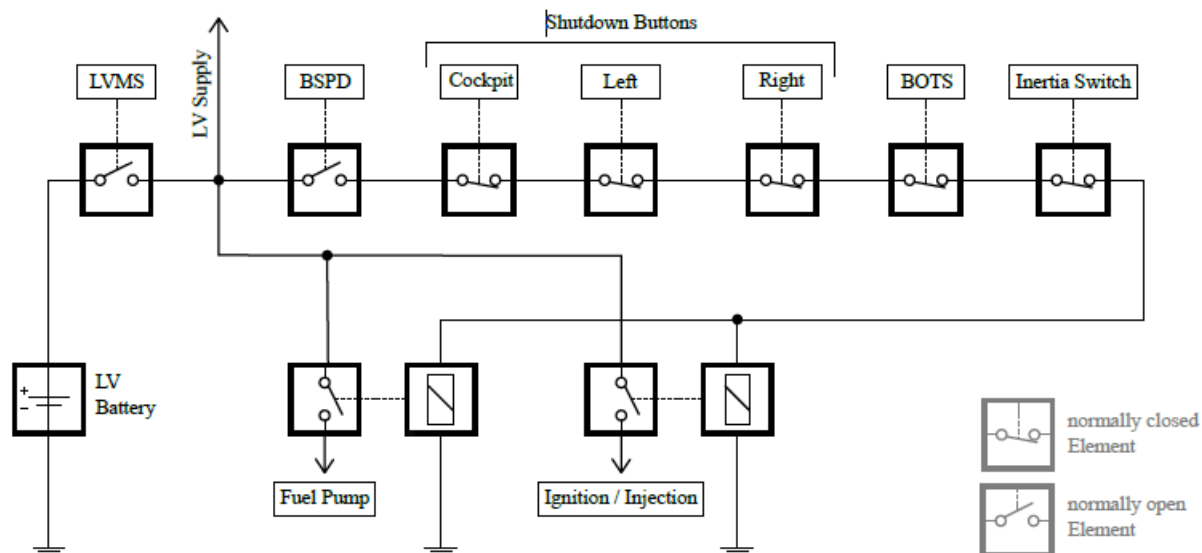


Figure 18: Explanatory example schematic of the required shutdown circuit

EV ELECTRIC VEHICLES

EV1 DEFINITIONS

EV1.1 Tractive System (TS)

EV1.1.1 Tractive System (TS) – every part that is electrically connected to the motor(s) and TS accumulators. The LVS may be supplied by the TS if a galvanic isolation between both systems is ensured.

EV1.1.2 TS enclosures – every housing or enclosure containing parts of the TS.

EV1.2 Galvanic Isolation

EV1.2.1 Galvanic Isolation – two electric circuits are defined as galvanically isolated, if all of the following are true:

- the resistance between both circuits is $\geq 500 \Omega/V$, related to the maximum TS voltage of the vehicle, at a test voltage of maximum TS voltage or 250 V, whichever is higher.
- The isolation voltage RMS, AC for 1 min, between both circuits is higher than three times the maximum TS voltage or 750 V, whichever is higher.
- The working voltage of the isolation barrier, if specified in the datasheet, is higher than the maximum TS voltage.

Capacitors that bridge galvanic isolation must be class-Y capacitors.

EV1.2.2 High Current Path– any path of a [TS circuitry](#) that, during normal operation, carries more than 1A.

EV2 ELECTRIC POWERTRAIN

EV2.1 Motors

EV2.1.1 Only electric motors are allowed.

EV2.1.2 Motor attachments must follow [T10](#).

EV2.1.3 Motor casings must follow [T7.3](#).

EV2.1.4 The motor(s) must be connected to the accumulator through a motor controller.

EV2.2 Power Limitation

EV2.2.1 The TS power at the outlet of the TSAC must not exceed 80 kW.

EV2.2.2 [The TS current at the outlet of the TSAC must not exceed 500A.](#)

EV2.2.3 Regenerating energy is allowed and unrestricted [with regard to power.](#)

EV2.2.4 Wheels must not be spun in reverse.

EV3 GENERAL REQUIREMENTS

EV3.1 Grounding

EV3.1.1 TS enclosures, see [EV1.1.2](#), must consist of either

- a grounded solid layer made of at least 0.5 mm thick electrically conductive material, aluminum or better, see [EV3.1.3](#) or
- be made out of electrically insulating materials providing an isolation resistance of at least $2M\Omega$, measured at a voltage of 500V, between the inside and the outside of the enclosure. The TS enclosure must be rigid and must prevent possible mechanical penetrations. Protruding electrically conductive parts, such as fasteners or connectors, must follow [EV3.1.2](#)

The TSAC might use at least 0.9 mm thick steel layer as the grounded layer.

EV3.1.2 Electrically conductive seat, driver harness, and firewall mounting points as well as the aluminum layer of the TS firewall, see T4.8, and the LVS ground measuring point, see [EV4.7.8](#), must be grounded, see [EV3.1.3](#).

EV3.1.3 An electrically conductive part is grounded if its resistance to LVS ground is below $100m\Omega$, measured with a current of 1 A, and the grounding wire is able to continuously carry at least 10 % of the TS accumulator main fuse current rating.

EV3.1.4 Parts of the vehicle which are or may become electrically conductive within 100 mm of any TS component must have a resistance below 100Ω to LVS ground

EV3.1.5 The rotating part of the wheels may not be grounded.

EV3.2 Overcurrent Protection

EV3.2.1 All electrical systems must have appropriate overcurrent protection.

EV3.2.2 The continuous current rating of the overcurrent protection must not be greater than the continuous current rating of any electrical component, for example wire, busbar or other conductor that it protects.

i.e. if multiple pins of a connector are used to carry currents in parallel, each pin must be appropriately protected.

EV3.2.3 All overcurrent protection devices must have an interrupt current rating which is higher than the theoretical short circuit current of the system that it protects.

EV3.2.4 All overcurrent protection devices must be rated for the highest voltage in the systems they protect. All devices used must be rated for DC.

EV3.2.5 All overcurrent protection devices which are part of the TS must not rely on programmable logic. The overcurrent protection function of motor controllers/inverters for the motor outputs may rely on programmable logic.

EV3.2.6 The overcurrent protection must be designed for the expected surrounding temperature range but at least for 0°C to 85°C .

EV3.2.7 The TS high current path through the accumulator(s) must be fused.

EV4 TRACTIVE SYSTEM (TS)

EV4.1 General Requirements

EV4.1.1 The maximum permitted voltage that may occur between any two electrical connections is 600 VDC and for motor controller and Accumulator Management System (AMS) internal low power control signals 630 VDC.

EV4.1.2 All components in the TS must be rated for the maximum TS voltage.

The TS area of a PCB (see [EV4.3.6](#)), is considered as one component. Every input connected to the TS must be rated to the maximum TS voltage.

EV4.1.3 All components must be rated for the maximum possible temperature which may occur during usage.

EV4.2 TS System Enclosures

EV4.2.1 TS enclosures, see [EV1.1.2](#), must be labeled with (a) reasonably sized sticker(s) according to “ISO 7010-W012” (triangle with black lightning bolt on yellow background). The sticker must also contain the text “High Voltage” if the voltage is more than 60 VDC or 50 VAC RMS.

EV4.3 Separation of Traction System and Grounded Low Voltage System

EV4.3.1 The entire TS and LVS must be completely galvanically isolated (see [EV1.2.1](#) and [IN4.1.1](#)).

EV4.3.2 All connections from a TS component to external devices, such as laptops must include galvanic isolation, see [EV1.2.1](#).

EV4.3.3 TS and LVS circuits must be physically segregated such that they are not running through the same conduit or connector, except for interlock circuit connections.

EV4.3.4 Where both TS and LVS are present within an TS enclosure, they must be separated by barriers made of moisture-resistant insulating materials or maintain 20 mm spacing through air, or over a surface.

EV4.3.5 **Components, wires and cables, either inside or outside a casing, capable of movement must be positively restrained to maintain spacing.**

EV4.3.6 If TS and LVS are on the same PCB, they must be on separate well defined areas of the board, meeting the spacing requirements in Table 5, each area clearly marked with “TS” or “LV”. The outline of the area required for spacing must be marked.

Grooves and cut-outs must have a minimum width of 1.5 mm to influence the creepage path. “Conformal coating” refers to a coating insulator on a PCB. Solder resist is not a coating.

Voltage	Clearance Distance	Creepage Distance	
		General	Conformal Coating
0 VDC to 50 VDC	1.0 mm	4 mm	1.0 mm
50 VDC to 150 VDC	1.0 mm	5 mm	1.0 mm
150 VDC to 300 VDC	1.5 mm	10 mm	2.0 mm
300 VDC to 600 VDC	3.0 mm	20 mm	4.0 mm

Table 5: Spacing required between TS and LV.

EV4.3.7 Teams must be prepared to demonstrate spacing on team-built equipment. For inaccessible circuitry, fully assembled spare boards must be available.

EV4.4 Positioning of Tractive System Parts

EV4.4.1 Except what is permitted according to EV4.4.3, all parts belonging to the TS including cables and wiring must be located within the rollover protection envelope, see [T1.1.16](#). “Part” is the whole device such as the complete HVD.

EV4.4.2 Any part of the TS that is less than 350 mm above the ground, except what is permitted according to [EV4.4.3](#), must be protected from impacts, see [T3.14](#). Impact protection must follow [T3.15](#) when having bolted attachments. TS wiring in front of the front hoop may alternatively be shielded by the front bulkhead support structure according to [T3.13](#). The TSAC must be no higher than the top of the impact structure as mentioned in [EV5.5.1](#).

EV4.4.3 Outboard wheel motors are allowed only if

- an interlock is routed along the TS wiring such that the shutdown circuit, see [EV6](#), is opened before the TS wiring or its clamping fails
- an interlock is routed along a suspension member such that the shutdown circuit, see [EV6](#), is opened if the suspension fails
- TS wiring is not be able to reach the cockpit opening or the driver regardless of where it breaks
- wiring outside of the rollover protection envelope, see [T1.1.15](#) is minimum length
- wiring outside of an impact structure or front bulkhead support structure (see [EV4.4.2](#)), is minimum length
- Minimum length is the shortest distance plus extra wiring caused by bending radius.

EV4.5 Tractive System Insulation, Wiring and Conduit

EV4.5.1 All live parts of the TS must be protected from being touched. This must include team members working on or inside the vehicle. This is tested with a 100 mm long, 6 mm diameter insulated test probe when the TS enclosures are in place.

EV4.5.2 Insulation material that is rated for the maximum TS voltage must be used. Using only insulating tape or rubber-like paint for insulation is prohibited.

EV4.5.3 The temperature rating for TS wiring, connections, and insulation must be appropriate for the expected surrounding temperatures but at least 85°C.

EV4.5.4 TS components and containers must be protected from moisture in the form of rain or puddles (see [IN8](#)).

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- EV4.5.5 It must be possible to clearly assign and prove the wire gauge, temperature rating, and insulation voltage to each used TS wire.
- EV4.5.6 All TS wiring must be completed to professional standards with appropriately sized conductors and terminals and with adequate strain relief and protection from loosening due to vibration etc.
- EV4.5.7 TS wiring must be located out of the way of possible snagging or damage.
- EV4.5.8 All TS wiring that runs outside of TS enclosures must:
- Be enclosed in separate orange non-conductive conduit or use an orange shielded cable. The conduit must be securely anchored to the vehicle, but not to wire, at least at each end.
 - Be securely anchored at least at each end so that it can withstand a force of 200 N without straining the cable end crimp.
- Body work is not sufficient to meet this enclosure requirement.
- EV4.5.9 Any shielded TS cable must have the shield grounded.
- EV4.5.10 Every TS connector outside TS enclosures must include a pilot contact/interlock line which is part of the shutdown circuit. Housings only used to avoid interlocks are prohibited.
- EV4.5.11 All TS connections must be designed so that they use intentional current paths through conductors such as copper or aluminum and should not rely on steel bolts to be the primary conductor.
- EV4.5.12 All materials used in a stack-up or as a fastener of a TS connection must ensure equal pressure across the full contact area over the entire temperature range. Screw type connectors are allowed to be used inside the TSAC if only it is part of an OEM product given that the OEM product is automotive grade or is tested for appropriate vibrations.
- EV4.5.13 All electrical connections, including bolts, nuts, and other fasteners, in the high current path, see [EV1.2.2](#), of the TS must be secured from unintentional loosening. Fasteners must use positive locking mechanisms, see [T10.2](#), that are suitable for high temperatures, see [EV4.5.3](#).
- Components, e.g. inverters, certified for automotive use might be allowed without positive locking feature, if connections are completed as recommended by the manufacturers' datasheet and no positive locking is possible.
- EV4.5.14 Teams must be prepared to demonstrate positive locking. For inaccessible connections, appropriate photographs must be available.
- EV4.5.15 Soldered connections in the TS high current path, see [EV1.2.2](#), are only allowed if all of the following are true:
- connections on PCBs
 - the connected devices are not cells or wires
 - the devices are additionally mechanically secured against loosening
-

EV4.6 Data Logger

- EV4.6.1 A calibrated data logger will be provided by the officials and must be inserted during competition. The data logger measures TS voltage and TS current.
- EV4.6.2 The data logger must be in an easily accessible location so that it is possible to insert, remove or replace it within 15 min in ready to race condition.
- EV4.6.3 The data logger must not be placed within the accumulator container.
- EV4.6.4 All current supplying the TS must run through the data logger. The data logger must be inserted in the negative TS supply between the most negative AIR(s) and the inverters.
- EV4.6.5 The TS voltage sense connection of the data logger must be directly connected, see [T1.3.1](#), to the most positive AIR on the vehicle side. **If overcurrent protection is used, it must be a fuse.**
- EV4.6.6 The specification of the data logger will be available on the competition website.

EV4.7 Tractive System Measuring point (TSMP)

- EV4.7.1 Two TSMPs must be installed directly next to the master switches (see [EV6.2](#)).
- EV4.7.2 The TSMPs must be directly connected (see [T1.3.1](#)) to the intermediate circuit capacitors even if the HVD has been opened or the TS accumulator is disconnected.
- EV4.7.3 4 mm shrouded banana jacks rated for 600V CAT III or better must be used for the TSMPs.
- EV4.7.4 The TSMPs must be directly connected to the positive and negative motor controller/inverter supply. The TSMPs must be marked “TS+” and “TS-” and exclusively mounted on an orange background.
- EV4.7.5 The TSMPs must be protected by a non-conductive housing that can be opened without tools. The cover must always be mechanically linked to the vehicle.
- EV4.7.6 Each TSMP must be secured with a current limiting resistor according to the following table. Fusing of the TSMPs is prohibited. The resistors power rating must be chosen such that they are able to continuously carry the current if both TSMPs are short circuited.

Maximum TS Voltage	Resistor Value
$U_{max} \leq 200 \text{ VDC}$	5 k Ω
$200 \text{ VDC} < U_{max} \leq 400 \text{ VDC}$	10 k Ω
$400 \text{ VDC} < U_{max} \leq 600 \text{ VDC}$	15 k Ω

- EV4.7.7 All electric connections needed to connect the TSMP to the intermediate circuit capacitors, including bolts, nuts and other fasteners, must be secured from unintentional loosening by the use of positive locking mechanisms. Bolted connections must follow [T10.2](#), soldered connections [EV4.5.15](#).
- EV4.7.8 Next to the TSMPs a LVS ground measuring point must be installed. A 4 mm black shrouded banana jack must be connected to LVS ground and must be marked “GND”.

EV4.8 High Voltage Disconnect (HVD)

- EV4.8.1 It must be possible to disconnect at least one pole of the TS accumulator by quickly removing an unobstructed and directly accessible element, fuse or connector. It must be possible to disconnect the HVD without removing any bodywork. The HVD must be above 350 mm from the ground. Remote actuation of the HVD through a long handle, rope, or wire is not permitted.
- EV4.8.2 Any ESO must be able to remove the HVD within 10 s when the vehicle is in ready-to-race condition.
- EV4.8.3 A dummy connector or similar may be required to restore the system's isolation (see EV4.5). The dummy connector must be attached to the pushbar (see T13.1), if not in use.
- EV4.8.4 The HVD must be clearly marked with "HVD".
- EV4.8.5 No tools must be necessary to open the HVD. An interlock is required (see EV4.5.10).

EV4.9 Discharge Circuit

- EV4.9.1 If a discharge circuit is required to meet EV6.1.5, it must be designed to handle the maximum tractive system voltage permanently.
- After three subsequent discharges within 15s in total, the discharge time specified in EV6.1.5 may be exceeded. Full discharging functionality must be given after a reasonable time with a deactivated discharge circuit.
- EV4.9.2 The discharge circuit must be wired in a way that it is always active whenever the shutdown circuit is open. Furthermore, the discharge circuit must be fail-safe such that it still discharges the intermediate circuit capacitors if the HVD has been opened or the TS accumulator is disconnected.
- EV4.9.3 Fusing of the discharge main current path is prohibited.
- EV4.9.4 No part of the discharge circuit must be inside the TSAC.
- EV4.9.5 *A discharge circuit must also be present in the handcart unless the charger already has it.*

EV4.10 Tractive System Active Light (TSAL)

- EV4.10.1 The vehicles must include a single TSAL that must indicate the TS status. The TSAL must not perform any other functions. A TSAL with multiple LEDs in one housing is allowed.
- EV4.10.2 The TSAL itself must have a red light, flashing continuously with a frequency between 2 Hz and 5 Hz and a duty cycle of 50%, active if and only if the LVS is active and the voltage across DC-link capacitors exceeds
- 60 VDC or 50 VAC RMS
 - Half the nominal TS voltage
- whichever is lower.
- EV4.10.3 The TSAL itself must have a green light, continuously on, active if and only if the LVS is active and ALL of the following conditions are true:

- All AIRs are opened.
 - The pre-charge relay, see [EV5.7.2](#), is opened.
 - The voltage at the vehicle side of the AIRs inside the TSAC does not exceed 60 VDC or 50 VAC RMS.
- EV4.10.4 The mentioned voltage detection must be performed inside the respective TS enclosure.
- EV4.10.5 The mentioned states of the relays (opened/closed) are the actual mechanical states. The mechanical state can differ from the intentional state, i.e. if a relay is stuck. Any circuitry detecting the mechanical state must meet [EV5.6.2](#).
- EV4.10.6 The voltage detection circuit of the red light and the relay state and voltage detection circuit of the green light must be independent. Any plausibility check between both lights is not allowed. A TSAL state with both lights simultaneously active might occur and must not be prevented.
- EV4.10.7 The TSAL must:
- Be located lower than the highest point of the main hoop and including the mounting within the rollover protection envelope, see [T1.1.15](#).
 - Be no lower than 75 mm from the highest point of the main hoop.
 - Not be able to contact the driver's helmet in any circumstances.
- EV4.10.8 The entire illuminated surface of the TSAL must be clearly visible:
- Except for angles less than 10° on each side which are blocked by the main hoop.
 - From a point 1.60 m vertically from ground level, within 3 m horizontal radius from the TSAL.
 - In direct sunlight.
- EV4.10.9 The TSAL and all needed circuitry must be hard wired electronics. Software control is not permitted.
- EV4.10.10 A green indicator light in the cockpit that is easily visible even in bright sunlight and clearly marked with "TS off" must light up if TSAL green light is on, see [EV4.10.3](#).
- EV4.10.11 [Signals influencing the TSAL and the indicator according to EV4.10.10 are SCS, see T11.9. The individual safe state of each of the TSAL lights is off. The TSAL's red light must not be illuminated for a visible check, see T11.9.6.](#)
- EV4.10.12 The TSAL's red light voltage detection circuit must include SCS Check, as required by [T11.9.2\(b\)](#), and turn TSAL red light off in case of sensor signal failure.
- EV4.10.13 TSAL green light must enter safe state:
- if TSAL's green light relay state detection circuit detects implausibility between the intentional and mechanical state of the relay (AIR and Precharge relay)
 - if TSAL's green light voltage detection circuit detects implausibility, as required by [T11.9.2\(b\)](#)
- EV4.10.14 A latching circuit is required to latch safe state caused by [EV4.10.13](#). It must not be triggered during normal operation conditions and must only be reset by power-cycling the LVS.

EV4.11 Activating the Tractive System

- EV4.11.1 The TS is active if any of the AIRs or the pre-charge relay is closed.
- EV4.11.2 The driver must be able to activate and deactivate the TS from within the cockpit without the assistance of any other person.
- EV4.11.3 The ASR must be able to activate the TS from outside the vehicle with an external TS activation button in proximity to the TSMS if and only if the ASMS is in “On” position.
- EV4.11.4 Closing the shutdown circuit by any part defined in EV6.1.2 must not (re-)activate the tractive system. Additional action must be required.
- EV4.11.5 The AS must not be able to (re-)activate the TS.
- EV4.11.6 The vehicle is ready to drive as soon as the motor(s) will respond to the input of the APPS.
- EV4.11.7 After the TS has been activated, additional actions must be required by the driver to set the vehicle to ready-to-drive mode (e.g. pressing a dedicated start button). The transition to ready-to-drive mode must only be possible during the actuation of the mechanical brakes and a simultaneous dedicated additional action.
- EV4.11.8 The ready-to-drive mode must be left immediately when the shutdown circuit is opened.

EV4.12 Ready-To-Drive Sound

- EV4.12.1 The vehicle must make a characteristic sound, continuously for at least 1 s and a maximum of 3 s while entering R2D mode.
- EV4.12.2 The sound level must be a minimum of 80 dBA and a maximum of 90 dBA, fast weighting in a radius of 2 m around the vehicle.
- EV4.12.3 The used sound must be easily recognizable. No animal voices, song parts or sounds that could be interpreted as offensive will be accepted.
- EV4.12.4 The vehicle must not make any other sounds similar to the ready-to-drive sound.

EV5 TRACTIVE SYSTEM ENERGY STORAGE

EV5.1 Definitions

- EV5.1.1 Cell – a battery cell or super-capacitor.
- EV5.1.2 Cell Energy – the maximum cell voltage times the nominal capacity of the used cell.
- EV5.1.3 TS Accumulator – all cells that store the electrical energy to be used by the TS as a whole.
- EV5.1.4 TS Accumulator Container (TSAC) – a container which contains the TS accumulator.
- EV5.1.5 TS Accumulator Segments – sub-divisions of the TS accumulator.

EV5.2 Allowed Tractive System Accumulators

EV5.2.1 All types of accumulators except molten salt and thermal batteries are allowed.

EV5.2.2 Fuel cells are prohibited.

EV5.3 Tractive System Energy Storage – General Requirements

EV5.3.1 All cells which store the TS energy must be enclosed in (an) accumulator container(s).

EV5.3.2 Each accumulator segment must not exceed a maximum static voltage of 120 VDC and a maximum energy of 6 MJ ([see EV5.1.2](#)) and a maximum mass of 12 kg.

EV5.3.3 Spare cells must be stored in an electrically insulated container made of fire retardant material, see T1.2.1. [The cells must be securely fixed inside the container to prevent any movement.](#) The container must be labeled as specified in EV5.3.7. [If this container is placed within another container, the outer container must also be labeled in accordance with EV5.3.7.](#)

EV5.3.4 Spare accumulators and spare cells must be presented at Technical Inspection [in a container that complies with EV5.3.7.](#)

EV5.3.5 It must be possible to open the TSAC for technical inspection.

EV5.3.6 Each TSAC must be removable from the vehicle while still remaining rules compliant without the need to install extra components. A dummy connector or similar may be used to restore the system's isolation ([see EV4.5](#)).

EV5.3.7 The vehicle number, the university name, the ESO phone number(s), cell chemistry used and max TS voltage must be displayed and written in Roman Sans-Serif characters of at least 20 mm high on the lid of each accumulator container. The characters must be clearly visible and placed on a high contrast background.

EV5.4 Tractive System Energy Storage – Electrical Configuration

EV5.4.1 If the TSAC is made from an electrically conductive material, the insulation barrier must be adequately protected against conductive penetrations.

EV5.4.2 Every TSAC must contain at least one fuse and at least two AIRs ([see EV5.6](#) and [EV3.2.7](#)).

EV5.4.3 LVS must not be included in the TSAC except where inherently required. Exceptions include the AIRs, TS DC/DC converters, the Accumulator Management System(AMS), the Insulation Monitoring Device (IMD), the TSAL's green light circuitry, and cooling fans.

EV5.4.4 Maintenance plugs must allow electrical separation of all internal cell segments ([see EV5.3.2](#)). The separation must affect both poles of all segments including the first and last segment.

EV5.4.5 Maintenance plugs must

- not require tools to separate the TSAC segments.
- be non-conductive on surfaces that do not provide any electrical connection.

- be designed in a way that it is physically impossible to electrically connect them in any way other than the design intent configuration.
- protected against accidental reconnection.
- be designed such that it is clearly visible whether the connection is open or closed. Electrically controlled switches must not be used.
- should be completely detached from the TSAC when in a removed state i.e they should be kept in a separate box.

Upon removal of all maintenance plugs, no voltage should be present at the TSAC side of AIR.

- EV5.4.6 Each TS accumulator segment must be electrically insulated by the use of suitable and fire retardant (see [T1.2.1](#)), material between the segments and on top of the segment to prevent arc flashes caused by inter segment contact or by parts/tools accidentally falling into the TSAC during maintenance.
- EV5.4.7 Every wire used in a TSAC, regardless of whether it is part of the LVS or TS, must follow [EV4.5.2](#), [EV4.5.3](#), and [EV4.5.5](#), [the exception being wires used in the intent of EV5.8.11](#).
- EV5.4.8 Each TSAC must have a prominent indicator, a voltmeter, or a red LED visible even in bright sunlight. [If a red LED is used as an indicator, it must continuously illuminate at its full brightness](#) whenever a voltage greater than 60VDC or half the maximum TS voltage, whichever is lower, is present at the vehicle side of the AIRs.
- EV5.4.9 The indicator must be clearly visible while disconnecting the TSAC from the vehicles. The indicator must be clearly marked with “Voltage Indicator”.
- EV5.4.10 The indicator must be hard-wired electronics without software control, directly and only supplied by the TS from the vehicle side of the AIRs, and always working, even if the accumulator is disconnected from the LVS or removed from the vehicle.

EV5.5 Tractive System Energy Storage – Mechanical Configuration

- EV5.5.1 [All TSACs must lie fully within and be attached to the primary structure. In side view the TSACs may not exceed the height of the impact structure, see T3.14.](#)
- EV5.5.2 [The structures surrounding the TSACs must protect it from impacts, see T3.14. Impact protection must follow T3.15 when having bolted attachments. No part of the TSAC may be part of the primary structure.](#)
- EV5.5.3 All materials used inside the TSAC and the TSAC itself, as well as all structural parts used to comply with [EV5.5.8](#) must be fire retardant, see [T1.2.1](#). All calculations must be conducted for an ambient temperature of 60°C except for metallic materials and continuous fiber-reinforced laminates.
- EV5.5.4 The design of the TSAC and its contents, calculations and/or tests must be documented in the ASES. This includes materials used, drawings, images, fastener locations, segment weight, cell, and segment position.
- EV5.5.5 TSAC(s) must be constructed of steel or aluminum. With the following requirements:
- The bottom of the TSAC must be at least 1.25 mm thick if made from steel or 3.2 mm if made from aluminum.

- The internal and external vertical walls, covers and lids must be at least 0.9 mm thick if made from steel or 2.3 mm if made from aluminum.

Alternative materials are allowed with proof of equivalency per [T3.3](#) or for composite materials per [EV5.5.6](#). When alternative materials are used, test samples must be presented at technical inspection.

EV5.5.6 Composite TSAC(s) must satisfy the following requirements:

- Data obtained from the laminate perimeter shear strength test and three point bending test ([T3.5](#)), should be used to prove adequate strength is provided.
- Each attachment point requires steel backing plates with a minimum thickness of 2 mm. Alternate materials may be used for backing plates if equivalency is approved.
- The calculations and physical test results must be included in the ASES.

EV5.5.7 If any structural part of the TSAC is manufactured using additive processes, physical testing must verify compliance with [EV5.5.9](#). Test samples must be produced on the same machine with identical parameters as the final part. Calculations must account for anisotropic material properties, and all results must be documented in the ASES.

EV5.5.8 TS accumulator segments, see [EV5.3.2](#), must be separated by internal vertical walls which extend upwards until the lid. Openings must follow [EV5.5.14](#).

EV5.5.9 The TSAC, its mounting to the chassis, and the mounting of each cell to the TSAC must be designed to withstand the accelerations defined in [T9.3.1](#). All TSAC attachment points must follow [EV5.5.15](#). TSACs made of materials as stated in [EV5.5.5](#) or [EV5.5.6](#) may need further reinforcement to comply with this rule.

EV5.5.10 Pouch cells must be fixed using one or both of the large surface(s) only. Each used surface must be fixed on at least 80%. Tabs of pouch cells must not carry mechanical loads and must not press into the pouch.

EV5.5.11 To fulfil [EV5.5.9](#), each cell must not exceed 0.05MPa shear and 1.0MPa compressive stress, calculated using 80% of the nominal large cell surface area. Proof must be provided in the ASES. Without stress analysis, a maximum of 6 pouch cells may be mechanically attached in series.

EV5.5.12 Friction-based cell mounting requires physical testing. Mechanically representative test cells or cell mockups can be used for testing.

EV5.5.13 All fasteners used within or to mount the TSAC must comply with [T10](#). Fasteners within the TSAC used for non-structural parts, e.g. PCBs, do not have to follow [T10.1.2](#). Fasteners made of electrically non-conductive material within the TSAC used for non-structural parts do not have to follow [T10](#).

EV5.5.14 The AIRs and the main fuse (see [EV3.2.7](#)) must be separated with an electrically insulated and fire retardant material (see [T1.2.1](#)) from the TS accumulator. Air is not considered to be a suitable insulation material in this case.

EV5.5.15 The TSAC must be mounted using at least 4 attachment points. [T3.16.6](#) does not apply to the number of attachment points. Any brackets used must be made of either 1.6mm thick steel or 4mm thick aluminium and must have gussets to support bending loads.

- EV5.5.16 Holes, both internal and external, in the TSAC are only allowed for the wiring-harness, ventilation, cooling or fasteners. Holes can extend to the edge of a wall. The TSAC must still be compliant with all other rules, especially the ones concerning its structural requirements. The total cutout area must be below 25% of the area of the respective single wall. External holes must be sealed according to [EV4.5](#).
- EV5.5.17 External openings in the TSAC or mounted ducts must not point toward the driver or the operator of the accumulator hand cart even though there is a firewall, see [T4.8](#), in between. Ducts and their mounting must be fire retardant, see [T1.2.1](#), and properly sealed.
- EV5.5.18 A sticker according to “ISO 7010-W012” (triangle with black lightning bolt on yellow background) with triangle side length of at least 100 mm and the text “Always Energized” must be applied on every TSAC. The sticker must also contain the text “High Voltage” if the voltage is more than 60 VDC or 50 VAC RMS.
- EV5.5.19 Any TS accumulators that may vent an explosive gas must have a ventilation system to prevent the vented gas from reaching an explosive concentration.
- EV5.5.20 Every TSAC which is completely sealed must also have a pressure relief valve to prevent high-pressure in the container.

EV5.6 Accumulator Isolation Relays (AIRs)

- EV5.6.1 At least two AIRs must be fitted inside each accumulator container.
- EV5.6.2 The AIRs must open both poles of the accumulator. If the AIRs are open, no TS voltage may be present outside of the accumulator container and the vehicle side of the AIRs must be galvanically isolated from the accumulator side (see [EV1.2.1](#)).
- EV5.6.3 The AIRs must be mechanical relays of a “normally open” type.

EV5.7 Pre-Charge Circuit

- EV5.7.1 A circuit that ensures that the voltage at the vehicle side of the AIRs is pre-charged to at least 95% of the actual TS accumulator voltage before closing the second AIR must be implemented.
- EV5.7.2 The pre-charge circuit must use a mechanical, normally open type relay. All pre-charge current must pass through this relay.

EV5.8 Accumulator Management System (AMS)

- EV5.8.1 Each TS accumulator must be monitored by an AMS whenever the LVS is active or the accumulator is connected to a charger.
- EV5.8.2 Every TSAC must contain its full AMS including its own and AMS exclusive SDC power stage, see [EV6.1](#).
- EV5.8.3 The AMS must continuously measure
- all cell voltages
 - the TS current
 - the temperature of thermally critical cells

- d. for lithium based cells: the temperature of at least 30% of the cells equally distributed within the accumulator container(s)

EV5.8.4 The maximum cell temperature is 60°C or the limit stated in the cell data sheet, whichever is lower.

EV5.8.5 The temperature sensor must be mounted on the electrically conducting surface of the cell. The placement must represent the maximum temperature location on that conducting surface under:

- Maximum charge and discharge conditions
- Maximum cooling effectiveness
- 30°C ambient temperature

If the maximum temperature occurs on a non-conductive part of the cell body, the sensor must be placed on the electrically conducting surface closest to that point.

EV5.8.6 The temperature sensor must be in rigid, direct contact with the electrically exposed conducting surface of the cell. Use of thermal adhesive is permitted; however, if used, the maximum allowable monitored cell temperature must be derated, with justification provided in the ESF.

EV5.8.7 Teams must justify the sensor placement in the ESF/ASES, including:

- Estimation of measurement error between the sensor location and the true maximum temperature
- Consideration of thermal resistances, electrical measurement error and cooling effects

EV5.8.8 A single temperature sensor may monitor multiple cells only if all monitored cells fully comply with the above requirements.

EV5.8.9 The AMS must open the SDC, if a critical voltage, temperature, or current value according to the cell manufacturer's datasheet or these rules persistently occurs for more than:

- 500 ms for voltage and current values
- 1 s for temperature values

The accuracy and noise of the measurement must be taken into account for these.

EV5.8.10 AMS cell voltage measurement inputs, temperature measurement inputs, and supply voltage of decentralized AMS slaves may be rated below the maximum TS voltage.

EV5.8.11 A red indicator light in the cockpit that is easily visible from inside and outside the cockpit even in bright sunlight and clearly marked with the lettering "AMS" must light up if the AMS opens the shutdown circuit. It must stay illuminated until the error state has been manually reset (see [EV6.1.6](#)). Signals controlling this indicator are SCS (see [T11.9](#)).

EV5.8.12 AMS signals are System Critical Signals, see [T11.9](#). The loss of a measurement connection, that leads to a violation of [EV5.8.3](#), must result in an open SDC.

EV5.8.13 It must be possible to interrupt the measurement signal path such that violations in each part of the continuous monitoring of the AMS, see [EV5.8.3](#), can be individually triggered during Technical Inspection. It must be possible to demonstrate this check on all of the segments of the accumulator during the event.

- EV5.8.14 The AMS must be able to read and display all measured values according to [EV5.8.3](#) in a single overview e.g. by connecting a laptop to the AMS at any place and any time e.g. inside the dynamic area.

EV6 SHUTDOWN CIRCUIT AND SYSTEMS

EV6.1 Shutdown Circuit

- EV6.1.1 The shutdown circuit directly carries the current driving the AIRs, see [EV5.6](#), and the pre-charge circuitry, see [EV5.7](#).
- EV6.1.2 The shutdown circuit is defined as a series connection of at least two master switches, three shutdown buttons, the BOTS ([see T6.2](#)), the IMD, the inertia switch ([see T11.5](#)), the BSPD ([see T11.6](#)), all required interlocks and the AMS. An explanatory schematic of the required shutdown circuit, excluding any possible interlock circuitry, is shown in figure 19 and figure 20.
- EV6.1.3 All parts of the shutdown circuit defined in [EV6.1.2](#) must be on the high-side connection of the AIR coils.
- EV6.1.4 The Tractive System Master Switch (TSMS) ([see EV6.2](#)) must be the last switch before the AIRs except for pre-charge circuitry and hardwired interlocks.
- EV6.1.5 If the SDC is opened the TS must be shut down by opening all AIRs as well as the pre-charge relay and the voltage in the TS must drop to below 60 VDC and 50 VAC RMS in less than 5 s. All TS accumulator current flow must stop immediately.

The action of opening the AIRs may be delayed by ≤ 250 ms to signal the action to the motor controllers and reduce the TS current before the AIRs are opened. The AIR supply must be abruptly switched off before reaching the minimum AIR supply voltage.

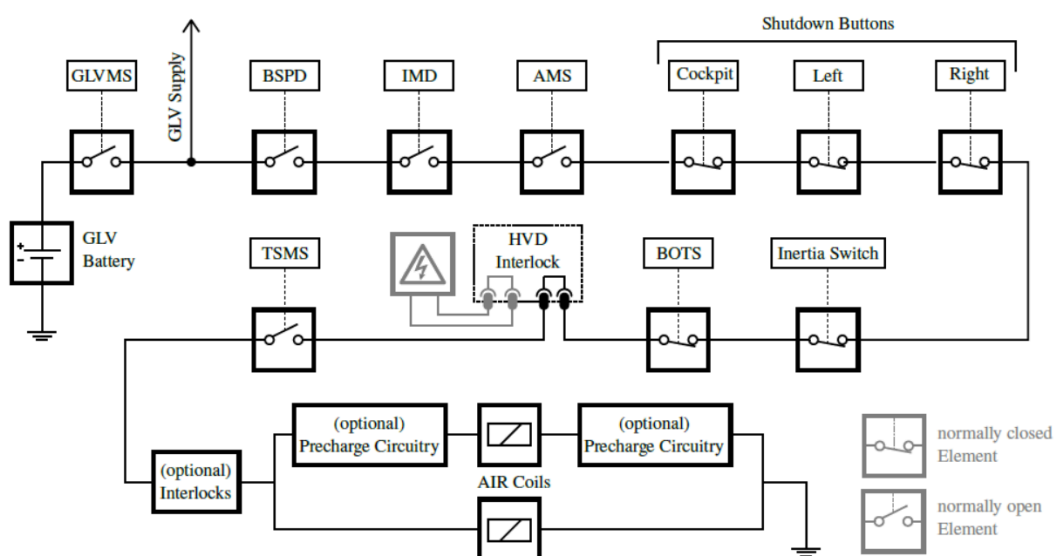


Figure 19: Explanatory example schematic of the required shutdown circuit

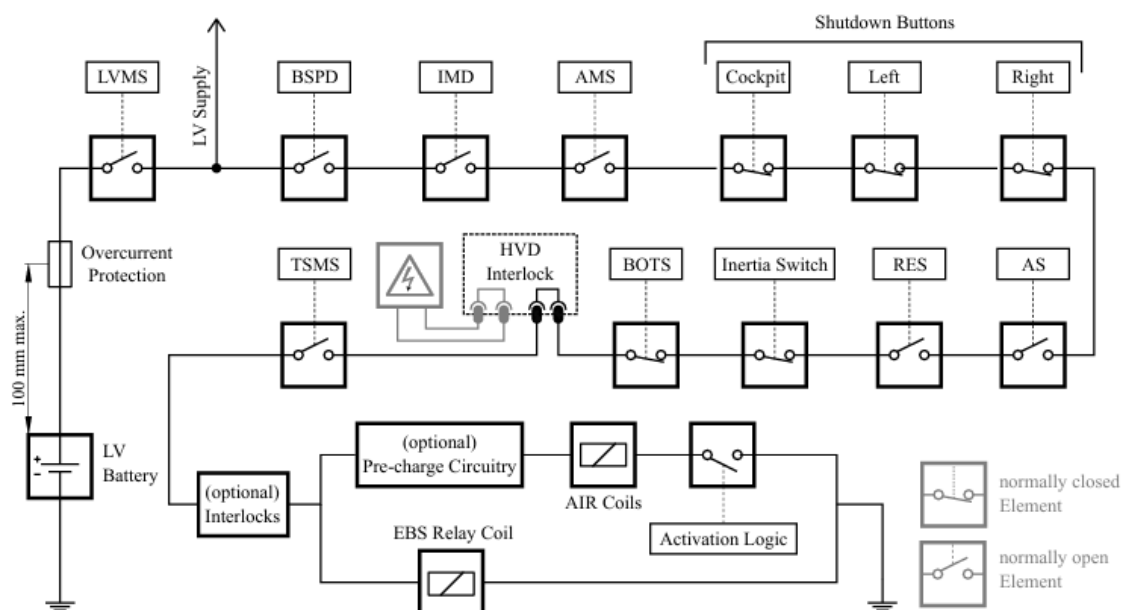


Figure 20 : [DC ONLY] Explanatory example schematic of the required shutdown circuit

- EV6.1.6 If the shutdown circuit is opened by the AMS or the IMD, it has to be latched open by a non-programmable logic that can only be manually reset by a person at the vehicle who is not the driver.
- EV6.1.7 All circuits that are part of the shutdown circuit must be designed in a way, that in the de-energized/disconnected state they are open such that each circuit will remove the current controlling the AIRs.
- EV6.1.8 It must be possible to demonstrate that all features of the shutdown circuit function correctly. This includes all interlocks.
- EV6.1.9 Every system that is required to or is able to open the shutdown circuit must have its own, non-programmable, power stage to achieve this. The respective power stages must be designed to be able to carry the shutdown circuit current, e.g. AIR inrush currents, and such that a failure cannot result in electrical power being fed back into the electrical shutdown circuit.
- EV6.1.10 The shutdown buttons, the BOTS, the TSMS and all interlocks must not act through any power stage.
- EV6.1.11 All signals influencing the shutdown circuit are SCSs (see T11.9).

EV6.2 Tractive System Master Switch (TSMS)

- EV6.2.1 A TSMS according to T11.2 must be part of the shutdown circuit (see EV6.1.2).
- EV6.2.2 The TSMS must be fitted with a “lockout/tagout” capability to prevent accidental activation of the TS. The ESO must ensure that it is locked in the off position whenever work is done on the vehicle or no ESO is present.

EV6.2.3 The TSMS must be mounted in the middle of a completely orange circular area of ≥ 50 mm diameter placed on a high contrast background.

EV6.2.4 The TSMS must be marked with “TS” and a symbol according to “ISO 7010-W012” (triangle with black lightning bolt on yellow background).

EV6.3 Insulation Monitoring Device (IMD)

EV6.3.1 Every vehicle must have an IMD installed in the TS system.

EV6.3.2 The IMD must be a Bender A-ISOMETER® iso-F1 IR155-3203, -3204, -4203, or -4204, or a Bender ISOMETER® iso165C-1, iso175, or equivalent IMD approved for automotive use. Equivalency may be approved by the officials’ based on the following criteria: robustness to vibration, operating temperature range, IP rating, availability of a direct output, a self-test facility and must not be powered by the system that is monitored.

EV6.3.3 The response value of the IMD must be set to $\geq 500 \text{ } \Omega/\text{V}$, related to the maximum TS operation voltage.

EV6.3.4 The response value must not be changed after electrical inspection.

EV6.3.5 The IMD must be connected on the vehicle side of the AIRs.

EV6.3.6 One IMD chassis ground measurement line must be connected to the grounded TSAC or the respective grounded enclosure of the IMD. The other chassis ground measurement line must be connected to the main hoop. Each connection must use a separate conductor, rated for at least maximum TS voltage. An open circuit in any of this ground measurement connections must result in an opened shutdown circuit.

EV6.3.7 In case of an insulation failure or an IMD failure, the IMD must open the shutdown circuit. This must be done without the influence of any programmable logic. See also [EV6.1.6](#) regarding the re-activation of the TS after an insulation fault.

EV6.3.8 A red indicator light in the cockpit that is easily visible from inside and outside the cockpit even in bright sunlight and clearly marked with the lettering “IMD” must light up if the IMD opens the shutdown circuit. It must stay illuminated until the error state has been manually reset ([see EV6.1.6](#)). Signals controlling this indicator are SCS ([see T11.9](#)).

EV7 CHARGERS

EV7.1 Chargers General Requirements

EV7.1.1 Only chargers presented and sealed at technical inspection are allowed. All connections of the charger(s) must be isolated and covered. No open connections are allowed.

EV7.1.2 Exposed conductive parts and the TSAC must be connected to protective earth (PE).

EV7.1.3 All chargers must either be accredited to a recognized standard e.g. CE, or when built by the team, must comply with all electrical requirements for the vehicle TS, e.g. [EV4.3](#), [EV3.1](#) and [EV4.2](#).

EV7.1.4 TS charging leads must be orange.

- EV7.1.5 When charging, the AMS must be active and must be able to open the SDC, see EV7.2.
- EV7.1.6 The charger must include a push-type emergency stop button which has a minimum diameter of 24mm. It must be clearly labeled [and easily visible](#).
- EV7.1.7 When charging, an IMD as described in [EV6.3](#) must be active and must be able to open the SDC, see [EV7.2](#). The second chassis ground measurement line must be connected to the casing of the charger.
- EV7.1.8 An IMD indicator light as defined in [EV6.3.7](#) must be available during charging.
- EV7.1.9 The Handcart must include a single TSAL that must indicate the TS status, see EV4.10.2, EV4.10.3. The TSAL must not perform any other functions. A TSAL with multiple LEDs in one housing is allowed.
- EV7.1.10 The charger must include TSMPs as described in EV4.7. The TSMPs must be connected to the TS output of the charger and placed behind the firewall away from fire direction.
- EV7.1.11 All the user interfaces (TSMP, ESS, indicators etc) should be towards and the operator side of the handcart

EV7.2 Charging Shutdown Circuit

- EV7.2.1 The charging Shutdown Circuit (SDC) consists of at least the charger shutdown button, see [EV7.1.6](#), the IMD, the AMS, and the charging leads' interlocks.
- EV7.2.2 If the SDC is opened the charging system must remain disabled and the SDC opened until it is manually reset. Closing the SDC by any part defined in [EV7.2.1](#) must not (re-)activate the TS, see [EV4.11.1](#), and the voltage at the TS connector has to remain below 60VDC or 50VACRMS.
- EV7.2.3 The charging shutdown systems must comply with [EV6.1.1](#), [EV6.1.5](#), [EV6.1.7](#), [EV6.1.8](#) and [EV6.1.9](#).
- EV7.2.4 All signals influencing the charger shutdown circuit are SCS ([see T11.9](#)).

EV8 TS ACCUMULATOR CONTAINER HAND CART

EV8.1 TS Accumulator Container Hand Cart

- EV8.1.1 A hand cart(s) must be used for transporting the accumulator container(s) around the competition site.
- EV8.1.2 The hand cart must have at least four wheels.
- EV8.1.3 The hand cart must have a brake which is always on and only released if someone pushes the handle, or similar.
- EV8.1.4 The brake must be capable of stopping the fully loaded accumulator container hand cart.
- EV8.1.5 The hand cart must be easily moved when the brake is released.

-
- EV8.1.6 The hand cart must be able to carry the load of the TSAC(s).
- EV8.1.7 The hand cart must provide a firewall to protect the person while moving the hand cart. The firewall must have the same width as the hand cart, appropriately protect the legs and body, and be at least 30 cm higher than the hand cart handle and the TSAC. The firewall must be made from a rigid, fire retardant material, see T1.2.1, and be transparent from 1.3 m above the ground.
- EV8.1.8 The TSAC(s) must be mechanically fixed to the hand cart to enable safe transportation.
- EV8.1.9 The TSAC(s) must not protrude the hand cart.
- EV8.1.10 The TSAC(s) must be protected from vibrations and shocks during normal operation of the cart, e.g. by the use of air tires.
- EV8.1.11 The hand cart itself must have a label according to EV5.3.7 on its firewall maximum 1.3 m above the ground. The label should be facing towards the person moving the handcart.
- EV8.1.12 The overall floor space used by the fully loaded hand cart must not exceed 1200mm x 800mm. If the TSAC is larger than the allowed floor space, exceptions may be approved by the officials prior to the competition.

EV9 ELECTRICAL SYSTEM FORM (ESF) and FAILURE MODES AND EFFECTS ANALYSIS (FMEA)

EV9.1 [EV ONLY] Electrical System Form (ESF)

- EV9.1.1 Prior to the competition, all teams must submit clearly structured documentation of their entire electrical system (including control and TS) called the ESF.
- EV9.1.2 Templates of the ESF shall be made available on the competition website.

EV9.2 [EV ONLY] Failure Modes and Effects Analysis (FMEA)

- EV9.2.1 Teams must submit a complete failure modes and effects analysis (FMEA) of the TS by the deadline published on the competition website.
- EV9.2.2 A template including required failures to be described will be made available online. Pictures, schematics and data sheets referenced in the FMEA must be included in the FMEA on additional table pages.

IN TECHNICAL INSPECTIONS

IN1 GENERAL

IN1.1 Technical Inspection Process

IN1.1.1 The technical inspection is divided into the following parts:

- Pre-Inspection
- [EV ONLY] Accumulator Inspection
- [EV ONLY] Electrical Inspection
- Mechanical Inspection
- [DC ONLY] Autonomous System Inspection
- Tilt Test
- Vehicle Weighing
- [CV ONLY] Noise Test
- [EV ONLY] Rain Test
- Brake Test
- [DC ONLY] EBS

IN1.1.2 The technical inspection process will be held in the following order:

Combustion	Electric	DC
Pre-Inspection	Pre-Inspection	Pre-Inspection
Mechanical	Accumulator Inspection	Accumulator Inspection
Inspection	Electrical / Mechanical Inspection*	Electrical / Mechanical Inspection*
Tilt Test	Tilt Test	Autonomous System Inspection
Noise Test	Rain Test	Tilt Test
Brakes Test	Brakes Test	Rain Test
		Brake Test
		EBS

*Electrical Inspection and Mechanical Inspection can be conducted in either order.

IN1.2 General Rules

IN1.2.1 Each vehicle must pass all parts of technical inspection before it may participate in any dynamic event.

IN1.2.2 [DC ONLY] Each vehicle must pass all parts of Technical Inspection including Autonomous System Inspection and EBS Test before it may run in Autonomous Mode.

IN1.2.3 Passing the technical inspections is not a certification of complete rules compliance of the vehicle.

IN1.2.4 The technical inspection sheet includes several inspection points and will be provided on the competition website prior to the competition. It must always stay with the push bar.

- IN1.2.5 The officials may inspect other points not mentioned on the technical inspection sheet to ensure compliance with the rules.
- IN1.2.6 Teams are responsible for confirming that their vehicle and the required equipment satisfies the requirements and restrictions of the rules before presenting it for technical inspection.
- IN1.2.7 Vehicles must be presented for technical inspection in ready-to-race condition.
- IN1.2.8 Safety-uncritical violations without team benefit that cannot be changed at the event may result in at least 20 penalty points, depending on officials' decision. *If such violations give a minor benefit, at least 40 points may be applied. Penalties are deducted from the team's overall score.*
- IN1.2.9 All items on the inspection sheet must be clearly visible for the officials without using instruments such as endoscopes or mirrors. Visible access may be provided by removing body panels or by providing removable access panels.
- IN1.2.10 The vehicle must maintain all required specifications throughout the competition.
- IN1.2.11 Officials reserve the right to ask the team for drilling of additional inspection holes to check the chassis for compliance with the rules.
- IN1.2.12 Officials will mark or seal various different approved parts. Removal of, or damage to the seals will void the inspection approval.
- IN1.2.13 Once the vehicle is approved for competition, any damage to the vehicle that requires repair(s) will void the inspection approval. After completion of the repair(s), the vehicle must be re-submitted to technical inspection for re-approval.
- IN 1.2.14 The officials reserve the right to prohibit the use of parts that could pose a safety risk to drivers, track marshalls or the environment.

IN1.3 Technical Inspection Sticker

- IN1.3.1 The competition technical inspection stickers will be placed on the nose of the vehicle.
- IN1.3.2 If a vehicle is no longer in compliance with the rules, the officials will set the vehicle's technical inspection status to fail, remove the respective inspection sticker(s) from the vehicle and note the reason for revoking the technical inspection approval in the technical inspection sheet.

IN1.4 Inspection Responsible Person (IRP)

- IN1.4.1 To accelerate the Technical Inspection process, the team must appoint one team member as Technical Inspection responsible person. *For the Autonomous System Inspection, see IN6, this has to be an ASR. [EV ONLY] For Electrical Inspection and Accumulator Inspection this has to be an ESO.*
- IN1.4.2 This inspection responsible person must be:
- Familiar with the vehicle.
 - Able to show compliance of the vehicle with all points mentioned on the technical inspection sheet.

- Able to perform the technical inspection autonomously observed by the officials, when asked.

IN1.4.3 Should the inspection responsible person be unable to perform one of these requirements, or the vehicle and all necessary items are not ready, the technical inspection will be aborted and the team will be asked to leave the technical inspection area.

IN1.5 Modifications and Repairs

IN1.5.1 After technical inspection, the only modifications permitted to the vehicle are:

- Adjustment of belts, chains and clutches
- Adjustment of the brake bias
- Adjustment of the driver restraint system, head restraint, seat and pedal assembly
- Substitution of the head restraint or seat insert for different drivers
- Adjustment to engine operating parameters, e.g. fuel mixture and ignition timing
- Adjustment of mirrors
- Adjustment of the suspension where no part substitution, other than springs, sway bars and shims, is required
- Adjustment of tire pressure
- Adjustment of winglet angles, but not the position of the complete aerodynamic device in relation to the vehicle
- Replenishment of fluids
- Replacement of defective tires or brake pads. Replacement tires and brake pads must be identical in material/composition/size to those presented and approved at technical inspection.
- Changing wheels and tires for “wet” or “damp” conditions as allowed in [D3.2](#) and [D7.5](#)
- Software calibration changes
- Recharging low voltage batteries
- Recharging TS accumulators
- [\(De-\)Coupling of actuators and sensor repositioning as allowed in T14.7](#)
- Installing and removing protective sensor covers, if approved during technical inspection
- [Replacement of LVbatteries. Replacement LV batteries must be identical and approved at Technical Inspection](#)

[Any other modification must be either approved by or performed under the supervision of Technical Inspection officials.](#)

IN2 PRE-INSPECTION

IN2.1 Pre-Inspection Required Items

IN2.1.1 The following items must be presented for pre-inspection:

- All helmets
- All driver’s equipment and other safety gear
- [Two unused and in date fire extinguishers](#)
- One set of four tires on rims for wet conditions
- One set of four tires on rims for dry conditions

The tire type/rim type combination presented during pre-inspection must be the same during the whole competition. The rims for dry tires and wet tires may be different, but for all dry tire sets and all wet tire sets the same.

IN3 [EV ONLY] ACCUMULATOR INSPECTION

IN3.1 Accumulator Inspection General Definitions

IN3.1.1 Cell modules or stacks do not need to be disassembled when AIRs, fuses, pre- and discharge circuit and positive locking mechanism of the maintenance plugs are reachable and visible for the officials.

IN3.1.2 The accumulator charger will be inspected and sealed.

IN3.1.3 The set of basic tools will be checked.

IN3.2 Accumulator Inspection Required Items

IN3.2.1 The following items must be presented at accumulator inspection:

- All TS accumulators
- Accumulator hand cart
- Accumulator charger
- Basic Tools (see IN3.2.2)
- Tools needed for the (dis)assembly of parts
- Samples of self designed PCBs that are part of the tractive system and inside the accumulator container (see EV4.3.6)
- Data sheets for all parts used in the accumulator
- Original delivery notes for material without serial number printed on according to T1.2.1
- Print-outs of rule questions (if applicable)
- Print-out or digital version of ASES

IN3.2.2 The following basic tools in good condition must be presented:

- Insulated cable shears
- Insulated screwdrivers
- Multimeter with protected probe tips and two 4 mm banana plug test leads rated for 600 V CAT III or better
- Insulated tools, if screwed connections are used in the tractive system
- Face shield
- at least two pairs of HV insulating gloves (not expired)
- Two HV insulating blankets of at least 1.0 m² each
- Safety glasses with side shields for all team members that might work on the tractive system or accumulator

All electrical safety items must be rated for at least the maximum tractive system voltage.

IN4 [EV ONLY] ELECTRICAL INSPECTION

IN4.1 Electrical Inspection General Definitions

IN4.1.1 The insulation resistance between the TS and LVS ground will be measured. Vehicles with a maximum TS voltage less than or equal to 250 V will be probed with 250 V and vehicles with a maximum TS voltage greater than 250 V with 500 V.

To pass this test, the measured insulation resistance must be at least 500 Ω /V related to the maximum TS voltage of the vehicle.

IN4.1.2 The IMD will be tested by connecting a resistor between the TSMP ([see EV4.7](#)) and LVS ground connector ([see EV4.7.8](#)).

The test is passed if the IMD shuts down the TS within 30 s at a fault resistance of 50% below the response value which corresponds to 250 Ω /V.

IN4.1.3 The BSPD will be tested by sending an appropriate signal that represents the current, to achieve ≤ 5 kW whilst pressing the brake pedal. This test must prove the functionality of the complete BSPD except for any commercially available current sensors. Ends of a current transducer's auxiliary winding must be insulated.

IN4.1.2 The team must prove the function of the BSPD current sensor functionality during a technical inspection, simulate ≤ 5 kW by scaling current or coiling wire (e.g., 20 turns for 1/20th current).

IN4.2 Electrical Inspection Required Items

IN4.2.1 The following items must be presented at electrical inspection:

- Vehicle with mounted TS accumulator
- Jack and push bar
- Samples of self designed PCBs that are part of the tractive system and are outside of the accumulator container ([see EV4.3.7](#))
- Tools needed for the BSPD check ([see T11.6.9](#))
- Data sheets for all parts used in the tractive system
- Original delivery notes for material without serial number printed on according to [T1.2.1](#)
- Tools needed for the (dis)assembly of parts for electrical inspection
- Print-outs of rule questions (if applicable)
- The connectors needed
 - to safely close the SDC while the HVD is removed and
 - to safely supply the TS using the same shrouded receptacles as used for TSMP when the TS accumulator is unconnected
- [LV Battery \(if applicable\), and possible replacement batteries](#)

IN5 MECHANICAL INSPECTION

IN5.1 Mechanical Inspection Required Items

IN5.1.1 The following items must be presented at mechanical inspection:

- Jacks and push bar
- The tallest driver of the team
- Copies of any safety structure equivalency forms
- Copies of any impact attenuator data requirement
- Print-outs of rule questions (if applicable)
- Physically tested IA assembly, including (representative) test fixture (if applicable)
- Teams with a monocoque: laminate test specimen(s)
- Teams using alloyed steel: test specimen(s)
- Only tools needed for the (dis)assembly of parts for mechanical inspection

IN6 AUTONOMOUS SYSTEM INSPECTION

IN6.1 Autonomous System Inspection Required Items

IN6.1.1 The following items are required at Autonomous System Inspection

- Data sheets for all perception sensors
- Documents which proof that all perception sensors meet local legislation
- RES remote control
- ASF
- Tools needed for the (dis)assembly of parts for Autonomous System Inspection
- Print-outs of rule questions (if applicable)

IN7 TILT TEST

IN7.1 Tilt Test Procedure

IN7.1.1 The tilt test will be conducted with the tallest driver fully strapped in normal driving position.

IN7.1.2 The tilt test will be conducted with all vehicle fluids at their maximum fill level.

IN7.1.3 The vehicle will be placed upon the tilt table and to an angle of 60°. There must be no fluid leaks and all wheels must remain in contact with the tilt table surface.

IN8 VEHICLE WEIGHING

IN8.1 Vehicle Weighing Procedure

IN8.1.1 All vehicles must be weighed in ready-to-race condition.

IN8.1.2 All fluids must be at their maximum fill level for weighing.

IN9 [EV ONLY] RAIN TEST

IN9.1 Rain Test General Definitions

IN9.1.1 Vehicles must have passed electrical inspection, see [IN4](#), to attempt the rain test.

IN9.2 Rain Test Procedure

IN9.2.1 The vehicle must be in ready-to-race condition. All components and constructions used to protect the vehicle from water during the rain test must be used during the entire competition. [Any fans \(if applicable\) must be switched on.](#)

IN9.2.2 The tractive system must be active during the rain test.

IN9.2.3 The vehicle must be jacked up using the jacks ([see T13.2](#)) and all driven wheels must be removed.

IN9.2.4 The vehicle must not be in ready-to-drive mode ([see EV4.11](#)).

IN9.2.5 The test will be conducted without a driver.

IN9.2.6 Water will be sprayed at the vehicle from any possible direction. The water spray is similar to a vehicle driving in rain and not a direct high-pressure stream of water.

IN9.2.7 The test is passed if the IMD is not triggered [and the TS is still active, see EV4.11.1](#), while water is sprayed at the vehicle for 120s and 120s after the water spray has stopped.

IN10 [CV ONLY] NOISE TEST

IN10.1 Noise Test Procedure

IN10.1.1 The sound level will be measured during a static test.

IN10.1.2 The vehicle must be compliant at all engine speeds up to the maximum test speed ([see CV3.2.1](#)).

IN10.1.3 Teams must bring a laptop to indicate the engine speed measured by the Electronic Control Unit (ECU).

IN10.1.4 Measurements will be made with a free-field microphone placed free from obstructions at the exhaust outlet level, 0.5 m from the end of the exhaust outlet, at an angle of 45° with the outlet in the horizontal plane.

IN10.1.5 Where more than one exhaust outlet is present, the test will be repeated for each exhaust and the highest reading will be used.

IN10.1.6 If the exhaust has any form of active tuning or throttling device or system, it must be compliant with the rules in all positions. Manually adjustable tuning devices must require tools to change them and must not be moved or modified after the noise test is passed. The position of the device must be visible to the officials and manually operable by the officials during the noise test.

- IN10.1.7 The test will be run with the gearbox in neutral. During this test the vehicle must be jacked up using the jack ([see T13.2](#)).
- IN10.1.8 After passing the noise test the function of the master switch, the cockpit-mounted shutdown button and the inertia switch will be tested.
- IN10.1.9 After passing [IN9.1.8](#) the air tightness of the intake system will be tested by closing off the inlet after which the engine must stall.

IN11 BRAKE TEST

IN11.1 Brake Test Procedure

- IN11.1.1 Lock all four wheels and stop the vehicle in a straight line at the end of an acceleration run specified by the officials.
- IN11.1.2 [CV ONLY] The vehicle must come to standstill without stalling the engine.
- IN11.1.3 [EV ONLY] After accelerating, the TS must be switched off by the driver, using the cockpit mounted shutdown button, and the driver must brake using only the mechanical brakes.
- IN11.1.4 [After the Brake Test, the vehicle must be able to continue driving under its own power without external assistance.](#)
- IN11.1.5 The brake light and TSAL illumination will be checked and the officials will verify if the illumination is satisfactory for external observation.
- IN11.1.6 [EV ONLY] The ready-to-drive sound will be checked and the officials will verify if the sound level is satisfactory.
- IN11.1.7 [During the Brake Test, tire pressure may be max. 1.0bar above cold pressure before any dynamic event and must never exceed the manufacturer's limit.](#)
- IN11.1.8 If the vehicle is unable to pass the test after three attempts, the vehicle must be repaired and then brought back for retesting.

IN11.2 EBS Test

- IN11.2.1 The EBS performance will be tested dynamically and must demonstrate the performance described in [T15.4](#).
- IN11.2.2 The test will be performed in a straight line marked with cones similar to Acceleration.
- IN11.2.3 During the Brake Test, the vehicle must accelerate in autonomous mode up to at least 40km/h within 20m. From the point where the RES is triggered, the vehicle must come to a safe stop within a maximum distance that corresponds to the Emergency Brake System Performance requirements according to [T15.4](#).
- IN11.2.4 In case of wet track conditions, the stopping distance will be scaled by the officials dependent on the friction level of the track.
- IN11.2.5 The EBS Test is conducted after all other elements of [IN11](#) have been passed.

IN12 POST EVENT INSPECTION

IN12.1 Post Event Inspection Procedure

IN12.1.1 The officials reserve the right to impound any vehicle at any time during or after any of the dynamic events to check for compliance with the rules. If necessary the vehicle will remain with the officials until the violation of the rule was discussed with the team.

IN12.1.2 After the endurance event, the vehicle must be placed in parc fermé where no team member may access the vehicle.

IN12.1.3 During and after any dynamic event, the vehicle must be in compliance with the rules.

IN12.1.4 For each violation of the rules, the team receives a separate penalty as follows:

- Group A: Violation of the rules without advantage for the team
- Group B: Violation of the rules with advantage for the team (e.g. aerodynamic device)

	Acceleration	Skidpad	Autocross	Endurance
Group A	0.3 s	0.2 s	2 s	30 s
Group B	1 s	0.6 s	5 s	2 min

The penalty applies to all runs since the vehicle entered the dynamic area the last time.

IN12.1.5 Violation of the rules concerning safety or the environment (e.g. BOTS, safety harness issues, ground clearance, fluid leaks, other than plain water, noise) results in a Disqualified (DQ) since entering the dynamic area.

IN12.1.6 Losing a part of the vehicle on the track will result in DQ for that particular run.

IN12.1.7 Changes in vehicle weight of more than ± 5 kg compared to the official technical inspection weight (see [IN7](#)) results in a 20 point penalty for each kg the tolerance is exceeded by. E.g. a weight difference of ± 6.2 kg results in a 40 point penalty.

IN12.1.8 [DC ONLY] If the [Autonomous System Inspection](#) is passed, the vehicle must be able to fully execute the [Inspection Mission](#), see [T14.10.2](#), during Post Inspection after any dynamic discipline. To achieve this, the vehicle may be modified according to [IN1.5.1](#). Violation will result in DQ.

IN12.1.9 [EV ONLY] The vehicle must be able to enter R2D mode, see EV4.11, for all tests requiring this mode during Post Inspection. [This capability must be maintained under all circumstances for at least 15min and without any modifications to the vehicle.](#) Violation will result in DQ.

IN12.1.10 [EV ONLY] Directly after endurance and leaving parc fermé, every team must come back to the charging area to disassemble the temperature logging device (if used by the competition) from the TS accumulator (see [EV5.8.5](#)).

IN12.1.11 Directly after endurance and leaving parc fermé, the data logger see [EV4.6](#) or [T14.2](#) will be disassembled from the vehicle.

S STATIC EVENTS

S1 BUSINESS PLAN PRESENTATION EVENT (BPP)

S1.1 Business Plan Presentation Objective

S1.1.1 The objective of the BPP is to assess the team's ability to develop, present and defend a comprehensive business model [in an investment round scenario](#). The business model must offer a product or a service integrating the team's specific prototype vehicle or a specific component of it. [The business model must be for-profit and provide a rewarding business opportunity](#). Therefore, non-profit organisations are not permitted.

S1.1.2 The following are not eligible as components of the team's current prototype vehicle for the creation of the business model:

- Consumables and fasteners
- Offcuts, scrap, or other by-products arising from the manufacturing process
- Tooling or tools used in the manufacturing process
- Off-vehicle infrastructure or equipment not permanently integrated into the vehicle
- Test, measurement, and calibration equipment not permanently integrated into the vehicle
- Personal protective equipment and driver apparel
- Standalone services, data, or cloud solutions that do not rely on a team-developed on-vehicle component or software
- Using, rebranding, or reselling bought parts without the team's involvement in design, production, or further processing
- Items that are not part of the team's current prototype vehicle

S1.1.3 During the presentation, teams must explicitly show how they derived their proposed product or service from the team's current prototype vehicle.

S1.1.4 The judges should be treated as if they were potential investors or partners for the presented business model. [A team may assign the judges specific roles for their BPP, i.e. as representatives of a specific company as strategic investors](#). Nonetheless, these roles have to allow the judges to pose as potential investors or partners.

S1.1.5 The quality of the actual prototype will not be considered as part of the BPP judging.

S1.2 Business Plan Procedure

S1.2.1 Prior to the competition, the team must submit a pitch video and participate in an online web presentation delivery at the Business Plan Presentation Prelims ([See S1.4](#)).

S1.2.2 The top teams are then judged by all business judges in the BPP finals at the competition.

S1.3 Pitch Video (BPPV)

S1.3.1 To convince the potential investors or partners that the team's presentation is worthy of their time, it is required that a Business Plan Pitch Video (BPPV) is submitted before the

competition. The pitch video should be considered as the first impression of the business idea and has to contain the relevant investment figures.

- S1.3.2 The BPPV must be submitted online via the competition website, not later than the deadline specified in the competition website. The BPPV must be uploaded on a video sharing platform of your choice (Youtube, Vimeo etc.). Please do not upload on Google Drive. The video link must be publicly accessible. If uploaded on Youtube, the privacy setting of the video must be changed to 'Unlisted' so that only those with the link can view the video and it will not be displayed publicly on your channel as well.
- S1.3.3 The link to the BPPV should be submitted within a document consisting of only 1 (one) A4 page. The document must only include the BPPV link. No additional external links to be added.
- S1.3.4 The BPPV must not exceed a length of 60 seconds.
- S1.3.5 A missing pitch video submission will forfeit the team's participation in the prelims. Additionally, if the judges find the pitch video to be of 'poor form', the judges may choose to cancel the team's presentation preliminaries time slot.

S1.4 Executive & Financial Summary

- S 1.4.1 To prepare the potential investors or partners for your team's upcoming presentation, it is required that a Business Plan Executive & Financial Summary (BPEFS) is submitted before the event.
- S 1.4.2 Further requirements regarding the BPEFS will be published on the competition website.
- S 1.4.2 The BPEFS must be submitted online in PDF format via the competition website, not later than the deadline specified on the competition website.

S1.5 Deep Dive Topic

- S 1.5.1 Prior to the competition a specific deep dive topic, which has to be part of the ten minutes presentation, will be published on the competition website.

S1.6 Presentation Prelims

- S1.6.1 The presentation prelims shall take place virtually.
- S1.6.2 Presentations are limited to a maximum of 10min. Anything presented after the 10 min time limit will not be considered for the scoring. The judges may stop any presentation exceeding 10 min.
- S1.6.3 The presentation will not be interrupted by questions. Immediately following the presentation there will be a question and answer session.
- S1.6.4 One or more team members may present the business plan.
- S1.6.5 All team members involved in the BPP must have their video feed turned on, failing to do so will lead to de-registration from the Business Plan Presentation event. The presentation by team members must be delivered 'live' (i.e. not recorded, with the exception of any

promotional video showcased as a part of the delivery). The team members who have been introduced may answer the judges' questions even if they were not actually presenting.

- S1.6.6 For the online presentation, teams must ensure reliability of internet connection and positioning of media transmission devices for best focus and sound. Lighting of space, choice of background and external noise must be paid attention to, to avoid interruptions during the presentation. Teams may choose to switch between live presentation and media display (PowerPoint, video etc.) at any point, during the course of the presentation delivery.

S1.7 Business Plan Presentation Finals

- S1.7.1 Some teams may be chosen to participate in the BPP finals to determine the BPP winner. The BPP finals will be held separately from the initial judging and teams will be informed about their participation during the event.
- S1.7.2 The BPP finals will be a public event at the competition.
- S1.7.3 *Handouts, prototypes, or other supporting material will only be evaluated if explicitly referenced during the presentation or Q&A. All items must be removed by the team after their presentation.*
- S1.7.4 Data projectors or screens with VGA or HDMI (type A) input connectors and audio jacks will be provided for video and audio signal transmission. Teams planning to use other presentation equipment are responsible for bringing it themselves. The competition organizers shall not be liable for mishaps that may occur while connecting personal devices to data projectors or other equipment.

S1.8 Business Plan Presentation Scoring

- S1.8.1 The BPP will be evaluated on the categories specified in the following table:

Category	Points
Executive & Financial Summary	10
Pitch Video	10
Novelty	10
Content	20
Finances	15
Deep Dive Topic	10
Demonstration, Structure and Delivery	10
Q&A - Delivery and General Impression	8
Q&A - Content	7
Total	100

- S1.8.2 If the business model presented fails to meet any requirement from [S1.1.1](#) or [S1.1.2](#), 15 penalty points will be deducted from the team's BPP scoring.
- S1.8.3 The scoring of the BPP is based on the average of the scores given by each of the judges.

S1.8.4 The scoring for the non-finalist is calculated as followed:

$$BPP\ Score = 70 \times \frac{P_{Team}}{P_{Max}}$$

P_{Team} is the score awarded to the team

P_{max} is the highest score awarded to any team not participating in the finals

S1.8.5 The scoring of the BPP finalists will vary from 71 to 75 points and is scored immediately after the BPP finals by all judges.

S2 COST AND MANUFACTURING EVENT

S2.1 Cost and Manufacturing Objective

S2.1.1 The objective of the cost and manufacturing event is to evaluate the team's understanding of the manufacturing processes and costs associated with the construction of a prototype race car. This includes trade off decisions between content and cost, make or buy decisions and understanding the differences between prototype and mass production.

S2.2 Cost and Manufacturing Procedure

S2.2.1 Prior to the competition, three Cost Report Documents (CRD) (see S2.3) must be submitted to the competition website by the deadline specified by the competition. The Cost and Manufacturing event will have 2 rounds: Round 1: Preliminary Judging (Virtual), Round 2: Finals (On-site).

S2.2.2 Round 1: Cost and Manufacturing Prelims will take place virtually. Teams will have to showcase a live view of their physical prototype. This preliminary round shall determine which teams will move to the finals. During the virtual judging session, participants must ensure that their video is turned on and also provide a live feed of the vehicle; failure to comply with either requirement will result in disqualification from the Cost and Manufacturing Event. For the teams to be eligible to participate in round 1, see S2.8.

S2.2.3 Round 2: Cost and Manufacturing Finals will be held at the competition on-site. Teams are allowed to bring electronic, handwritten, or printed handouts, flip charts or similar to the event, but the space available may be limited. Teams may also bring any photographs, drawings, charts, spare parts or other material that they believe are supportive to the event, but the space provided for cost and manufacturing event judging may be limited. The teams must present their vehicle at the designated time to the judges. Teams that miss their time slot will be disqualified from the Cost and Manufacturing event.

S2.2.4 During the competition, a discussion with the judges will take place, next to the team's vehicle. The discussion is split into three parts:

S2.2.5 Part1 "Bill of Material (BOM) Discussion":

A discussion to evaluate the team's ability to prepare an accurate engineering and manufacturing BOM for the complete vehicle. The team must prove the following:

- The specification of the vehicle in the CRD accurately reflects the vehicle brought to the competition.
- The costs and emissions within the Costed Carbonized Bill of Material (CCBOM) part of the BOM are correct and realistic.

- The manufacturing feasibility of the vehicle.

S2.2.6 Part 2 “Cost Understanding”:
A discussion to evaluate the general cost and manufacturing knowledge of the team.

S2.2.7 Part 3 “Real Case”:
A specific task to evaluate the cost and manufacturing knowledge in a certain field. The real case task will be published on the competition website prior to the competition.

S2.3 Cost Report Documents (CRD)

S2.3.1 The CRD consists of the following documents:

- The BOM, including CCBOM, created online on the www.formulastudent.de website.
- The supporting material file uploaded as a pdf file to the competition website
- The cost explanation file uploaded as a pdf file to the competition website

S2.3.2 Changes to the CRD will not be permitted after the deadline.

S2.3.3 A changelog of the changes on the vehicle since the submission deadline of the CRD can be presented to the judges at the BOM discussion [as a separate document](#). Missing parts of the BOM cannot be part of the changelog.

S2.3.4 It is compulsory for at least one set of CRD to be presented as hard copy, printed double-sided, to the event discussion. Any additional sets can be provided as either hard copy or digital versions.

S2.3.5 The website tool will provide a printable pdf version of the BOM.

S2.3.6 The hard copy must be in a ring binder with DIN A4 pages (or similar).

S2.3.7 The team must ensure that two judges are able to review the CRD independently.

S2.3.8 The presented CRD must be identical with the uploaded versions, otherwise five penalty points will be deducted from the team’s cost event score.

S2.4 Bill of Material (BOM)

S2.4.1 The BOM is a sorted list of all vehicle parts.

S2.4.2 The BOM is structured as follows:

- The BOM is broken down into “systems” ([see S2.4.7](#)), which are defined by the website tool.
- Each “system” is broken down into “assemblies”, which are defined by the website tool.
- Each “assembly” can optionally be broken down into “sub-assemblies”, which must be defined by the team.
- Each “assembly” or “sub-assembly” is broken down into “parts”, which must be defined by the team.

S2.4.3 The BOM must list all parts fitted to the prototype vehicle at any time during the competition.

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- S2.4.4 Only dry tires and wheels per [T2.6.1](#) need to be included in the BOM.
- S2.4.5 “Fasteners” are additional items necessary to assemble a “part” or “(sub-)assembly” and should not be listed.
- Examples for “fasteners”: DIN-931 hexagon screws, glue, zip ties, tape used for mounting or fixing, ...
 - Examples that are considered “parts” and not “fasteners”: Tape used for edge protection, safety wire, ...
- S2.4.6 A self-manufactured “fastener” is considered a “part”.
- S2.4.7 The “systems” are:
- ~~Autonomous System~~
 - Brake System
 - Chassis and Body
 - Drivetrain
 - Engine and Tractive System
 - Grounded Low Voltage System
 - Miscellaneous, Fit and Finish
 - Steering System
 - Suspension System
 - Wheels, Wheel Bearings and Tires
- S2.4.8 When adding “parts” to the BOM, the comments section should be reviewed thoroughly. The chosen part name must clearly describe what is included. For example, if a spring is included, the “part” should not be called “damper”.
- S2.4.9 Only metric units must be used within the BOM.
- S2.5 Costed Carbonized Bill of Material**
- S2.5.1 The manufacturing and assembly processes, actual costs and greenhouse gas emissions of all parts from one or two systems of the BOM must be listed in the CCBOM.
- S2.5.2 For 2027, a CCBOM for the “[Engine and Tractive System](#)” must be submitted.
- S2.5.3 Additional to the BOM, the CCBOM must:
- Include the actual associated manufacturing processes for each part as used for the prototype vehicle.
 - Include tooling (e.g. welding jigs, molds, patterns and dies).
 - Include the actual costs of materials, fabrication, bought parts and assembly for each part as used for the prototype vehicle.
 - Include the system’s carbon footprint for each part, manufacturing and assembly process of the prototype vehicle in Carbon Dioxide Equivalents (CO₂e).
- S2.5.4 Each constituent “part” of an “assembly” must be classified as “bought” or “made”.
- S2.5.5 Additional to the structure of the BOM, in the CCBOM each “part” is broken down into
- “materials” used (e.g. aluminium)
 - “processes” that describe the necessary steps for manufacturing
-

- “fasteners” used to assemble the part (if applicable)
- “tooling” required for manufacturing (if applicable)
- “overhead” for additional associated costs and emissions of manufacturing and assembly of a part that are not covered by the points above

- S2.5.6 “Tooling” is the necessary tools used for transforming the “material” into the desired shape. If production tooling is associated with processes that are specific to the part geometry, it must be included. For example the dies to stamp out a chassis bracket are tooling.
- S2.5.7 Hand or power tools must not be included.
- S2.5.8 For “bought parts”, only fasteners have to be included [in the emissions calculation](#) (if applicable). [Fasteners themselves are not considered bought parts](#). If the part was modified, the associated “processes” and “tooling” (if applicable) for this modification must also be included.
- S2.5.9 [A modified bought part is considered a make part, where the bought part acts as the material. Associated “processes”, “tooling”, “fasteners” and “overhead” \(if applicable\) for the modifications must be included. However, the “carbon footprint” of the bought part itself should not be considered.](#)
- S2.5.10 The cost calculations must be completed as realistically as possible and exclude research, development and capital expenditures for real estates (e.g. plant or development hours of the team).
- S2.5.11 All costs must be displayed in EUR. For calculating the prices in EUR from other currencies, the team must provide the exchange rates used.
- S2.5.12 There is no maximum cost. Receipts are not required for any items.
- S2.5.13 The calculations for machine hourly rates, wages (including overheads) etc. must be shown.
- S2.5.14 The “carbon footprint” is calculated in CO₂e based on a “Cradle-to-Gate” Life Cycle Assessment (LCA) of the selected “system(s)”.
- S2.5.15 The “Cradle-to-Gate” analysis includes extraction of raw materials needed, manufacturing of “made parts”, “sub-assemblies” and “assemblies” as well as manufacturing of the “system” (ready to be mounted into the final prototype vehicle).
- S2.5.16 For “bought parts”, only the “carbon footprint” of modifications has to be included.
- S2.5.17 The “carbon footprint” for the transportation between the steps must be included.
- S2.5.18 The “carbon footprint” must exclude research and development.

S2.6 Supporting Material File

- S2.6.1 The supporting material file is a document containing additional information which allows the judges to understand the BOM. It should include drawings, exploded view drawings and/or pictures of the vehicle and the parts included in the BOM.

S2.7 Cost and Emissions Explanation File

- S2.7.1 The cost and emissions explanation file is a document which allows the judges to understand the costs and greenhouse gas emissions within the CCBOM part of the BOM.
- S2.7.2 The cost model used must be pointed out, especially which specific costs are included, e.g. machine operation costs.
- S3.7.3 The methodology used for determining the Global Warming Potential (GWP) in CO₂e also needs to be explained. The specific carbon emission factors associates with e.g one material type need to be included.

S2.8 Cost and Manufacturing Prelims Eligibility

- S2.8.1 To be eligible for the Cost and Manufacturing Prelims, the vehicle must be in a ready-to-race condition. This will be determined based on the team's Photo Submission. The submission will be reviewed, and comments will be provided within 7 days for the team to address. After the rectification deadline (as mentioned on the website), only teams with a complete Photo Submission will qualify for the Cost and Manufacturing Prelims.
- S2.8.2 The completeness of the Photo Submission will be determined at the officials' discretion, and their decision will be final.

S2.9 Cost and Manufacturing Vehicle Condition

- S2.9.1 Vehicles must be presented for cost and manufacturing judging in finished condition, fully assembled, complete, ready-to-race.
- S2.9.2 The judges will not evaluate any vehicle that is presented at the cost and manufacturing event, in what they consider to be an unfinished state and will award zero points for the entire event.
- S2.9.3 Vehicles may be presented for judging without having passed technical inspection, even if final tuning and setup is in progress.
- S2.9.4 Covers and/or parts may be removed during the judging to facilitate access and presentation of components or concepts.

S2.10 Cost and Manufacturing Scoring

- S2.10.1 The cost and manufacturing event will be evaluated on the categories specified in the following table:

Category	Points
Part 1 "BOM Discussion"	50
Part 2 "Cost Understanding"	25
Part 3 "Real Case"	25
Total	100

- S2.10.2 If items are missing from the BOM, points are deducted until 0 points are scored for "BOM and BOM discussion".

- S2.10.3 Some teams may be chosen to participate in the cost and manufacturing finals to determine the cost and manufacturing event winner. The cost and manufacturing finals will be held separately from the initial judging and teams will be informed about their participation during the event.
- S2.10.4 In case of finals the highest achievable scoring for the non-finalists is 80 points.

S3 ENGINEERING DESIGN EVENT

S3.1 Engineering Design Objective

- S3.1.1 The concept of the Engineering Design event is to evaluate the student's engineering process and effort that went into the design of a vehicle, meeting the intent of the competition.
- S3.1.2 Proprietary components and systems that are incorporated into the vehicle design as finished items are not evaluated as a student designed unit, but are only assessed on the team's selection and application of that unit.
- S3.1.3 The Engineering Design event will be split into two rounds. The preliminary judging round and a finals round both will be held at the competition site.

S3.2 Technical Vehicle System Documentation (TVSD)

- S3.2.1 The TVSD should provide the judges with a structured overview of the vehicle and the team's design approach. The level of detail must allow the evaluation of design decisions, engineering methods, and validation procedures according to the engineering categories listed in the [engineering design rubric as published on competition websites](#). Any information required to scope, explain or highlight design features, concepts, methods or objectives that demonstrate the value and performance of the vehicle should be included at the team's discretion.
- S3.2.2 A completed TVSD must be submitted online via the competition website, not later than the deadline specified on the [competition website](#).
- S3.2.3 The TVSD must include three technical drawings (no renderings) showing the vehicle from the front, top and side. Each drawing must contain the overall vehicle dimensions (length, width, and height) clearly annotated at least once.
- S3.2.4 The TVSD will be used to sort the teams into appropriate Engineering Design queues and to support the evaluation during Engineering Design.
- S3.2.5 Any evidence or supplementary documentation referenced in the TVSD must be available at the event and presented to the judges upon request.

S3.3 Engineering Design Event Procedure

- S3.3.1 The design event starts with the submission of the [TVSD](#) and their review respectively.
- S3.3.2 At the competition, teams will present their knowledge and their vehicle to the judges, which will evaluate the teams' performance following the design objectives stated in chapter [S3.1](#).

- S3.3.3 Some teams may be chosen to participate in the design finals at the competition site to determine the engineering design event winner. Teams that have been selected for the finals will be informed about their participation during the event.
- S3.3.4 Teams may showcase any photographs, drawings, charts, spare parts or other material that they believe are supportive to the design event. For the finals, participating teams must note that the space provided will be limited.
- S3.3.5 [EV ONLY] Only sealed accumulator containers which passed the accumulator inspection may be presented or mounted in the vehicle. They must not be opened.
- S3.3.6 [EV ONLY] At the finals, only discharged, below 5% State of Charge, accumulator cells each having all terminals electrically isolated may be presented.
- S3.3.7 [DC ONLY] There is no extra Engineering Design presentation for the DC, however scoring will be adapted, and there will be more judges during the normal presentation focusing on the AS.
- S3.3.8 [DC ONLY] There will be separate Engineering Design finals for the DC

S3.4 Engineering Design Vehicle Condition

- S3.4.1 Vehicles must be presented for design judging in finished condition, fully assembled, complete and ready-to-race.
- S3.4.2 The judges will not evaluate any vehicle that is presented at the design event in what they consider to be an unfinished state and will award zero points for the entire design event.
- S3.4.3 The vehicles may be presented for design judging without having passed technical inspection, even if final tuning and setup is in progress.
- S3.4.4 Covers and/or parts may be removed during the design judging to facilitate access and presentation of components or concepts.

S3.5 Engineering Design Judging Criteria

- S3.5.1 The judges will evaluate the engineering effort based upon the team's TVSD responses to questions and an inspection of the vehicle.
- S3.5.2 The judges will inspect the vehicle to determine if the design concepts are adequate and appropriate for the application (relative to the objectives set forth in the rules).
- S3.5.3 The judges may deduct points if the team cannot adequately explain the engineering and construction of the vehicle.
- S3.5.4 The rubrics for each round will be made available on the competition website.

S3.6 Engineering Design Scoring

- S3.6.1 The overall engineering design event maximum scoring is 150 points for CV/EV. The total points will be distributed among Engineering Design Prelims and Finals.
- S3.6.1 The overall engineering design event maximum scoring is 150 points for DC.

D DYNAMIC EVENT REGULATIONS

D1 DYNAMIC EVENTS GENERAL

D1.1 Definitions

- D1.1.1 Starting – crossing the light barrier at the starting line starts a lap.
- D1.1.2 Finishing – crossing the light barrier at the finish line ends a lap. Can be the same as the starting line.
- D1.1.3 Lap – a complete trip on a defined track, beginning at the start line and ending at the finish line.
- D1.1.4 Run – a single attempt to compete in a dynamic event. Runs consist of one or more lap(s). A run is started when receiving a GREEN FLAG or the go-signal by the officials.
- D1.1.5 Did Not Finish (DNF) – starting a run without finishing it. For the purpose of scoring, a DNF is treated as a DQ.
- D1.1.6 Disqualified (DQ) – being removed from the scoring due to a rule violation.
- D1.1.7 Valid Run – a run that is neither DNF nor DQ.
- D1.1.8 Running Order – the sequence in which teams take part in a dynamic event. If not defined otherwise, the running order is queue based and teams on their first run receive priority.
- D1.1.9 To score points in any dynamic event, a team must have at least one valid run in that event.
- D1.1.10 [Turning Diameter](#) – the diameter of the smallest circle within which a vehicle can be turned round completely

D1.2 Driver Limitations

- D1.2.1 In total, a minimum of three and a maximum of four drivers are allowed for each team.
- D1.2.2 An individual driver must not drive in more than two dynamic events.
- D1.2.3 The endurance and efficiency event is considered a single event.
- D1.2.4 An individual driver must not drive more than two runs in each dynamic event.
- D1.2.5 No more than two drivers are allowed to drive in each dynamic event.

D1.3 Number of Runs in the Dynamic Events

D1.3.1 Table 6 defines the number of runs for each dynamic event.

Dynamic Event	Runs	DC
Acceleration	4	-
[DC Only] Acceleration	-	2
Autocross	4	-
[DC Only] Autocross	-	2
Skidpad	4	-
[DC Only] Skidpad	-	2
Endurance	1	-
[DC Only] Trackdrive	-	1

Table 6: Number of runs for each dynamic event

D1.4 Dynamic Area and Dynamic Vests

- D1.4.1 The technical inspections and all dynamic events are held in the dynamic area.
- D1.4.2 Four dynamic vests are handed out to each team by the officials and must be worn in the dynamic area.
- D1.4.3 Only four members per team, including the driver, may enter the dynamic area.
- D1.4.4 Drivers must not wear the dynamic vest when sitting in the vehicle. The driver's vest must be attached to the pushbar.
- D1.4.5 The number of tools that may be used in this area is restricted to those which can be safely carried by the four team members in one trip.

D1.5 Track Walks

- D1.5.1 There will be track walks before the Autocross and Endurance event and [DC ONLY] [Trackdrive](#).
- D1.5.2 During the track walk only analogue measurement devices may be used.

D2 DRIVING RULES

D2.1 Flags

- D2.1.1 Flag signals are commands that must be obeyed immediately and without question.
- D2.1.2 [There will be no flag signs for vehicles in autonomous mode](#)
- D2.1.3 BLACK FLAG – The driver must pull into the driver change area for discussion with the officials concerning an incident. A time penalty may be assessed.
- D2.1.4 BLACK FLAG WITH ORANGE DOT – Mechanical black flag. The driver must pull into the driver change area for a mechanical inspection of the vehicle, something has been observed that requires a closer inspection.

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- D2.1.5 BLUE FLAG – The driver must pull into the designated passing zone to be passed by a faster competitor. The driver must obey the track marshals' signals at the end of the passing zone.
- D2.1.6 CHECKERED FLAG – The session has been completed. The driver must exit the course at the first opportunity.
- D2.1.7 GREEN FLAG –
- a. The session has started, the driver may enter the course under direction of the track marshals. In case of stalling, the vehicle can be restarted, but the driver has to await another green flag as the opening in traffic may have closed.
 - b. The driver is clear to re-enter the track after using the slow lane to let a faster vehicle pass.
 - c. The driver may pick up speed again after a yellow flag was displayed.
- D2.1.8 RED FLAG – The driver must come to an immediate safe and controlled stop on the course and must follow track marshals directions.
- D2.1.9 YELLOW FLAG – Danger, the driver must slow down, something has happened beyond the flag station, no overtaking unless directed by the track marshals.
- D2.1.10 RED AND YELLOW STRIPED FLAG – The track is slippery or something is on the racing surface that should not be there. The driver must be prepared for evasive maneuvers to avoid the situation.

D2.2 Driving Under Power

- D2.2.1 During driving, the mechanical integrity of the vehicle must be maintained.
- D2.2.2 Vehicles must not be driven in reverse.
- D2.2.3 The vehicle must be capable of starting and restarting without external assistance / batteries at all times.
- D2.2.4 Push starts are prohibited.
- D2.2.5 Vehicles may only be driven under power when running in a dynamic event, on the practice track and during brake test.
- D2.2.6 Driving off-site is absolutely prohibited. Teams found to have driven their vehicle at an off-site location during the period of the competition will be excluded from the competition.
- D2.2.7 Burnouts prior to and during the events are prohibited.
- D2.2.8 When driving in autonomous mode, an ASR has to be present at the race control with the RES. Additionally, a single monitoring device (laptop, tablet, ...) may be brought (no complicated antenna construction or similar!).

D2.3 Ground Clearance

- D2.3.1 Sliding skirts or other aerodynamic devices that by design, fabrication or as a consequence of moving, contact the track surface are prohibited. Any violation may be penalized by a mechanical black flag.

D2.3.2 The ground clearance may be tested at any time.

D2.4 Cones & Markings for Autonomous Running

D2.4.1 Details of the cones used and more detailed track layout figures will be shared on the competition website.

D2.5 Start-up Procedure for Autonomous Running

D2.5.1 No additional equipment (e.g. laptop, jack-up device, pressure tank, etc.) is allowed to start up the vehicle at the staging/starting line.

D2.5.2 If the vehicle does not enter “AS Ready” state within 1min after being staged, the team may be sent back to the preparation area by the officials. In this case any priority is lost.

D2.5.3 The vehicle may only be staged with the steering system in straight position.

D2.5.4 The ASMS may only be switched on by the ASR after approval from an official at the starting line.

D2.6 Vehicle Break Downs and Usage of RES during Autonomous Running

D2.6.1 If a vehicle comes to standstill for any reason, it may have up to 30s to attempt to continue to drive. If the vehicle doesn't restart within 30s, it will be disqualified.

D2.6.2 If the average speed of the first three laps in Trackdrive (after completing the third lap) is below 2.5m/s or the average speed of any of the following laps is below 3.5m/s, the vehicle will be disqualified.

D2.6.3 If a traceable signal loss of the RES appears and doubtless proof can be brought by the team that it was not self-inflicted, a re-run may be granted for the respective autonomous run, if the officials are notified before the affected dynamic discipline closes. The data logger data must be uploaded before notifying the officials.

D2.6.4 At the direction of the officials, team members may be instructed to retrieve broken-down vehicles. This recovery may only be done under the control of the officials.

D2.7 Procedure After Completing a Driverless (DV) Dynamic Discipline

D2.7.1 After approval from the officials the ASR has to deactivate the vehicle using the RES.

D2.7.2 The vehicle must be collected by the ASR and an additional team member immediately after approval from the officials.

D2.8 Practice Track

D2.4.1 A practice track for testing and tuning vehicles is available.

D2.9 Handling of Re-Runs

D2.5.1 A re-run may be granted in following cases:

- Usage of YELLOW FLAG in Autocross.
- Usage of RED FLAG
- After official's decision due to technical or organizational issues.

D2.5.2 The team will be informed about a granted re-run either by a sign at the exit or by direct communication through an official.

D2.5.3 It is at the discretion of the team whether they accept the re-run. If the re-run is accepted, it must be executed immediately and the corresponding initial run is removed from the scoring.

D3 WEATHER CONDITIONS

D3.1 Operating Conditions

D3.1.1 The following track conditions are recognized:

- Dry
- Damp
- Wet

D3.1.2 The operating condition is decided by the officials and may change at any time.

D3.1.3 The current operating condition will be prominently displayed at the dynamic area.

D3.2 Tires Allowed

D3.2.1 Teams must run the tires allowed for each operating condition:

Operating Condition	Tires Allowed
Dry	Dry
Damp	Dry or Wet
Wet	Wet

D3.2.2 When the operating condition is damp, teams may change between dry tires and wet tires:

- Any time during the acceleration, skidpad, and autocross events.
- Any time before taking the GREEN FLAG to start endurance and according to D7.6

D3.2.3 If an event had varied operating conditions, the minimum performance levels to score points may be adjusted if deemed appropriate by the officials.

D3.2.4 Only one set of tires per type (dry/wet) may be used during all the dynamic disciplines. The Brake Test, practice area, [DC ONLY] [Driverless Autocross](#), [DC ONLY] [Trackdrive](#) and the static events are excluded from this regulation.

D4 SKIDPAD EVENT

D4.1 Skidpad Track Layout

- D4.1.1 The skidpad course consists of two pairs of concentric circles in a figure of eight pattern.
- D4.1.2 The centers of these circles are 18.25 m apart. The inner circles are 15.25 m in diameter and the outer circles are 21.25 m in diameter.
- D4.1.3 17 cones are placed around the inside of each inner circle. 13 cones are positioned around the outside of each outer circle, in the pattern shown in the skidpad layout diagram.
- D4.1.4 Each circle is marked with a line, outside the inner circle and inside the outer circle.
- D4.1.5 The driving path is the 3 m wide path between the inner and outer circles. The vehicles enter and exit through gates on a 3 m wide path that is tangent to the circles where they meet.
- D4.1.6 The line between the centers of the circles defines the start/finish line. A lap is defined as traveling around one of the circles, starting and ending at the start/finish line.

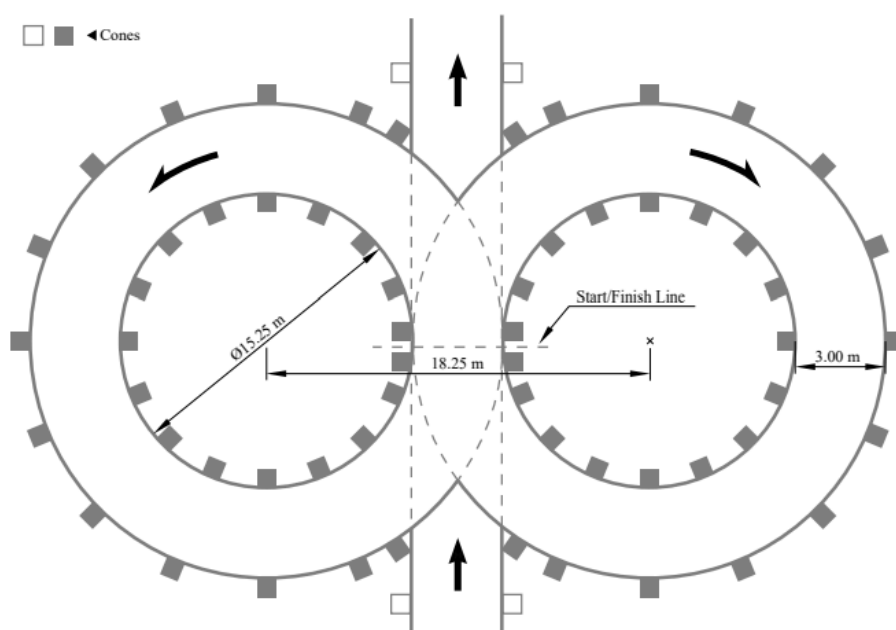


Figure 20: Skidpad Track Layout

D4.2 Skidpad Procedure

- D4.2.1 A run consists of a sequence of laps in exactly the following order. The vehicle must enter perpendicular to the figure of eight and must take one full lap on the right circle to establish the turn. The next lap must be on the right circle and will be timed. Immediately following the second lap, the vehicle must enter the left circle for the third lap. The fourth lap must be on the left circle and will be timed. Immediately upon finishing the fourth lap, the vehicle must exit the track perpendicular to the figure of eight and moving in the same direction as entered.

- D4.2.2 The run time is the average time of the timed left and the timed right circle plus penalties which are added after the averaging.
- D4.2.3 [MANUAL MODE ONLY] The driver has the option to make the second run immediately after their first run.
- D4.2.4 [AUTONOMOUS MODE ONLY] The foremost part of the vehicle is staged 15m in front of the starting line.
- D4.2.5 [AUTONOMOUS MODE ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 25m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in T14.8.

D5 ACCELERATION EVENT

D5.1 Acceleration Track Layout

- D5.1.1 The acceleration track is a straight line with a length of 75 m from starting line to finish line. The track is at least 3 m wide. Cones are placed along the track at intervals of about 5 m.

D5.2 Acceleration Procedure

- D5.2.1 A run consists of a single lap.
- D5.2.2 [MANUAL MODE ONLY] The driver has the option to make the second run immediately after their first run.
- D5.2.3 The foremost part of the vehicle is staged 0.30m behind the starting line.
- D5.2.4 [AUTONOMOUS MODE ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 75m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in T14.8.

D6 AUTOCROSS EVENT

D6.1 Autocross Track Layout

- D6.1.1 The autocross track layout is a handling course built to the following guidelines:
- Straights: No longer than 80 m
 - Slaloms: Cones in a straight line with 7.5 m to 12 m spacing
 - Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
 - The minimum track width is 3 m.
 - The minimum required turning diameter according to [D1.1.10](#) is 9 m.
- D6.1.2 The length of the autocross track is less than 1.5 km
- D6.1.3 [DC ONLY] The Autocross track layout is instead built to the guidelines defined in D8.1.

D6.2 Autocross Procedure

- D6.2.1 A run consists of a single lap.
- D6.2.2 [MANUAL MODE ONLY] Each driver has the option to make the second run immediately after their first run.
- D6.2.3 [AUTONOMOUS MODE ONLY] The vehicle is staged such that the front wheels are 6m in front of the starting line on the track.
- D6.2.4 [AUTONOMOUS MODE ONLY] After finishing the run, the foremost part of the vehicle must come to a full stop within 30m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in T14.8.

D7 ENDURANCE AND EFFICIENCY EVENT

D7.1 Endurance Track Layout

- D7.1.1 The endurance track layout is a closed lap circuit built to the following guidelines:
- Straights: No longer than 80 m
 - Slaloms: Cones in a straight line with 9 m to 15 m spacing
 - Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
 - The minimum track width is 3 m
 - The minimum required turning diameter is 9 m
- D7.1.2 The length of one lap of the endurance track is approximately 1 km.
- D7.1.3 The length of the complete endurance is approximately 22 km.

D7.2 Endurance Procedure

- D7.2.1 Before entering the event, each CV's fuel tank must be filled to the fuel level line, see [CV2.6.4](#), at the fueling station. During fueling, once filled to the scribe line, no shaking or tilting of the tank, the fuel system or the entire vehicle is allowed.
- D7.2.2 A run consists of a multiple laps driven by two different drivers.
- D7.2.3 The first driver will drive for 11 km and will then be signaled into the driver change area.
- D7.2.4 After the driver change specified in [D7.5](#), the second driver will drive for an additional 11 km and will be signaled to exit the track after crossing the finish line.
- D7.2.5 Each lap of Endurance is individually timed. The corrected elapsed time is determined by subtracting the extra-long lap for the driver change from the total time and adding any penalty times.
- D7.2.6 [CV ONLY] The second driver will proceed directly to the fueling station. The tank will be filled to the refill mark and the amount of fuel will be recorded.

D7.3 Endurance Running Order

- D7.3.1 The running order for the endurance will be established in a way that vehicles with similar speeds shall be on the track together.
- D7.3.2 The running order will be published as announced on the competition website.
- D7.3.3 The procedure for handling the running order will be defined on the competition website.

D7.4 Passing

- D7.4.1 During the endurance event, overtaking is only permissible in the designated passing zones and under the control of the track marshals.
- D7.4.2 Passing zones have two parallel lanes, a slow lane only used by the vehicles that are being overtaken and a fast lane for the vehicles that are overtaking.
- D7.4.3 Passing zones may be situated on either the left or right of the fast lane.
- D7.4.4 The passing zone procedure will be as follows:
- A slower leading vehicle will be shown the blue flag and must drive into the slow lane and decelerate.
 - The following faster vehicle will continue in the fast lane to pass the slow vehicle.
 - The vehicle that has been overtaken may re-enter the track when the track marshal who is in charge of that passing zone is showing the green flag.
- D7.4.5 The passing rules do not apply to vehicles that are stopped on track or vehicles that have spun out and are not moving. When passing a stationary vehicle, it is critical to slow down, drive cautiously and to follow the instructions from the track marshals.

D7.5 Endurance Driver Change Procedure

- D7.5.1 Only three team members including the driver may enter the driver change area. They may only bring the tools required to adjust the vehicle to accommodate the second driver and/or change tires.
- D7.5.2 [EV ONLY] One of the team members must be an ESO.
- D7.5.3 During the driver change, the team may:
- perform changes to accommodate the second driver
 - operate the master switch(es)
 - change tires as per D7.6.
- No other work may be performed during the driver change.
- D7.5.4 Each team is given three minutes to change their driver.
- D7.5.5 The driver change time will start once the vehicle is stopped in the driver change area and the first driver has turned off the [CV ONLY] engine / [EV ONLY] TS.
- D7.5.6 [EV ONLY] The TSMS must be switched off by the ESO and the TSAL must have changed to green color, before anyone is allowed to touch the vehicle or to climb out the vehicle.

- D7.5.7 The first driver will climb out the vehicle and any necessary adjustments will be made to the vehicle to fit the second driver (seat cushions, head restraint, pedal position, etc.). The second driver will then be secured in the vehicle.
- D7.5.8 When the second driver is fully secured in the vehicle, the vehicle has restarted and is ready-to-drive again, the driver change time is stopped.
- D7.5.9 If the driver change takes longer than three minutes, the extra time is included in the final time.

D7.6 Endurance Event Tire Changes

- D7.6.1 All tire changes after a vehicle has received the GREEN FLAG to start the endurance event must take place in the driver change area.
- D7.6.2 If the operating condition changes to wet during endurance, the track will be RED FLAGGED or vehicles will be BLACK FLAGGED and brought into the driver change area.
- D7.6.3 If a team wants to change tires, the officials must be informed beforehand.
- D7.6.4 Tire changes can be carried out during the driver change or during an extra stop.
- D7.6.5 If the tire change is carried out during the driver change, in deviation from D7.5, four team members may enter the driver change area.
- D7.6.6 The allowed tire changes and associated conditions are given in table 10.

Existing Operating Conditions	Currently Running on	Condition changes to Dry	Condition changes to Damp	Condition changes to Wet
Dry	Dry Tires	–	A	B
Damp	Dry Tires	–	A	B
Damp	Wet Tires	C	-	–
Wet	Wet Tires	C	C	–

	Requirement	Time Allowed
A	May change from dry to wet	Note 1
B	Must change from dry to wet	Note 1
C	May change from wet to dry	Note 2

Note 1: Any time in excess of 10 minutes without driver change, or 13 minutes with driver change, is added to the team's total time for endurance.

Note 2: The time used to change to dry tires is added to the team's total time for endurance.

- D7.6.7 Teams that have incurred a puncture during the endurance event due to external factors (e.g. debris on track) may change the tire with no time penalty, if the vehicle reaches the driver change area under its own power without external assistance.
- D7.6.8 The wheel will be inspected by the officials. If the deflation was not caused by external factors, the vehicle will be disqualified.

- D7.6.9 Deflation caused by running Off-Course (OC) or impacting barriers or other objects due to driver error will not be regarded as external factors.

D7.7 Endurance Specific Regulations

- D7.7.1 Teams are prohibited from working on or fueling their vehicles during the run.
- D7.7.2 Wheel-to-wheel racing is prohibited.
- D7.7.3 If a vehicle stalls out on the track, it is allowed one lap by the vehicle that is following it (approximately one minute) to restart.
- D7.7.4 If a vehicle has a restart problem at the driver change or after a RED FLAG, it is allowed 2 min to restart the engine or to enter R2D. The 2 min start from the time the driver first tries to restart the engine or to enter R2D. The time is added to the endurance time.
- D7.7.5 If restarts are not accomplished within the above times, the vehicle is disqualified for the run.
- D7.7.6 If a vehicle breaks down it will be removed from the course and will not be allowed to re-enter the course.
- D7.7.7 Team members may be instructed by the officials to retrieve broken down vehicles. This recovery may only be carried out under the control of the officials.

D7.8 Endurance Restart Procedure After Red Flag

- D7.8.1 When the cause of the RED FLAG has been rectified, all vehicles on the track will be guided to the driver change area under YELLOW FLAG in order to receive further instructions from the officials.
- D7.8.2 The lap in which the RED FLAG occurred is completely ignored for endurance and efficiency scoring and repeated after the restart.

D7.9 Endurance Scoring

- D7.9.1 The post event technical inspection (see IN11) must be passed to score points in the endurance event.
- D7.9.2 10% of the maximum points for the event according to table 3 are awarded to every team that finishes endurance without DNF or DQ.

D7.10 Efficiency Procedure

- D7.10.1 Efficiency is measured during the endurance event.
- D7.10.2 Only vehicles which scored minimum $P_{\min}$ points in Endurance are considered for Efficiency scoring.
- D7.10.2 Only vehicles which meet all of the following requirements are considered for efficiency scoring:
- the vehicle received points for the endurance event

- the uncorrected elapsed endurance time does not exceed 1.1 times the uncorrected elapsed time of the fastest vehicle
- [CV ONLY] the consumed fuel mass does not exceed 15 kg / 100 km 91 Octane

D7.10.3 [CV ONLY] Fuel pumps must be on and fuel valves must be opened during refueling.

~~D7.10.4 [CV ONLY] The measured fuel mass of vehicles using E85 fuel is divided by 1.45 to be comparable to the vehicles using 98 RON.~~

D7.10.5 [EV ONLY] The endurance energy is calculated as the time integrated value of the measured voltage multiplied by the measured current logged by the data logger, see [EV4.6](#). Regenerated energy is multiplied by 0.9 and subtracted from the used energy.

D8 [DC ONLY] TRACKDRIVE

D8.1 Trackdrive Tracklayout

D8.1.1 The Trackdrive layout is a closed loop circuit built to the following guidelines:

- Straights: No longer than 80m
- Miscellaneous: Chicanes, multiple turns, decreasing radius turns, hairpin turns, etc.
- The minimum track width is 3m
- The minimum required turning diameter according to D1.1.10 is 9 m

D8.1.2 The length of one lap is approximately 200 m to 500 m.

D8.2 Trackdrive Running Order

D8.2.1 The running order will be defined by the officials.

D8.2.2 The running order will be published as announced on the competition website.

D8.2.3 The procedure for handling the running order will be defined on the competition website.

D8.3 Trackdrive Procedure

D8.3.1 A run consists of ten laps

D8.3.2 The vehicle is staged such that the front wheels are 6m in front of the starting line on the track.

D8.3.3 After ten laps, the foremost part of the vehicle must come to a full stop within 30m behind the finish line, all four wheels must remain within the track boundaries and the vehicle must enter the finish-state described in T14.8.

D8.3.4 There will be no last lap signal i.e. the vehicle should count laps itself.

D8.3.5 Each lap of Trackdrive is individually timed. The corrected elapsed time is determined by adding any penalty times.

D9 DYNAMIC EVENTS SCORING

D9.1 Default Scoring

D9.1.1 Points are given based on the following formula:

$$SCORE = (P_{max} - P_{min}) \left(\frac{T_{max} - T_{team}}{T_{max} - T_{min}} \right)^2 + P_{min}$$

P_{max} is the maximum points for the event according to table 3

P_{min} is the minimum points for the event according to table 7

T_{team} is the team's best time including penalties. T_{team} is capped to T_{max}

T_{max} is defined in table 7.

T_{min} is the time of the fastest vehicle including penalties.

Dynamic Event	T_{max}	P_{min}
Skidpad	$1.35 T_{min}$	$0.05 P_{max}$
Acceleration	$1.7 T_{min}$	$0.05 P_{max}$
Autocross	$1.25 T_{min}$	$0.1 P_{max}$
Endurance	$1.1 T_{min}$	$0.1 P_{max}$
[DC ONLY] Skidpad	$1.7 T_{min}$	$0.05 P_{max}$
[DC ONLY] Skidpad	$2.25 T_{min}$	$0.05 P_{max}$

Table 7: Tmax and Pmin for each dynamic event

D9.2 [DC ONLY] Driverless Cup Scoring

D9.2.1 Points for Skidpad and Acceleration are given according to D9.1.1.

D9.2.2 Points for Autocross are given based on the following formula:

$$AUTOCROSS_SCORE = 0.9P_{max} \left(\frac{T_{max} - T_{team,total}}{T_{max} - T_{min}} \right) + 0.1P_{max}$$

$$T_{team,total} = \min(T_{team,1}, \text{avg}(T_{team,1}, T_{team,2}))$$

$P_{\max}$ is the maximum points for the event according to table 3

$T_{\text{team},1}$ is the team's autonomous mode time including penalties of run 1.

$T_{\text{team},2}$ is the team's autonomous mode time including penalties of run 2.

$T_{\max}$ is the time for driving the lap with 6m/s.

$T_{\min}$ is the fastest autonomous mode $T_{\text{team},\text{total}}$ of all teams.

For runs that are DNF or DQ or for runs with a $T_{\text{team},i}$ above $T_{\max}$, $T_{\text{team},i}$ set to $T_{\max}$

D9.2.3 Points for Trackdrive are given based on the following formula:

$$\text{TRACKDRIVE_SCORE} = 0.75P_{\max} \left(\frac{T_{\max}}{T_{\text{team}}} - 1 \right)$$

$P_{\max}$ is the maximum points for the event according to table 3

T_{team} is the team's corrected elapsed time. T_{team} is capped at $T_{\max}$.

$T_{\max}$ is 2 times of the corrected elapsed time of the fastest vehicle.

D9.2.4 In deviation of D1.1.9, an additional 2.5% of the maximum points for the discipline according to table 3 are given for every completed lap.

D9.3 Efficiency Scoring

D9.3.1 Points for Efficiency are given based on the following formula:

$$\text{EFFICIENCY SCORE} = P_{\max} \left(\frac{EF_{\max} - EF_{\text{Team}}}{EF_{\max} - EF_{\min}} \right)^2$$

with,

$P_{\max}$ is the maximum points for the Efficiency according to table 3

EF_{Team} the team's Efficiency Factor

$EF_{\min}$ the lowest Efficiency Factor of all teams which were considered for Efficiency

$EF_{\max}$ is defined as $2 \cdot EF_{\min}$

D9.3.2 The Efficiency Factor is calculated based on the following formula:

$$EF = T^2 \cdot E$$

with

T uncorrected elapsed driving time

E [CVONLY] corrected used fuel mass / [EV ONLY] used energy

D10 DYNAMIC EVENTS PENALTIES

D10.1 General Penalties

D10.1.1 Penalties will not be assessed for accident avoidance or other reasons deemed sufficient by the officials.

D10.1.2 A cone is Down or Out (DOO) if the cone has been knocked over or the entire base of the cone lies outside the box marked around the cone in its undisturbed position.

D10.1.3 The DOO penalty ([see D9.1.7](#)) is added for each DOO including entry and exit gate cones before the start and after the finish line that occur on that particular run.

D10.1.4 Off-course (OC)

- An OC occurs when the vehicle has all four wheels outside the course boundary as indicated by edge marking.
- Missing one or more gates of a given slalom at autocross or endurance is counted as one OC per occurrence.
- When an OC occurs, the driver must re-enter the track at the next possible point.
- When re-entering the driver needs to wait for a gap and follow the instructions of the track marshals.

D10.1.6

	Acceleration	Skidpad	Autocross	Endurance
DOO	2 s	0.2 s	2 s	2 s
OC	DQ	DQ	10 s	10 s

D10.1.7 DNF equals zero points for that run.

D10.1.8 DQ equals zero points for that run.

D10.1.9 If a team Did Not Attempt (DNA) an event the score is zero points.

D10.1.10 Each run with an incorrect number of laps at skidpad is classified as DNF.

D10.1.11 Failure to obey a flag: 1 minute time penalty.

D10.1.12 Reckless or aggressive driving or “Over Driving”: BLACK FLAG ([see D2.1.3](#))

D10.1.13 Vehicle to vehicle contact: 2 minutes up to DQ depending on the nature of the incident.

D10.1.14 If a vehicle stalls before passing the finish line and cannot restart without external assistance, the vehicle is DNF for that run.

D10.1.15 If a vehicle is not able to start the run, within 30 s after receiving a GREEN FLAG or go-signal, the vehicle is DQ for that run.

D10.2 Endurance Penalties

D9.2.1 Teams that are running out-of-order as defined on the competition website, are penalized by two minutes and may then run at the discretion of the officials.

-
- D10.2.2 Mechanical Problem: The time taken for mechanical inspection under a “mechanical black flag” (see [D2.1.4](#)) is considered officials’ time and is not included in the team’s total time. If the inspection reveals a mechanical integrity problem the vehicle is scored DNF.
- D10.2.3 If it is impossible to impose a penalty by a stop under a black flag, e.g. not enough laps left, the officials may add an appropriate time penalty to the team’s elapsed time.
- D10.2.4 A team may be disqualified if their vehicle is too slow or being driven without proper control. A vehicle is considered to be too slow if the average speed, based on lap times, for two consecutive laps is below 8 m/s in dry conditions or below 5 m/s in wet or damp conditions.
- D10.2.5 Any violation to the procedure of driver change ([D7.5.3](#)) will lead to a time penalty of up to two minutes.
- D10.2.6 All rules and penalties from [IN12](#) are also applicable during driver change.
- D10.3 [CV ONLY] Efficiency Penalties**
- D10.3.1 Should the fuel level change after refueling due to effects such as described in [CV2.7.1](#), then the difference in fuel level will be measured and twice the amount will be added to the official fuel consumption figure.
- D10.4 [EV ONLY] Power and Voltage Violation Penalties**
- D10.4.1 A violation is defined as using more than the maximum power (see [EV2.2](#)) or exceeding the specified voltage (see [EV4.1.1](#)) after a moving average over 500 ms is applied to the respective data logger signal (see [EV4.6](#)).
- D10.4.2 Each violation is a DQ for the fastest valid run of the event. If a violation occurred during two runs, the two fastest valid runs will be DQ etc.
- D10.4.3 The respective data and the resulting decision of violations may be made public.
- D10.4.4 Non-availability of data logger data due to the team’s fault will be treated as a violation.